

Table 19.10 Organization of *Bergey's Manual of Systematic Bacteriology*

Taxonomic Rank	Representative Genera	Textbook Coverage
Volume 1. The Archaea and the Deeply Branching and Phototrophic Bacteria		
Domain Archaea		
Phylum Crenarchaeota	<i>Thermoproteus</i> , <i>Pyrodictium</i> , <i>Sulfolobus</i>	pp. 456–58
Phylum Euryarchaeota		
Class I. <i>Methanobacteria</i>	<i>Methanobacterium</i>	pp. 458–61
Class II. <i>Methanococci</i>	<i>Methanococcus</i>	
Class III. <i>Halobacteria</i>	<i>Halobacterium</i> , <i>Halococcus</i>	pp. 461–63
Class IV. <i>Thermoplasmata</i>	<i>Thermoplasma</i> , <i>Picrophilus</i>	p. 463
Class V. <i>Thermococci</i>	<i>Thermococcus</i> , <i>Pyrococcus</i>	p. 463
Class VI. <i>Archaeoglobi</i>	<i>Archaeoglobus</i>	p. 463
Class VII. <i>Methanopyri</i>	<i>Methanopyrus</i>	p. 458
Domain Bacteria		
Phylum Aquificae	<i>Aquifex</i> , <i>Hydrogenobacter</i>	p. 467
Phylum Thermotogae	<i>Thermotoga</i> , <i>Geotoga</i>	pp. 467–68
Phylum Thermodesulfobacteria	<i>Thermodesulfobacterium</i>	
Phylum “Deinococcus-Thermus”	<i>Deinococcus</i> , <i>Thermus</i>	p. 468
Phylum Chrysiogenetes	<i>Chrysogenes</i>	
Phylum Chloroflexi	<i>Chloroflexus</i> , <i>Herpetosiphon</i>	p. 470
Phylum Thermomicrobia	<i>Thermomicrobium</i>	
Phylum Nitrospira	<i>Nitrospira</i>	
Phylum Deferribacteres	<i>Geovibrio</i>	
Phylum Cyanobacteria	<i>Prochloron</i> , <i>Synechococcus</i> , <i>Pleurocapsa</i> , <i>Oscillatoria</i> , <i>Anabaena</i> , <i>Nostoc</i> , <i>Stigonema</i>	pp. 471–76
Phylum Chlorobi	<i>Chlorobium</i> , <i>Pelodictyon</i>	pp. 470–71
Volume 2. The Proteobacteria		
Phylum Proteobacteria		
Class I. Alphaproteobacteria	<i>Rhodospirillum</i> , <i>Rickettsia</i> , <i>Caulobacter</i> , <i>Rhizobium</i> , <i>Brucella</i> , <i>Nitrobacter</i> , <i>Methylobacterium</i> , <i>Beijerinckia</i> , <i>Hyphomicrobium</i>	pp. 487–95
Class II. Betaproteobacteria	<i>Neisseria</i> , <i>Burkholderia</i> , <i>Alcaligenes</i> , <i>Comamonas</i> , <i>Nitrosomonas</i> , <i>Methylophilus</i> , <i>Thiobacillus</i>	pp. 495–98
Class III. Gammaproteobacteria	<i>Chromatium</i> , <i>Leucothrix</i> , <i>Legionella</i> , <i>Pseudomonas</i> , <i>Azotobacter</i> , <i>Vibrio</i> , <i>Escherichia</i> , <i>Klebsiella</i> , <i>Proteus</i> , <i>Salmonella</i> , <i>Shigella</i> , <i>Yersinia</i> , <i>Haemophilus</i>	pp. 498–507
Class IV. Deltaproteobacteria	<i>Desulfovibrio</i> , <i>Bdellovibrio</i> , <i>Myxococcus</i> , <i>Polyangium</i>	pp. 507–13
Class V. Epsilonproteobacteria	<i>Campylobacter</i> , <i>Helicobacter</i>	p. 514
Volume 3. The Low G + C Gram-Positive Bacteria		
Phylum Firmicutes		
Class I. Clostridia	<i>Clostridium</i> , <i>Peptostreptococcus</i> , <i>Eubacterium</i> , <i>Desulfotomaculum</i> , <i>Heliobacterium</i> , <i>Veillonella</i>	pp. 523–25
Class II. Mollicutes	<i>Mycoplasma</i> , <i>ureaplasma</i> , <i>Spiroplasma</i> , <i>Acholeplasma</i>	pp. 518–21
Class III. Bacilli	<i>Bacillus</i> , <i>Caryophanon</i> , <i>Paenibacillus</i> , <i>Thermoactinomyces</i> , <i>Lactobacillus</i> , <i>Streptococcus</i> , <i>Enterococcus</i> , <i>Listeria</i> , <i>Leuconostoc</i> , <i>Staphylococcus</i>	pp. 525–33
Volume 4. The High G + C Gram-Positive Bacteria		
Phylum Actinobacteria		
Class Actinobacteria	<i>Actinomyces</i> , <i>Micrococcus</i> , <i>Arthrobacter</i> , <i>Corynebacterium</i> , <i>Mycobacterium</i> , <i>Nocardia</i> , <i>Actinoplanes</i> , <i>Propionibacterium</i> , <i>Streptomyces</i> , <i>Thermomonospora</i> , <i>Frankia</i> , <i>Actinomadura</i> , <i>Bifidobacterium</i>	pp. 539–49
Volume 5. The Planctomycetes, Spirochaetes, Fibrobacteres, Bacteroidetes, and Fusobacteria		
Phylum Planctomycetes	<i>Planctomyces</i> , <i>Gemmata</i>	p. 477
Phylum Chlamydiae	<i>Chlamydia</i>	pp. 477–78
Phylum Spirochaetes	<i>Spirochaeta</i> , <i>Borrelia</i> , <i>Treponema</i> , <i>Leptospira</i>	pp. 479–81
Phylum Fibrobacteres	<i>Fibrobacter</i>	
Phylum Acidobacteria	<i>Acidobacterium</i>	
Phylum Bacteroidetes	<i>Bacteroides</i> , <i>Porphyromonas</i> , <i>Prevotella</i> , <i>Flavobacterium</i> , <i>Sphingobacterium</i> , <i>Flexibacter</i> , <i>Cytophaga</i>	
Phylum Fusobacteria	<i>Fusobacterium</i> , <i>Streptobacillus</i>	pp. 481–83
Phylum Verrucomicrobia	<i>Verrucomicrobium</i>	
Phylum Dictyoglomi	<i>Dictyoglomus</i>	

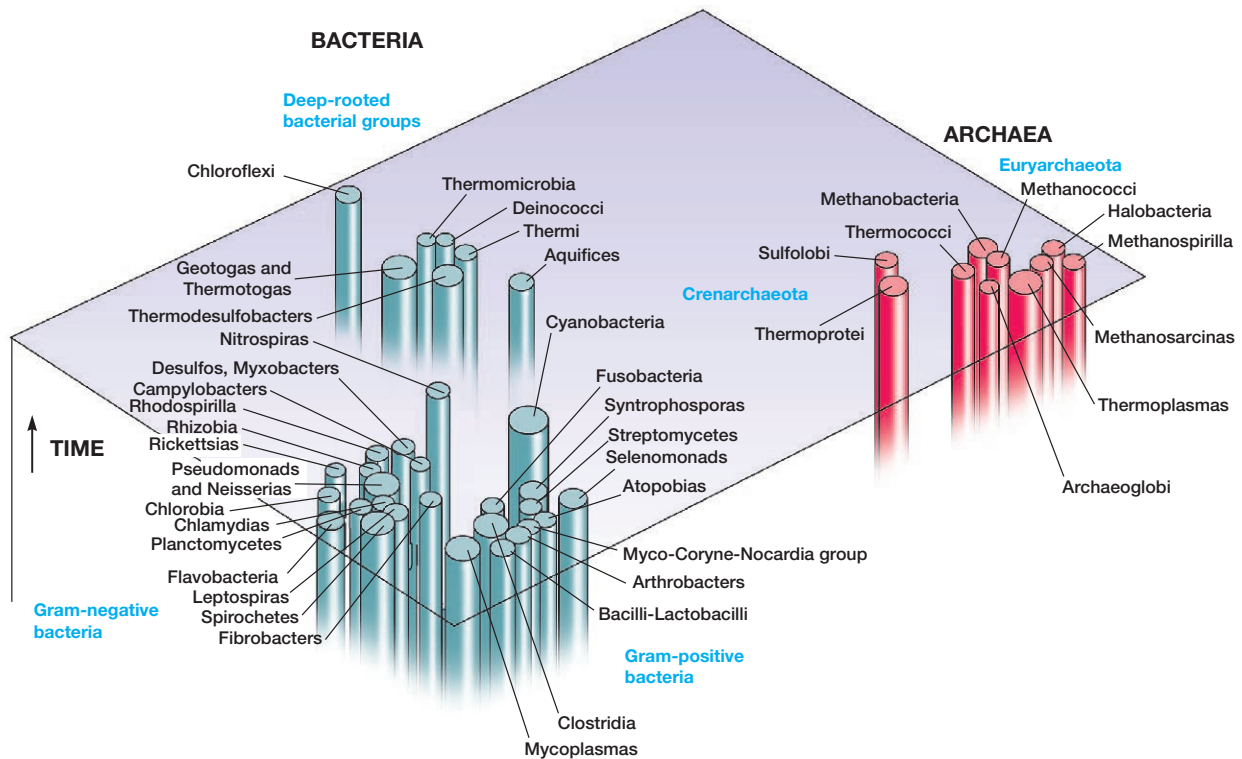


Figure 19.13 Major Prokaryotic Groups and Their Relatedness. Disk size is roughly proportional to the relative number of sequenced prokaryotes in each group. Closely related prokaryotic groups are clustered together. Note that the two prokaryotic domains (*Bacteria* and *Archaea*) are clearly separate. The cylinders faded out to indicate that the antiquity of these groups is uncertain.

19.9 A Survey of Prokaryotic Phylogeny and Diversity

Before beginning a detailed introduction to prokaryotic diversity, it might be best to very briefly survey the major groups in the order they are discussed in the second edition of *Bergey's Manual*. This overview is meant only as a general survey of prokaryotic diversity. The second edition places prokaryotes into 25 phyla, only some of which will be mentioned here. Many of these groups will be discussed in much more detail in chapters 20 through 24. Recall that all organisms may be placed in one of three domains or empires as depicted in the universal phylogenetic tree (figure 19.3). We are concerned only with the prokaryotic domains, *Archaea* and *Bacteria*, in this survey.

Volume 1 contains a wide diversity of prokaryotes in two domains: the *Archaea* and the *Bacteria*. The *Archaea* differ from *Bacteria* in many ways as summarized in table 19.8. At present, they are divided into two phyla based on rRNA sequences (figure 19.14). The phylum *Crenarchaeota* contains thermophilic and hyperthermophilic sulfur-metabolizing organisms of the orders *Thermoproteales*, *Desulfurococcales*, and *Sulfolobales*. However, recently many other *Crenarchaeota* have been discovered. Some are inhibited by sulfur; others grow in the oceans at low temperatures as picoplankton. The phylum clearly is more diverse than first thought. The second

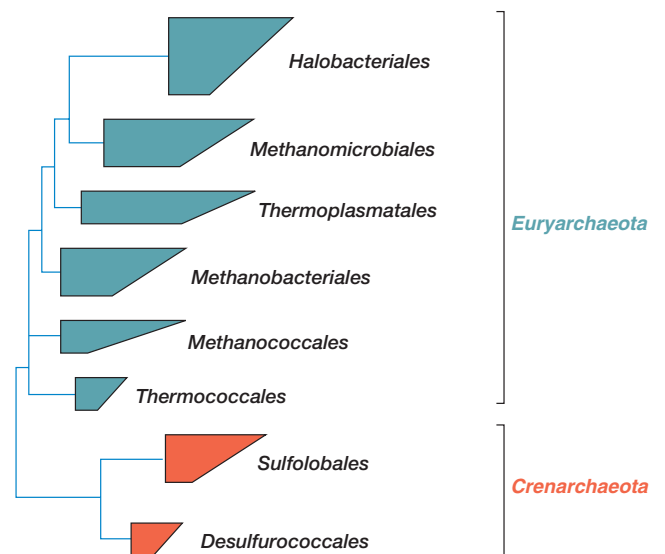


Figure 19.14 Phylogeny of the Archaea. The tree is based on 16S rRNA data and shows relationships between the better-studied orders. Each tetrahedron represents a group of related organisms; its horizontal edges indicate the shortest and longest branches in the group. See text for discussion.

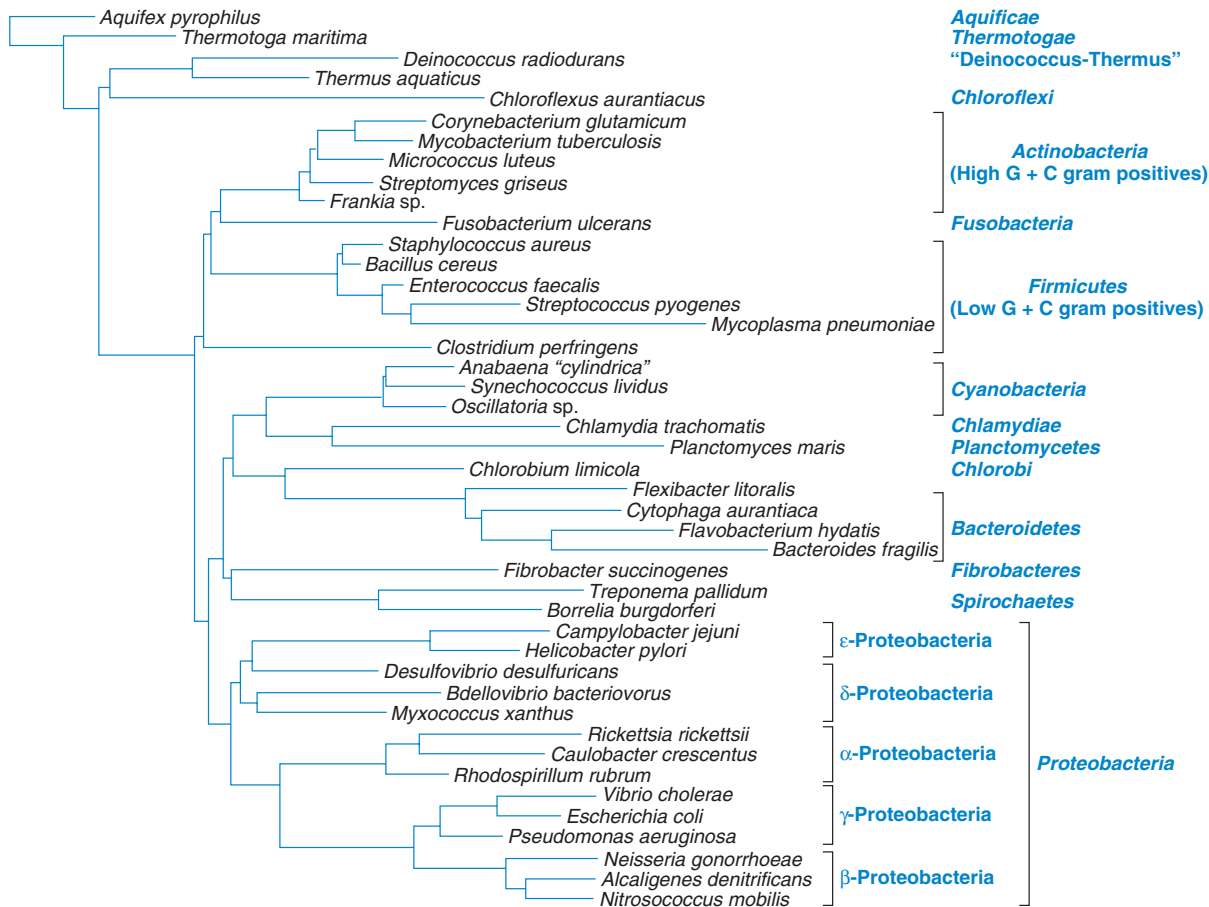


Figure 19.15 Phylogeny of the Bacteria. The tree is based on 16S rRNA comparisons. See text for discussion. Source: *The Ribosomal Database Project*.

phylum, the *Euryarchaeota*, contains primarily methanogenic procaryotes and halophilic procaryotes; thermophilic, sulfur-reducing organisms (the thermoplasmas and thermococci) also are in this phylum. The two phyla are divided into eight classes and 12 orders.

The bacteria are an extraordinarily diverse assemblage of procaryotes that have been divided into 23 phyla (figure 19.15). In volume 1 are placed deeply branching bacterial groups and phototrophic bacteria. The more important phyla are described in the following sections.

1. Phylum *Aquificae*. The phylum *Aquificae* contains autotrophic bacteria such as *Aquifex* and *Hydrogenobacter* that can use hydrogen for energy production. *Aquifex* (meaning "water maker") actually produces water by using hydrogen to reduce oxygen. This group contains some of the most thermophilic organisms known and is the deepest or earliest branch of the bacteria.
2. Phylum *Thermotogae*. This phylum is composed of one class and five genera. *Thermotoga* and other members of the class *Thermotogae* are anaerobic, thermophilic, fermentative, gram-negative bacteria that have unusual fatty

acids and resemble *Aquifex* with respect to their ether-linked lipids.

3. Phylum "Deinococcus Thermus." The order *Deinococcales* contains bacteria that are extraordinarily radiation resistant. The genus *Deinococcus* is gram positive. It has high concentrations of carotenoid pigments, which may protect it from radiation, and unique lipids.
4. Phylum *Chloroflexi*. The phylum *Chloroflexi* has one class and two orders. Many members of this gram-negative group are called green nonsulfur bacteria. *Chloroflexus* carries out anoxygenic photosynthesis and is a gliding bacterium; in contrast, *Herpetosiphon* is a nonphotosynthetic, respiratory gliding bacterium. Both genera have unusual peptidoglycans and lack lipopolysaccharides in their outer membranes.
5. Phylum *Cyanobacteria*. The oxygenic photosynthetic bacteria are placed in the phylum *Cyanobacteria*, which contains the class *Cyanobacteria* and five subsections. Cyanobacteria have chlorophyll *a* and almost all species possess phycobilins. These bacteria can be unicellular or filamentous, either branched or unbranched. The

cyanobacteria in the subsections differ from each other in general morphological characteristics and reproduction. Cyanobacteria incorporate CO₂ photosynthetically through use of the Calvin cycle just like plants and many purple photosynthetic bacteria.

6. Phylum *Chlorobi*. The phylum *Chlorobi* contains anoxygenic photosynthetic bacteria known as the green sulfur bacteria. They can incorporate CO₂ through the reductive tricarboxylic acid cycle rather than the Calvin cycle and oxidize sulfide to sulfur granules, which accumulate outside the cell.

Volume 2 of the second edition is devoted completely to the gram-negative proteobacteria, often called the purple bacteria. The phylum *Proteobacteria* is a large and extremely complex group that currently contains over 1,300 species in 384 genera. Even though they are all related, the group is quite diverse in morphology, physiology, and life-style. All major nutritional types are represented: phototrophy, heterotrophy, and chemolithotrophy of several varieties. Many species important in medicine, industry, and biological research are proteobacteria. Obvious examples are the genera *Escherichia*, *Neisseria*, *Pseudomonas*, *Rhizobium*, *Rickettsia*, *Salmonella*, and *Vibrio*. The phylum is divided into five classes based on rRNA data. Because photosynthetic bacteria are found in the α , β , and γ classes of the proteobacteria, many believe that the whole phylum arose from a photosynthetic ancestor. Presumably many strains lost photosynthesis when adapting metabolically to new ecological niches.

1. Class I—*Alphaproteobacteria*. The α -proteobacteria include most of the oligotrophic forms (those capable of growing at low nutrient levels). *Rhodospirillum* and other purple nonsulfur bacteria are photosynthetic. Some genera have unusual metabolic modes: methylotrophy (e.g., *Methylobacterium*), chemolithotrophy (*Nitrobacter*), and nitrogen fixation (*Rhizobium*). *Rickettsia* and *Brucella* are important pathogens. About half of the microbes in this group have distinctive morphology such as prosthecae (*Caulobacter*, *Hyphomicrobium*).
2. Class II—*Betaproteobacteria*. The β -proteobacteria overlap the α subdivision metabolically. However the β -proteobacteria tend to use substances that diffuse from organic decomposition in the anaerobic zone of habitats. Some of these bacteria use such substances as hydrogen (*Alcaligenes*), ammonia (*Nitrosomonas*), methane (*Methylobacillus*), or volatile fatty acids (*Burkholderia*).
3. Class III—*Gammaproteobacteria*. The γ -proteobacteria compose a large and complex group of thirteen orders and 20 families. They often are chemoorganotrophic, facultatively anaerobic, and fermentative. However, there is considerable diversity among the γ -proteobacteria with respect to energy metabolism. Some important families such as *Enterobacteriaceae*, *Vibrionaceae*, and *Pasteurellaceae* use the Embden-Meyerhof pathway and the pentose phosphate pathway. Others such as the *Pseudomonadaceae* and *Azotobacteriaceae* are aerobes and have the Entner-Doudoroff and pentose phosphate pathways. A few are photosynthetic (e.g., *Chromatium* and *Ectothiorhodospira*), methylotrophic (*Methylococcus*), or sulfur-oxidizing (*Beggiatoa*).

4. Class IV—*Deltaproteobacteria*. The δ -proteobacteria contain seven orders, and 17 families. Many of these bacteria can be placed in one of three groups. Some are predators on other bacteria as the class name implies (e.g., *Bdellovibrio*). The order *Myxococcales* contains the fruiting myxobacteria such as *Myxococcus*, *Stigmatella*, and *Polyangium*. The myxobacteria often also prey on other bacteria. Finally, the class has a variety of anaerobes that generate sulfide from sulfate and sulfur while oxidizing organic nutrients (*Desulfovibrio*).
5. Class V—*Epsilonproteobacteria*. This section is composed of only one order, *Campylobacteriales*, and two families. Despite its small size two important pathogenic genera are ϵ -proteobacteria: *Campylobacter* and *Helicobacter*.

Volume 3 of *Bergey's Manual* surveys the gram-positive bacteria with low G + C content in their DNA, which are members of the phylum *Firmicutes*. The dividing line is about 50% G + C; bacteria with a mol% lower than this value are in volume 3. Most of these bacteria are gram positive and heterotrophic. However because of their close relationship to low G + C gram-positive bacteria, the mycoplasmas are placed here even though they lack cell walls and stain gram negative. There is considerable variation in morphology: some are rods, others are cocci, and mycoplasmas are pleomorphic. Endospores may be present. The phylum contains three classes.

1. Class I—*Clostridia*. This class contains three orders and 11 families. Although they vary in morphology and size, the members tend to be anaerobic. Genera such as *Clostridium*, *Desulfotomaculum*, and *Sporohalobacter* form true bacterial endospores; many others do not. *Clostridium* is one of the largest bacterial genera.
2. Class II—*Mollicutes*. The class *Mollicutes* contains five orders and six families. Members of the class often are called mycoplasmas. These bacteria lack cell walls and cannot make peptidoglycan or its precursors. Because mycoplasmas are bounded by the plasma membrane, they are pleomorphic and vary in shape from cocci to helical or branched filaments. They are normally nonmotile and stain gram negative because of the absence of a cell wall. In contrast with almost all other bacteria, most species require sterols for growth. The genera *Mycoplasma* and *Spiroplasma* contain several important animal and plant pathogens.
3. Class III—*Bacilli*. This large class comprises a wide variety of gram positive, aerobic or facultatively anaerobic, rods and cocci. The class *Bacilli* has two orders, *Bacillales* and *Lactobacillales*, and 16 families. As with the members of the class *Clostridia*, some genera (e.g., *Bacillus*, *Sporosarcina*, *Paenibacillus*, and *Sporolactobacillus*) form true endospores. The section contains many medically and industrially important genera: *Bacillus*, *Lactobacillus*, *Streptococcus*, *Lactococcus*, *Enterococcus*, *Listeria*, and *Staphylococcus*.

Volume 4 is devoted to the high G + C gram positives, those bacteria with mol% values above 50 to 55%. All bacteria in this volume are placed in the phylum *Actinobacteria* and class *Actinobacteria*. There is enormous morphological variety among these prokaryotes. Some are cocci, others are regular or irregular rods. High G + C gram positives called actinomycetes often form

complex branching hyphae. Although none of these bacteria produce true endospores, many genera do form a variety of asexual spores and some have complex life cycles. There is considerable variety in cell wall chemistry among the high G + C gram positives. For example, the composition of peptidoglycan varies greatly. Mycobacteria produce large mycolic acids that distinguish their walls from those of other bacteria.

The taxonomy of these bacteria is very complex. There are five subclasses, six orders, 14 suborders, and 40 families. Genera such as *Actinomyces*, *Arthrobacter*, *Corynebacterium*, *Micrococcus*, *Mycobacterium*, and *Propionibacterium* were placed in volume 2 of the first edition. They are now in the new volume 4 within the suborders *Actinomycineae*, *Micrococcineae*, *Corynebacterineae*, and *Propionibacterineae* because rRNA studies have shown them to be actinobacteria. The largest and most complex genus is *Streptomyces*, which contains over 500 species.

Volume 5 describes an assortment of nine phyla that are located here for convenience. The inclusion of these groups in volume 5 does not imply that they are directly related. Although they are all gram-negative bacteria, there is considerable variation in morphology, physiology, and life cycle pattern. Several genera are of considerable biological or medical importance. We will briefly consider four of the nine phyla.

1. Phylum *Planctomycetes*. The planctomycetes are related to the chlamydias according to their rRNA sequences. The phylum contains only one order, one family, and four genera. Planctomycetes are coccoid to ovoid or pear-shaped cells that lack peptidoglycan. Some have a membrane-enclosed nucleoid. Although they are normally unicellular, the genus *Isosphaera* will form chains. They divide by budding and may produce nonprosthecae appendages called stalks. Planctomycetes grow in aquatic habitats, and many move by flagella or gliding motility.
2. Phylum *Chlamydiae*. This small phylum contains one class, one order, and four families. The genus *Chlamydia* is by far the most important genus. *Chlamydia* is an obligately intracellular parasite with a unique life cycle involving two distinctive stages: elementary bodies and reticulate bodies. These bacteria do resemble planctomycetes in lacking peptidoglycan. They are small coccoid organisms with no appendages. Chlamydias are important pathogens and cause many human diseases.
3. Phylum *Spirochaetes*. This phylum contains helically shaped, motile, gram-negative bacteria characterized by a unique morphology and motility mechanism. The exterior boundary is a special outer membrane that surrounds the protoplasmic cylinder, which contains the cytoplasm and nucleoid. Periplasmic flagella lie between the protoplasmic cylinder and the outer membrane. The flagella rotate and move the cell even though they do not directly contact the external environment. These chemoheterotrophs can be free living, symbiotic, or parasitic. For example, the genera *Treponema* and *Borrelia* contain several important human pathogens. The phylum has one class, *Spirochaetes*, three families, and 13 genera.

4. Phylum *Bacteroidetes*. This phylum has three classes (*Bacteroides*, *Flavobacteria*, and *Sphingobacteria*), three orders, and 12 families. Some of the better-known genera are *Bacteroides*, *Flavobacterium*, *Flexibacter*, and *Cytophaga*. The gliding bacteria *Flexibacter* and *Cytophaga* are ecologically significant and will be discussed later.

Because *Bergey's Manual* is the principal resource in procaryotic taxonomy and is used by microbiologists around the world, we will follow *Bergey's Manual* in organizing the survey of procaryotic diversity, Chapters 20 through 24. In so far as possible, the organization of the second edition of *Bergey's Manual* will be employed. Chapter 20 is devoted to the *Archaea*. Chapter 21 covers the bacteria of volumes one and five except the *Archaea*. Chapter 22 is devoted to the proteobacteria. Chapters 23 and 24 deal with the low G + C and high G + C gram-positive bacteria, respectively. Chapter contents will follow the overall phylogenetic scheme of *Bergey's Manual*. Phylogenetic and organizational details may well change somewhat before publication of each volume, but the general picture should adequately reflect the second edition.

The classification in the first and second editions of *Bergey's Manual* are so different that two appendixes are provided to help in the transition. Appendix III gives the classification of procaryotes according to the first edition. Appendix IV describes the classification system the second edition of *Bergey's Manual* will employ.

Finally, it must be emphasized that procaryotic nomenclature is as much in flux as classification. The names of families and genera are fairly well established and stable in the new system (at least in the absence of future discoveries); in fact, many family and genus names remain unchanged in the second edition of *Bergey's Manual*. In contrast, the names of orders and higher taxa are not always completely settled in the second edition. The cautions expressed in Box 19.1 must always be kept in mind. An excellent way of keeping up to date is through the use of the *Bergey's Manual* Trust web page (www.cme.msu.edu/Bergeys/). The web page has a variety of information about such topics as *Bergey's Manual*, microbial databases, and culture collections. The web site also contains *Bergey's* revision of the Ribosomal Database Project (RDP) phylogenetic tree and lists of validly named procaryotic species. The phylogenetic tree and procaryotic species lists provide a current view of *Bergey's* phylogenetic procaryotic classification. Because the names of kingdoms, classes, and orders are still changing, their use will be kept to the minimum necessary for consistency with the use of taxa by the first edition, ease of communication, and student comprehension. Some higher taxonomic names may well change in the next few years, but we will still employ them here for the above reasons.

-
1. Briefly summarize the contents of each of the five volumes in the second edition.
 2. Give some ways in which the five classes of proteobacteria differ from each other.
 3. In what phyla (and classes of *Proteobacteria* and *Firmicutes*) are the following placed: cyanobacteria, green nonsulfur bacteria, *Rickettsia*, the *Enterobacteriaceae*, *Campylobacter*, *Clostridium*, the mycoplasmas, *Bacillus*, *Streptomyces* and *Mycobacterium*, *Chlamydia*, *Treponema*, and *Cytophaga*?
-

Summary

1. Taxonomy, the science of biological classification, is composed of three parts: classification, nomenclature, and identification.
 2. Living organisms can be divided into three domains: the *Eucarya*, the *Bacteria* and the *Archaea* (**figure 19.3**). The eucaryotic cell may have arisen from procaryotic cells by endosymbiotic events.
 3. The definition of species is different for sexually and asexually reproducing organisms. A bacterial species is a collection of strains that have many stable properties in common and differ significantly from other groups of strains.
 4. Microorganisms are named according to the binomial system.
 5. The two major types of natural classification are phylogenetic systems and phenetic systems.
 6. Classifications may be constructed by means of numerical taxonomy, in which the general similarity of organisms is determined using a computer to calculate and analyze association coefficients.
 7. Morphological, physiological, metabolic, and ecological characteristics are widely used in microbial taxonomy.
 8. The study of transformation and conjugation in bacteria is sometimes taxonomically useful. Plasmid-borne traits can cause errors in bacterial taxonomy if care is not taken.
 9. Proteins are direct reflections of mRNA sequences and may be used to compare genomes from different organisms.
 10. The G + C content of DNA is easily determined and taxonomically valuable because it is an indirect reflection of the base sequence.
 11. Nucleic acid hybridization studies are used to compare DNA or RNA sequences and thus determine genetic relatedness (**figure 19.7**).
 12. Nucleic acid sequencing is the most powerful and direct method for comparing genomes.
- The sequences of 16S and 5S rRNA are used most often in phylogenetic studies. Complete procaryotic chromosomes are now being sequenced and compared.
13. Phylogenetic relationships often are shown in the form of branched diagrams called phylogenetic trees (**figure 19.8**). Trees may be either rooted or unrooted and are created in several different ways.
 14. The sequences of rRNA, DNA, and proteins are used to produce phylogenetic trees. Often members of a group will have a unique characteristic rRNA sequence that distinguishes them from members of other taxonomic groups.
 15. Studies on the sequence of 16S rRNA and other molecular properties suggest that the procaryotes are divided into two very different groups: *Bacteria* and *Archaea*. *Archaea* differ from bacteria in cell wall composition, membrane lipids, tRNA structure, ribosomes, and many other properties (**table 19.8**).
 16. Although probably most microbiologists favor the three-domain system, there are alternatives such as the five-, six-, and eight-kingdom systems (**figure 19.12**).
 17. *Bergey's Manual of Systematic Bacteriology* gives the accepted system of procaryotic taxonomy.
 18. The first edition of *Bergey's Manual of Systematic Bacteriology* provides a primarily phenetic classification, and many taxa are not phylogenetically homogeneous. Easily determined features such as cell shape, Gram-staining characteristics, oxygen relationships, motility, and the mode of energy production are used to classify procaryotes (**table 19.9**).
 19. The second edition of *Bergey's Manual* is phylogenetic and distributes procaryotes between two domains and 25 phyla (**table 19.10** and **figure 19.13**). Comparisons of nucleic acid sequences, particularly 16S rRNA sequences, are the foundation of this new classification.
 20. The second edition will have five volumes. The general organization of the five volumes is summarized in table 19.10 and briefly outlined below.
 - (1) Volume 1: *The Archaea and the Deeply Branching and Phototrophic Bacteria*. This volume describes the archaea, cyanobacteria, green sulfur and nonsulfur bacteria, deinococci, and other deeply branching groups.
 - (2) Volume 2: *The Proteobacteria*. All of the proteobacteria (purple bacteria) are placed in this volume and are divided into five major groups based on rRNA sequences and other characteristics: α -proteobacteria, β -proteobacteria, γ -proteobacteria, δ -proteobacteria, and ϵ -proteobacteria.
 - (3) Volume 3: *The Low G + C Gram-Positive Bacteria*. This volume contains gram-positive bacteria with G + C content below about 50%. Some of the major groups are the clostridia, bacilli, streptococci, and staphylococci. Mycoplasmas also are placed here.
 - (4) Volume 4: *The High G + C Gram-Positive Bacteria*. Gram-positive bacteria with G + C content above around 50 to 55% are in this volume. Such groups as *Corynebacterium*, *Mycobacterium*, *Nocardia*, and the actinomycetes are located here.
 - (5) Volume 5: *The Planctomycetes, Spirochaetes, Fibrobacteres, Bacteroidetes, and Fusobacteria*. Volume 5 has a variety of different gram-negative bacterial groups. The most practically important examples are the chlamydias and the spirochetes.

Key Terms

Archaea 424
Bacteria 424
 binomial system 426
 biovar 425
 classification 422
 dendrogram 427
 domains 424
 endosymbiotic hypothesis 424
Eucarya 424
 evolutionary distance 433
 G + C content 429
 genus 426
 identification 422
 Jaccard coefficient (S_J) 427

melting temperature (T_m) 430
 molecular chronometers 432
 morphovar 425
 natural classification 426
 nomenclature 422
 nucleic acid hybridization 431
 numerical taxonomy 426
 oligonucleotide signature sequences 434
 phenetic system 426
 phenons 427
 phylogenetic or phyletic classification systems 428
 phylogenetic tree 433
 phylogeny 428
 polyphasic taxonomy 435

procaryotic species 425
 protists 438
 serovar 425
 similarity matrix 427
 simple matching coefficient (S_{SM}) 426
 species 425
 strain 425
 stromatolites 423
 systematics 422
 taxon 422
 taxonomy 422
 type strain 425

Questions for Thought and Review

1. Why are size and shape often less useful in characterizing procaryotic species than eucaryotic microbial species?
2. Why might a phylogenetic approach to microbial classification be preferable to a phenetic approach? Give arguments for the use of a phenetic classification. Which do you think is preferable and why?
3. Would a numerical taxonomist favor the phylogenetic or phenetic approach to obtaining a natural classification? Why might it be best to use unweighted characteristics in classification? Are there reasons to give some properties more weight or importance than others?
4. In what way does the simple matching coefficient (S_{SM}) differ from the Jaccard coefficient (S_J)? Give a reason why the former coefficient might be preferable to the latter.
5. What genetic feature of procaryotes makes it advisable to use as many characteristics as convenient in classifying and identifying an organism?
6. What properties of the molecules should be considered when selecting RNAs or proteins to sequence for the purpose of determining the relatedness of microorganisms that are only distantly related?
7. Discuss the problems in developing an accurate phylogenetic tree. Is it possible to create a completely accurate universal phylogenetic tree?
8. Why is the current procaryotic classification system likely to change considerably? How would one select the best features to use in identification of unknown procaryotes and determination of relatedness?

Additional Reading

General

- Achenbach, L. A., and Coates, J. D. 2000. Disparity between bacterial phylogeny and physiology. *ASM News* 66(12):714–15.
- Amann, R. I.; Ludwig, W.; and Schleifer, K.-H. 1995. Phylogenetic identification and in situ detection of individual microbial cells without cultivation. *Microbiol. Rev.* 59(1):143–69.
- Balows, A.; Truper, H. G.; Dworkin, M.; Harder, W.; and Schleifer, K.-H. 1992. *The procaryotes*, 2d ed. New York: Springer-Verlag.
- Bryant, T. N. 2000. Identification of bacteria, computerized. In *Encyclopedia of microbiology*, 2d ed., vol. 2, J. Lederberg, editor-in-chief, 709–21. San Diego: Academic Press.
- Goodfellow, M., and O'Donnell, A. G., editors. 1993. *Handbook of new bacterial systematics*. San Diego, Calif.: Academic Press.
- Hillis, D. M.; Moritz, C.; and Mable, B. K., editors. 1996. *Molecular Systematics*, 2d. ed. Sunderland, Mass.: Sinauer Associates.
- Hugenholtz, P.; Goebel, B. M.; and Pace, N. R. 1998. Impact of culture-independent studies on the emerging phylogenetic view of bacterial diversity. *J. Bacteriol.* 180(18):4765–74.
- Logan, N. A. 1994. *Bacterial systematics*. Boston: Blackwell Scientific.
- Margulis, L.; Corliss, J. O.; Melkonian, M.; and Chapman, D. J., editors. 1990. *Handbook of prototista*. Boston: Jones and Bartlett.
- Priest, F., and Austin, B. 1993. *Modern bacterial taxonomy*, 2d ed. New York: Chapman and Hall.
- Staley, J. T. 1999. Bacterial biodiversity: A time for place. *ASM News* 65(10):681–87.
- Vandamme, P.; Pot, B.; Gillis, M.; De Vos, P.; Kersters, K.; and Swings, J. 1996. Polyphasic taxonomy, a consensus approach to bacterial systematics. *Microbiol. Rev.* 60(2):407–38.

19.2 Microbial Evolution and Diversity

- Brown, J. R., and Doolittle, W. F. 1997. *Archaea* and the procaryote-to-eukaryote transition. *Microbiol. Mol. Biol. Rev.* 61(4): 456–502.
- Cavalier-Smith, T. 1987. The origin of eukaryote and archaebacterial cells. *Ann. N.Y. Acad. Sci.* 503:17–54.
- de Duve, C. 1996. The birth of complex cells. *Sci. Am.* 274(4):50–57.
- Gogarten, J. P. 1995. The early evolution of cellular life. *Trends Ecol. Evol.* 10(4):147–51.
- Gupta, R. S., and Golding, G. B. 1996. The origin of the eukaryotic cell. *Trends Biochem. Sci.* 21:166–71.
- Knoll, A. H. 1991. End of the proterozoic eon. *Sci. Am.* 265(4):64–73.
- Kurland, C. G., and Andersson, S. G. E. 2000. Origin and evolution of the mitochondrial proteome. *Micro. Mol. Bio. Rev.* 64(4):786–820.
- Margulis, L. 1971. Symbiosis and evolution. *Sci. Am.* 225(2):49–57.
- Saier, M. H., Jr. 2000. Bacterial diversity and the evolution of differentiation. *ASM News* 66(6):337–43.
- Vidal, G. 1984. The oldest eukaryotic cells. *Sci. Am.* 250(2):48–57.
- Woese, C. R.; Kandler, O.; and Wheelis, M. L. 1990. Towards a natural system of organisms: Proposal for the domains archaea, bacteria, and eucarya. *Proc. Natl. Acad. Sci.* 87:4576–79.

19.4 Classification Systems

- Sneath, P. H. A., and Sokal, R. R. 1973. *Numerical taxonomy: The principles and practice of numerical classification*. San Francisco: W. H. Freeman.
- Stackebrandt, E. 1992. Unifying phylogeny and phenotypic diversity. In *The procaryotes*, 2d ed. A. Balows et al., editors, 19–47. New York: Springer-Verlag.

Critical Thinking Questions

1. A potential candidate for a molecular chronometer is tRNA. Explain why it would be an appropriate choice. What are some of the drawbacks of this choice?
2. Form a group and collect a heterogeneous group of screws, nails, tacks, and other similar fasteners. Create a classification system that groups the items logically. Assign “genus and species” names to the items, attempting to indicate phylogenetic relationships. This exercise will acquaint you with the problems faced by taxonomists and will provide an opportunity to sharpen your powers of observation. Compare your phylogenetic models with those of other groups. How are they similar and what are the differences?

19.5 Major Characteristics Used in Taxonomy

- Johnson, J. L. 1984. Nucleic acids in bacterial classification. In *Bergey's manual of systematic bacteriology*, J. G. Holt, vol. 1, N. R. Krieg, editors, 8–11. Baltimore, Md.: Williams & Wilkins.
- Jones, D., and Krieg, N. R. 1984. Serology and chemotaxonomy. In *Bergey's manual of systematic bacteriology*, J. G. Holt, vol. 1, N. R. Krieg, editors, 15–18. Baltimore, Md.: Williams & Wilkins.

19.6 Assessing Microbial Phylogeny

- Doolittle, R. F. 1995. Of Archae and Eo: What's in a name? *Proc. Natl. Acad. Sci.* 92:2421–23.
- Forster, P. 1997. Protein versus rRNA: Problems in rooting the universal tree of life. *ASM News* 63(2):89–95.
- Ludwig, W., and Schleifer, K.-H. 1999. Phylogeny of *Bacteria* beyond the 16S rRNA standard. *ASM News* 65(11):752–57.
- Olsen, G. J., and Woese, C. R. 1993. Ribosomal RNA: A key to phylogeny. *FASEB J.* 7:113–23.
- Sneath, P. H. A. 1989. Analysis and interpretation of sequence data for bacterial systematics: The view of a numerical taxonomist. *Syst. Appl. Microbiol.* 12:15–31.
- Woese, C. R.; Olsen, G. J.; Ibba, M.; and Söll, D. 2000. Aminoacyl-tRNA synthetases, the genetic code, and the evolutionary process. *Micro. Mol. Biol. Rev.* 64(1):202–36.
- Zuckerkandl, E., and Pauling, L. 1965. Molecules as documents of evolutionary history. *J. Theoret. Biol.* 8:357–66.

19.7 The Major Divisions of Life

- Baldauf, S. L.; Roger, A. J.; Wenk-Siefert, I.; Doolittle, W. F. 2000. A kingdom-level phylogeny of eukaryotes based on combined protein data. *Science* 290:972–77.
- Cavalier-Smith, T. 1993. Kingdom protozoa and its 18 phyla. *Microbiol. Rev.* 57(4):953–94.

- Doolittle, W. F. 1999. Phylogenetic classification and the universal tree. *Science* 284:2124–28.
- Doolittle, W. F. 2000. Uprooting the tree of life. *Sci. Am.* 282(2):90–95.
- Gupta, R. S. 1998. Protein phylogenies and signature sequences: A reappraisal of evolutionary relationships among archaeobacteria, eubacteria, and eukaryotes. *Micro. Mol. Biol. Rev.* 62(4):1435–91.
- Kabnick, K. S., and Peattie, D. A. 1991. *Giardia*: A missing link between prokaryotes and eukaryotes. *American Scientist* 79:34–43.
- Mayr, E. 1998. Two empires or three? *Proc. Natl. Acad. Sci.* 95:9720–23.
- Olsen, G. J.; Woese, C. R.; and Overbeek, R. 1994. The winds of (evolutionary) change: Breathing new life into microbiology. *J. Bacteriol.* 176(1):1–6.
- Pace, N. R. 1996. New perspective on the natural microbial world: Molecular microbial ecology. *ASM News.* 62(9):463–70.
- Pace, N. R. 1997. A molecular view of microbial diversity and the biosphere. *Science* 276:734–40.
- Sogin, M. L.; Morrison, H. G.; Hinkle, G.; and Silberman, J. D. 1996. Ancestral relationships of the major eukaryotic lineages. *Microbiologia* 12(1):17–28.
- Williams, D. M., and Embley, T. M. 1996. Microbial diversity: Domains and kingdoms. *Ann. Rev. Ecol. Syst.* 27:569–95.
- Woese, C. R. 1981. Archaeobacteria. *Sci. Am.* 244(6):98–122.
- Woese, C. R. 1987. Bacterial evolution. *Microbiol. Rev.* 51(2):221–71.
- Woese, C. 1998. The universal ancestor. *Proc. Natl. Acad. Sci.* 95:6854–59.
- Zavarzin, G. A.; Stackebrandt, E.; and Murray, R. G. E. 1991. A correlation of phylogenetic diversity in the *Proteobacteria* with the influences of ecological forces. *Can. J. Microbiol.* 37:1–6.
- Zillig, W.; Palm, P.; Reiter, W.-D.; Gropp, F.; Pühler, G.; and Klenk, H.-P. 1988. Comparative evaluation of gene expression in archaeobacteria. *Eur. J. Biochem.* 173:473–82.

19.8 Bergey's Manual of Systematic Bacteriology

- Garrry, G. M., editor-in-chief. 2001. *Bergey's manual of systematic bacteriology*, 2d ed., vol. 1, D. R. Boone and R. W. Castenholz, editors. New York: Springer-Verlag.
- Holt, J. G., editor. 1984–1989. *Bergey's manual of systematic bacteriology*. 4 vols. Baltimore, Md.: Williams & Wilkins.
- Holt, J. G.; Krieg, N. R.; Sneath, P. H. A.; Staley, J. T.; and Williams, S. T. 1994. *Bergey's manual of determinative bacteriology*, 9th ed. Baltimore, Md.: Williams & Wilkins.

CHAPTER 20

The Archaea



Archaea are often found in extreme environments such as this geyser in Yellowstone National Park.

Outline

- 20.1 Introduction to the *Archaea* 451
 - Archaeal Cell Walls 451
 - Archaeal Lipids and Membranes 452
 - Genetics and Molecular Biology 453
 - Metabolism 453
 - Archaeal Taxonomy 455
- 20.2 Phylum *Crenarchaeota* 456
- 20.3 Phylum *Euryarchaeota* 458
 - The Methanogens 458
 - The Halobacteria 461
 - The Thermoplasms 463
 - Extremely Thermophilic S^0 -Metabolizers 463
 - Sulfate-Reducing Archaea 463

Concepts

1. *Archaea* differ in many ways from both bacteria and eucaryotes. These include differences in cell wall structure and chemistry, membrane lipid structure, molecular biology, and metabolism.
2. *Archaea* usually grow in a few restricted or specialized habitats: anaerobic, hypersaline, and high temperature.
3. *Bergey's Manual* currently divides the archaea into five major groups: methanogenic archaea, extreme halophiles, cell wall-less archaea, extremely thermophilic S^0 -metabolizers, and sulfate reducers.
4. The second edition of *Bergey's Manual* divides the archaea into two phyla, the *Crenarchaeota* and *Euryarchaeota*, each with several orders.
5. Methanogenic and sulfate-reducing archaea have unique cofactors that participate in methanogenesis.
6. *Archaea* have special structural, chemical, and metabolic adaptations that enable them to grow in extreme environments.

As is often the case, epoch-making ideas carry with them implicit, unanalyzed assumptions that ultimately impede scientific progress until they are recognized for what they are. So it is with the prokaryote-eukaryote distinction. Our failure to understand its true nature set the stage for the sudden shattering of the concept when a “third form of life” was discovered in the late 1970s, a discovery that actually left many biologists incredulous. Archaeobacteria, as this third form has come to be known, have revolutionized our notion of the prokaryote, have altered and refined the way in which we think about the relationship between prokaryotes and eukaryotes . . . and will influence strongly the view we develop of the ancestor that gave rise to all extant life.

C. R. Woese and R. S. Wolfe

Chapters 20 through 24 survey the procaryotes described in *Bergey's Manual of Systematic Bacteriology*. Where possible, the order of presentation will follow that of the second edition of *Bergey's Manual*. Chapters 20 and 21 cover the material contained in volumes 1 and 5 of the second edition. Chapter 20 describes the *Archaea*; chapter 21 focuses on the remaining groups in volumes 1 and 5. Chapter 22 covers the proteobacteria, which are located in volume 2. Volume 3 is devoted to the low G + C gram-positive bacteria and will be discussed in chapter 23. Finally, chapter 24 deals with the high G + C bacteria of volume 4.

This chapter begins with a general introduction to the *Archaea*. Then it briefly discusses the biology of each major archaeal group.

Comparison of the sequences of rRNA from a great variety of organisms shows that organisms may be divided into three major groups: the bacteria, archaea, and eucaryotes (see figures 19.3 and 19.13). Some of the most important differences are summarized in table 19.8 (see p. 436). Because archaea are different from both bacteria and eucaryotes, their most distinctive properties will first be described in some detail and compared with those of the latter two groups. [rRNA sequences and the archaea \(pp. 424, 435–36\)](#)

20.1 Introduction to the Archaea

As a group the *Archaea* [Greek *archaios*, ancient] are quite diverse, both in morphology and physiology. They can stain either gram positive or gram negative and may be spherical, rod-shaped, spiral, lobed, plate-shaped, irregularly shaped, or pleomorphic. Some are single cells, whereas others form filaments or aggregates. They range in diameter from 0.1 to over 15 μm , and some filaments can grow up to 200 μm in length. Multiplication may be by binary fission, budding, fragmentation, or other mechanisms. Archaea are just as diverse physiologically. They can be aerobic, facultatively anaerobic, or strictly anaerobic. Nutritionally they range from chemolithoautotrophs to organotrophs. Some are mesophiles; others are hyperthermophiles that can grow above 100°C.

Archaea often are found in extreme aquatic and terrestrial habitats. They are often present in anaerobic, hypersaline, or high-temperature environments. Recently archaea have been discovered

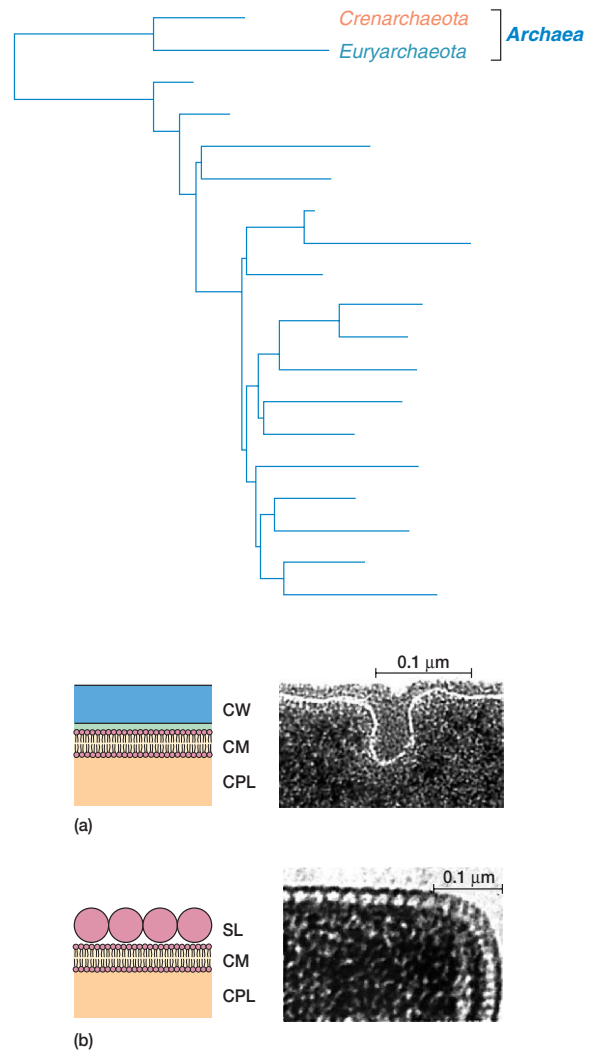


Figure 20.1 Cell Envelopes of Archaea. Schematic representations and electron micrographs of (a) *Methanobacterium formicicum*, a typical gram-positive organism, and (b) *Thermoproteus tenax*, a gram-negative archaeon. CW, cell wall; SL, surface layer; CM, cell membrane or plasma membrane; CPL, cytoplasm.

in cold environments. It appears that they constitute up to 34% of the procaryotic biomass in coastal Antarctic surface waters (see chapter 29). A few are symbionts in animal digestive systems.

Archaeal Cell Walls

Although archaea can stain either gram positive or gram negative depending on the thickness and mass of the cell wall, their wall structure and chemistry differ from that of the bacteria. There is considerable variety in archaeal wall structure. Many gram-positive archaea have a wall with a single thick homogeneous layer resembling that in gram-positive bacteria and thus stain gram positive (figure 20.1a). Gram-negative archaea lack the outer membrane and complex

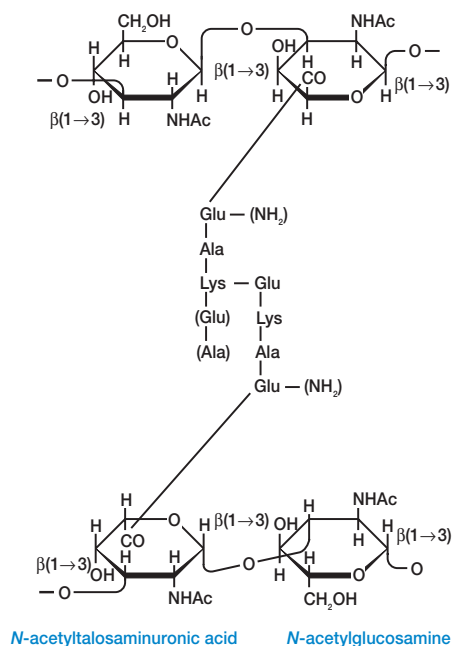


Figure 20.2 The Structure of Pseudomurein. The components in parentheses are not always present. Ac represents the acetyl group.

peptidoglycan network or sacculus of gram-negative bacteria. Instead they usually have a surface layer of protein or glycoprotein subunits (figure 20.1b). [Peptidoglycan structure and chemistry \(pp. 56–57\)](#)

The chemistry of archaeal cell walls is also quite different from that of the bacteria. None have the muramic acid and D-amino acids characteristic of bacterial peptidoglycan. Not surprisingly, all archaea resist attack by lysozyme and β -lactam antibiotics such as penicillin. Gram-positive archaea can have a variety of complex polymers in their walls. *Methanobacterium* and some other methanogens have walls containing **pseudomurein**, a peptidoglycanlike polymer that has L-amino acids in its cross-links, N-acetylglucosaminuronic acid instead of N-acetylmuramic acid, and $\beta(1\rightarrow3)$ glycosidic bonds instead of $\beta(1\rightarrow4)$ glycosidic bonds (**figure 20.2**). *Methanosarcina* and *Halococcus* lack pseudomurein and contain complex polysaccharides similar to the chondroitin sulfate of animal connective tissue. Other heteropolysaccharides are also found in gram-positive walls.

Gram-negative archaea have a layer of protein or glycoprotein outside their plasma membrane. The layer may be as thick as 20 to 40 nm. Sometimes there are two layers, a sheath surrounding an electron-dense layer. The chemical content of these walls varies considerably. Some methanogens (*Methanobolus*), *Halobacterium*, and several extreme thermophiles (*Sulfolobus*, *Thermoproteus*, and *Pyrodictium*) have glycoproteins in their walls. In contrast, other methanogens (*Methanococcus*, *Methanomicrobium*, and *Methanogenium*) and the extreme thermophile *Desulfurococcus* have protein walls.

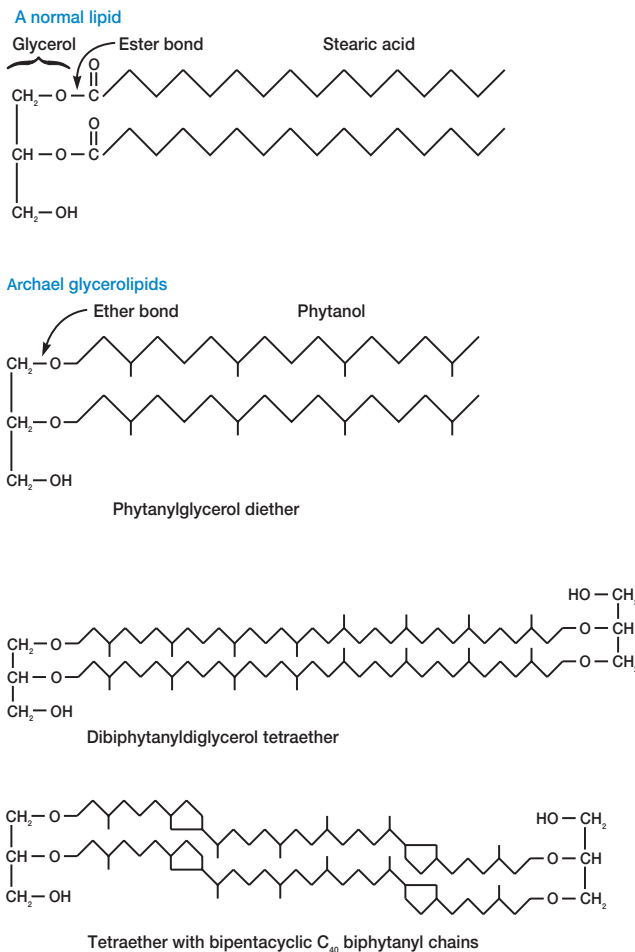


Figure 20.3 Archaeal Membrane Lipids. An illustration of the difference between archaeal lipids and those of bacteria. Archaeal lipids are derivatives of isopranyl glycerol ethers rather than the usual glycerol fatty acid esters. Three examples of common archaeal glycerolipids are given.

Archaeal Lipids and Membranes

As emphasized in table 19.8, a most distinctive feature of the archaea is the nature of their membrane lipids. They differ from both bacteria and eucaryotes in having branched chain hydrocarbons attached to glycerol by ether links rather than fatty acids connected by ester links (**figure 20.3**). Sometimes two glycerol groups are linked to form an extremely long tetraether. Usually the diether side chains are 20 carbons in length, and the tetraether chains are 40 carbons. Cells can adjust the overall length of the tetraethers by cyclizing the chains to form pentacyclic rings (figure 20.3), and biphytanyl chains may contain from one to four cyclopentyl rings. Polar lipids are also present in archaeal membranes: phospholipids, sulfolipids, and glycolipids. From 7 to 30% of the membrane lipids are nonpolar lipids, which usually are derivatives of squalene (**figure 20.4**). These lipids can be

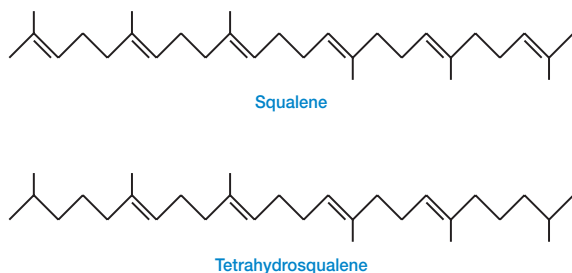


Figure 20.4 Nonpolar Lipids of Archaea. Two examples of the most predominant nonpolar lipids are the C_{30} isoprenoid squalene and one of its hydroisoprenoid derivatives, tetrahydrosqualene.

combined in various ways to yield membranes of different rigidity and thickness. For example, the C_{20} diethers can be used to make a regular bilayer membrane (**figure 20.5a**). A much more rigid monolayer membrane may be constructed of C_{40} tetraether lipids (**figure 20.5b**). Of course archaeal membranes may contain a mix of diethers, tetraethers, and other lipids. As might be expected from their need for stability, the membranes of extreme thermophiles such as *Thermoplasma* and *Sulfolobus* are almost completely tetraether monolayers.

Genetics and Molecular Biology

Some features of archaeal genetics are similar to those in bacteria. The archaeal chromosomes that have been studied are single closed DNA circles. However, the genomes of some archaeons are significantly smaller than the normal bacterium. *E. coli* DNA has a size of about 2.5×10^9 daltons, whereas *Thermoplasma acidophilum* DNA is about 0.8×10^9 daltons and *Methanobacterium thermoautotrophicum* DNA is 1.1×10^9 daltons. The variation in G + C content is great, from about 21 to 68 mol%, and is another sign of archaeal diversity. Archaea have few plasmids. Recently the genome of the archaeon *Methanococcus jannaschii* was completely sequenced and compared with the genomes of other organisms. About 56% of its 1,738 genes are unlike those in bacteria and eucaryotes. If this degree of difference is characteristic of the domain *Archaea*, these organisms are as distinctive genotypically as they are in other respects.

Archaeal mRNA appears similar to that of bacteria rather than to eucaryotic mRNA. Polygenic mRNA has been discovered, and there is no evidence for mRNA splicing. Archaeal promoters are similar to those in bacteria.

Despite these and other similarities, there are also many differences between archaea and other organisms. Unlike both bacteria and eucaryotes, the T ψ C arm (*see p. 266*) of archaeal tRNA lacks thymine and contains pseudouridine or 1-methylpseudouridine. The archaeal initiator tRNA carries methionine as does the eucaryotic initiator tRNA. Although archaeal ribosomes are 70S, similar to bacterial ribosomes, electron microscopic studies show that their shape is quite variable and sometimes differs from that of both bacterial and eucaryotic ribosomes. They do resemble eucaryotic ribosomes in their sensitivity to anisomycin and insensi-

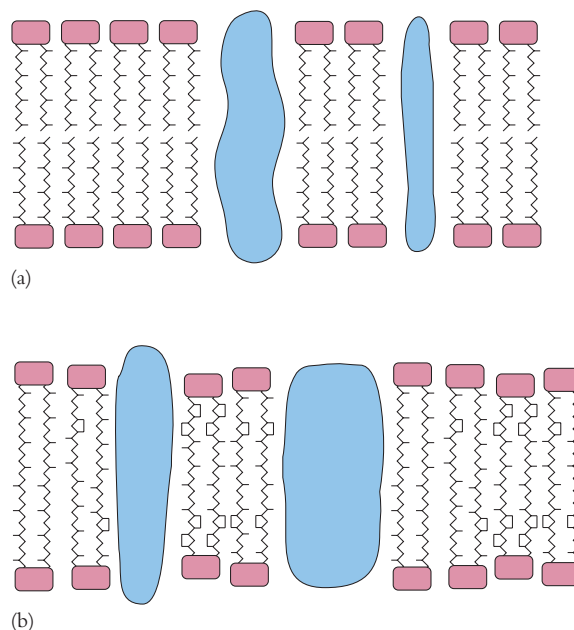


Figure 20.5 Examples of Archaeal Membranes. (a) A membrane composed of integral proteins and a bilayer of C_{20} diethers. (b) A rigid monolayer composed of integral proteins and C_{40} tetraethers.

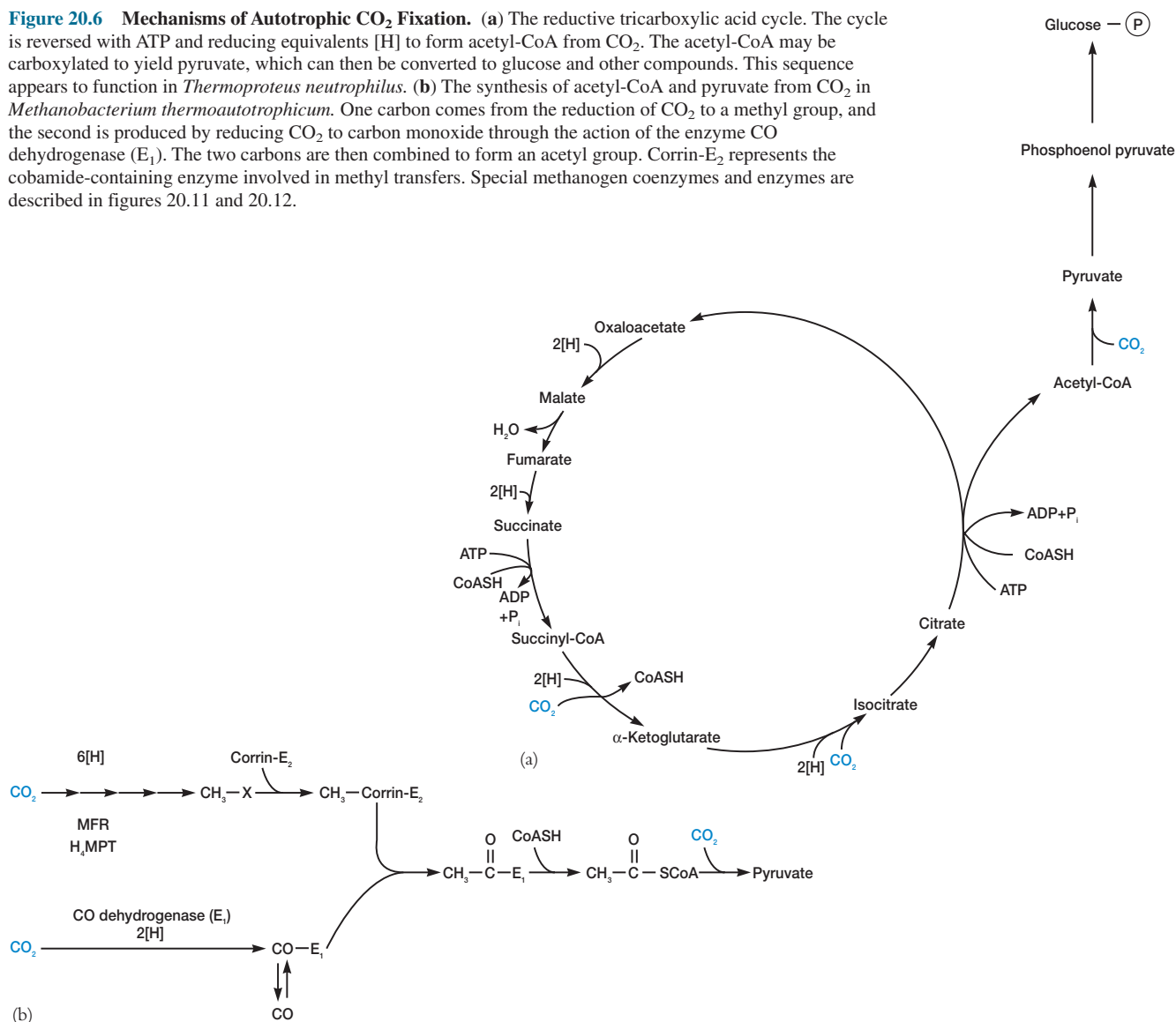
tivity to chloramphenicol and kanamycin. Furthermore, their elongation factor 2 reacts with diphtheria toxin just as the eucaryotic EF-2 does. Some archaea, such as many methanogens in the phylum *Euryarchaeota*, differ from other procaryotes in having histone proteins that bind with DNA to form nucleosome-like structures. Finally, archaeal DNA-dependent RNA polymerases resemble the eucaryotic enzymes, not the bacterial RNA polymerase. They are large, complex enzymes and are insensitive to the drugs rifampin and streptolydigin. These and other differences distinguish the archaea from both bacteria and eucaryotes. [Ribosomes and the mechanism of protein synthesis \(pp. 265–72\); DNA transcription \(pp. 261–63\)](#)

Metabolism

Not surprisingly in view of the variety of their life-styles, archaeal metabolism varies greatly between the members of different groups. Some archaea are organotrophs; others are autotrophic. A few even carry out an unusual form of photosynthesis.

Archaeal carbohydrate metabolism is best understood. The enzyme 6-phosphofructokinase has not been found in archaea, and they do not appear to degrade glucose by way of the Embden-Meyerhof pathway. Extreme halophiles and thermophiles catabolize glucose using a modified form of the Entner-Doudoroff pathway (*see p. 179 and appendix II*) in which the initial intermediates are not phosphorylated. The halophiles have slightly different modifications of the pathway than do the extreme thermophiles but still produce pyruvate and NADH or NADPH. Methanogens do not catabolize glucose to

Figure 20.6 Mechanisms of Autotrophic CO₂ Fixation. (a) The reductive tricarboxylic acid cycle. The cycle is reversed with ATP and reducing equivalents [H] to form acetyl-CoA from CO₂. The acetyl-CoA may be carboxylated to yield pyruvate, which can then be converted to glucose and other compounds. This sequence appears to function in *Thermoproteus neutrophilus*. (b) The synthesis of acetyl-CoA and pyruvate from CO₂ in *Methanobacterium thermoautotrophicum*. One carbon comes from the reduction of CO₂ to a methyl group, and the second is produced by reducing CO₂ to carbon monoxide through the action of the enzyme CO dehydrogenase (E₁). The two carbons are then combined to form an acetyl group. Corrin-E₂ represents the cobamide-containing enzyme involved in methyl transfers. Special methanogen coenzymes and enzymes are described in figures 20.11 and 20.12.



any significant extent. In contrast with glucose degradation, gluconeogenesis proceeds by a reversal of the Embden-Meyerhof pathway in halophiles and methanogens. All archaea that have been studied can oxidize pyruvate to acetyl-CoA. They lack the pyruvate dehydrogenase complex present in eucaryotes and respiratory bacteria and use the enzyme pyruvate oxidoreductase for this purpose. Halophiles and the extreme thermophile *Thermoplasma* do seem to have a functional tricarboxylic acid cycle. No methanogen has yet been found with a complete tricarboxylic acid cycle. Evidence for functional respiratory chains has been obtained in halophiles and thermophiles. [The Embden-Meyerhof pathway and tricarboxylic acid cycle](#) (pp. 176–77, 183–84, and appendix II)

Very little is known in detail about biosynthetic pathways in the archaea. Preliminary data suggest that the synthetic pathways for amino acids, purines, and pyrimidines are similar to those in other organisms. Some methanogens can fix atmospheric dinitrogen. Not only do many archaea use a reversal of the Embden-Meyerhof pathway to synthesize glucose, but at least some methanogens and extreme thermophiles employ glycogen as their major reserve material.

Autotrophy is widespread among the methanogens and extreme thermophiles, and CO₂ fixation occurs in more than one way. *Thermoproteus* and possibly *Sulfolobus* incorporate CO₂ by the reductive tricarboxylic acid cycle (**figure 20.6a**). This pathway is also present in the green sulfur bacteria (*see pp. 470–71*).

Table 20.1 Characteristics of the Major Archaeal Groups

Group	General Characteristics	Representative Genera
Methanogenic archaea	Strict anaerobes. Methane is the major metabolic end product. S^0 may be reduced to H_2S without yielding energy production. Cells possess coenzyme M, factors 420 and 430, and methanopterin.	<i>Methanobacterium</i> <i>Methanococcus</i> <i>Methanomicrobium</i> <i>Methanosarcina</i> <i>Archaeoglobus</i>
Archaea sulfate reducers	Irregular gram-negative coccoid cells. H_2S formed from thiosulfate and sulfate. Autotrophic growth with thiosulfate and H_2 . Can grow heterotrophically. Traces of methane also formed. Extremely thermophilic and strictly anaerobic. Possess factor 420 and methanopterin but not coenzyme M or factor 430.	
Extremely halophilic archaea	Coccoid or irregularly shaped rods. Gram-negative or gram-positive, primarily aerobic chemoorganotrophs. Require high sodium chloride concentrations for growth (≥ 1.5 M). Colonies are various shades of red. Neutrophilic or alkalophilic. Mesophilic or slightly thermophilic. Some species contain bacteriorhodopsin and use light for ATP synthesis.	<i>Halobacterium</i> <i>Halococcus</i> <i>Natronobacterium</i>
Cell wall-less archaea	Pleomorphic cells lacking a cell wall. Thermoacidophilic and chemoorganotrophic. Facultatively anaerobic. Plasma membrane contains a mannose-rich glycoprotein and a lipoglycan.	<i>Thermoplasma</i>
Extremely thermophilic S^0 -metabolizers	Gram-negative rods, filaments, or cocci. Obligately thermophilic (optimum growth temperature between 70–110°C). Usually strict anaerobes but may be aerobic or facultative. Acidophilic or neutrophilic. Autotrophic or heterotrophic. Most are sulfur metabolizers. S^0 reduced to H_2S anaerobically; H_2S or S^0 oxidized to H_2SO_4 aerobically.	<i>Desulfurococcus</i> <i>Pyrodicticum</i> <i>Pyrococcus</i> <i>Sulfolobus</i> <i>Thermococcus</i> <i>Thermoproteus</i>

Methanogenic archaea and probably most extreme thermophiles incorporate CO_2 by the reductive acetyl-CoA pathway (figure 20.6b). A similar pathway also is present in acetogenic bacteria and autotrophic sulfate-reducing bacteria.

Archaeal Taxonomy

As shown in figure 19.3 and 19.13 and in table 19.8 (see p. 436), the archaea are quite distinct from other living organisms. Within the domain, however, there is great diversity (see figures 19.13 and 19.14). The first edition of *Bergey's Manual* divided the archaea into five major groups based on physiological and morphological differences. **Table 20.1** summarizes some the characteristics of these five groups and gives representatives of each.

On the basis of rRNA data, the second edition of *Bergey's Manual* will divide the archaea into the phyla *Euryarchaeota* [Greek *eurus*, wide, and Greek *archaios*, ancient or primitive] and *Crenarchaeota* [Greek *crene*, spring or fount, and *archaios*]. The euryarchaeotes are given this name because they occupy many different ecological niches and have a variety of metabolic patterns. The phylum *Euryarchaeota* is very diverse with seven classes (*Methanobacteria*, *Methanococci*, *Halobacteria*, *Thermoplas-*

mata, *Thermococci*, *Archaeoglobi*, and *Methanopyri*), nine orders, and 15 families. The methanogens, extreme halophiles, sulfate reducers, and many extreme thermophiles with sulfur-dependent metabolism are located in the *Euryarchaeota*. Methanogens are the dominant physiological group.

The crenarchaeotes (figure 20.7) are thought to resemble the ancestor of the archaea, and almost all the well-characterized species are thermophiles or hyperthermophiles. The phylum *Crenarchaeota* is divided into one class, *Thermoprotei*, and three orders. The order *Thermoproteales* contains gram-negative anaerobic to facultative, hyperthermophilic rods. They often grow chemolithoautotrophically by reducing sulfur to hydrogen sulfide. Members of the order *Sulfolobales* are coccus-shaped thermoacidophiles. The order *Desulfurococcales* contains gram-negative coccoid or disk-shaped hyperthermophiles. They grow either chemolithotrophically by hydrogen oxidation or organotrophically by either fermentation or respiration with sulfur as the electron acceptor. The taxonomy of both phyla will undoubtedly undergo further revisions as more organisms are discovered. This is particularly the case with the crenarchaeotes because of the discovery of mesophilic forms in the ocean; these crenarchaeotes may constitute a significant fraction of the oceanic picoplankton.

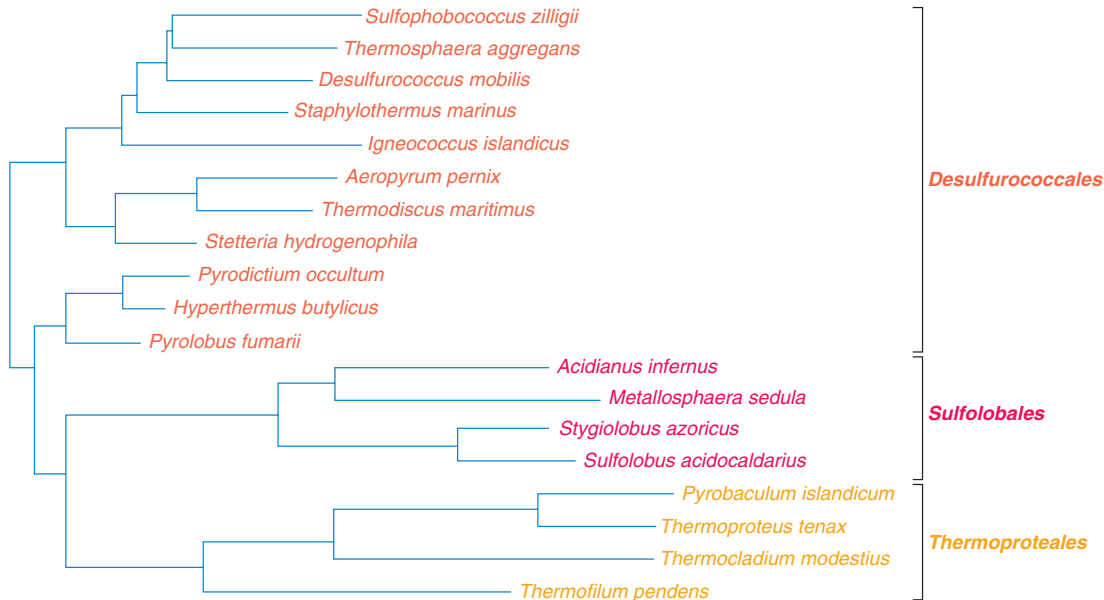


Figure 20.7 The Phylum Crenarchaeota. A phylogenetic tree developed with 16S rRNA data for crenarchaeotal-type species. The three orders are indicated.

1. What are the archaea? Briefly describe the major ways in which they differ from bacteria and eucaryotes.
2. How do archaeal cell walls differ from those of the bacteria? What is pseudomurein?
3. In what ways do archaeal membrane lipids differ from those of bacteria and eucaryotes? How might these differences lead to stronger membranes?
4. List the differences between archaea and other organisms with respect to tRNA, ribosome structure and behavior, EF-2 sensitivity, and RNA polymerases.
5. Briefly describe the way in which archaea degrade and synthesize glucose. In what two unusual ways do they incorporate CO₂?
6. Characterize the five different groups of archaea and distinguish between them. Distinguish between the phyla *Euryarchaeota* and *Crenarchaeota*.

20.2 Phylum Crenarchaeota

As mentioned previously, most of the crenarchaeotes that have been isolated are extremely thermophilic, and many are acidophiles and sulfur dependent. The sulfur may be used either as an electron acceptor in anaerobic respiration or as an electron source by lithotrophs. Almost all are strict anaerobes. They grow in geothermally heated water or soils that contain elemental sulfur. These environments are scattered all over the world. Examples are the sulfur-rich hot springs in Yellowstone National Park, Wyoming, and the waters surrounding areas of submarine volcanic activity (**figure 20.8**). Such habitats are sometimes called solfatara. These archaea can be very thermophilic and often are classified as hyperthermophiles (*see p. 126*). The most extreme example is *Pyrodictium*, an archaeon isolated from geothermally heated sea floors. *Pyrodictium* has a temperature minimum of 82°C, a growth optimum



(a)



(b)

Figure 20.8 Habitats for Thermophilic Bacteria. (a) The pump geyser in Yellowstone National Park. The orange color is due to the carotenoid pigments of thermophilic bacteria. (b) The sulfur cauldron in Yellowstone National Park. The water is at its boiling point and very rich in sulfur. *Sulfolobus* grows well in such habitats.

at 105°C, and a maximum at 110°C. Both organotrophic and lithotrophic growth occur in this group. Sulfur and H₂ are the most common electron sources for lithotrophs. At present, the phylum contains 69 genera; two of the better-studied genera are *Thermoproteus* and *Sulfolobus*.

Members of the genus *Sulfolobus* are gram-negative, aerobic, irregularly lobed spherical archaeons with a temperature optimum around 70 to 80°C and a pH optimum of 2 to 3 (figure 20.9a,b). For this reason, they are classed as **thermoacidophiles**, so called because they grow best at acid pH values and high temperatures. Their cell wall contains lipoprotein and carbohydrate but lacks peptidoglycan. They grow lithotrophically on sulfur granules in hot acid springs and soils while oxidizing the sulfur

to sulfuric acid (figures 20.8b and 20.9b). Oxygen is the normal electron acceptor, but ferric iron may be used. Sugars and amino acids such as glutamate also serve as carbon and energy sources.

Thermoproteus is a long thin rod that can be bent or branched (figure 20.9c). Its cell wall is composed of glycoprotein. *Thermoproteus* is a strict anaerobe and grows at temperatures from 70 to 97°C and pH values between 2.5 and 6.5. It is found in hot springs and other hot aquatic habitats rich in sulfur. It can grow organotrophically and oxidize glucose, amino acids, alcohols, and organic acids with elemental sulfur as the electron acceptor. That is, *Thermoproteus* can carry out anaerobic respiration. It will also grow chemolithotrophically using H₂ and S⁰. Carbon monoxide or CO₂ can serve as the sole carbon source.

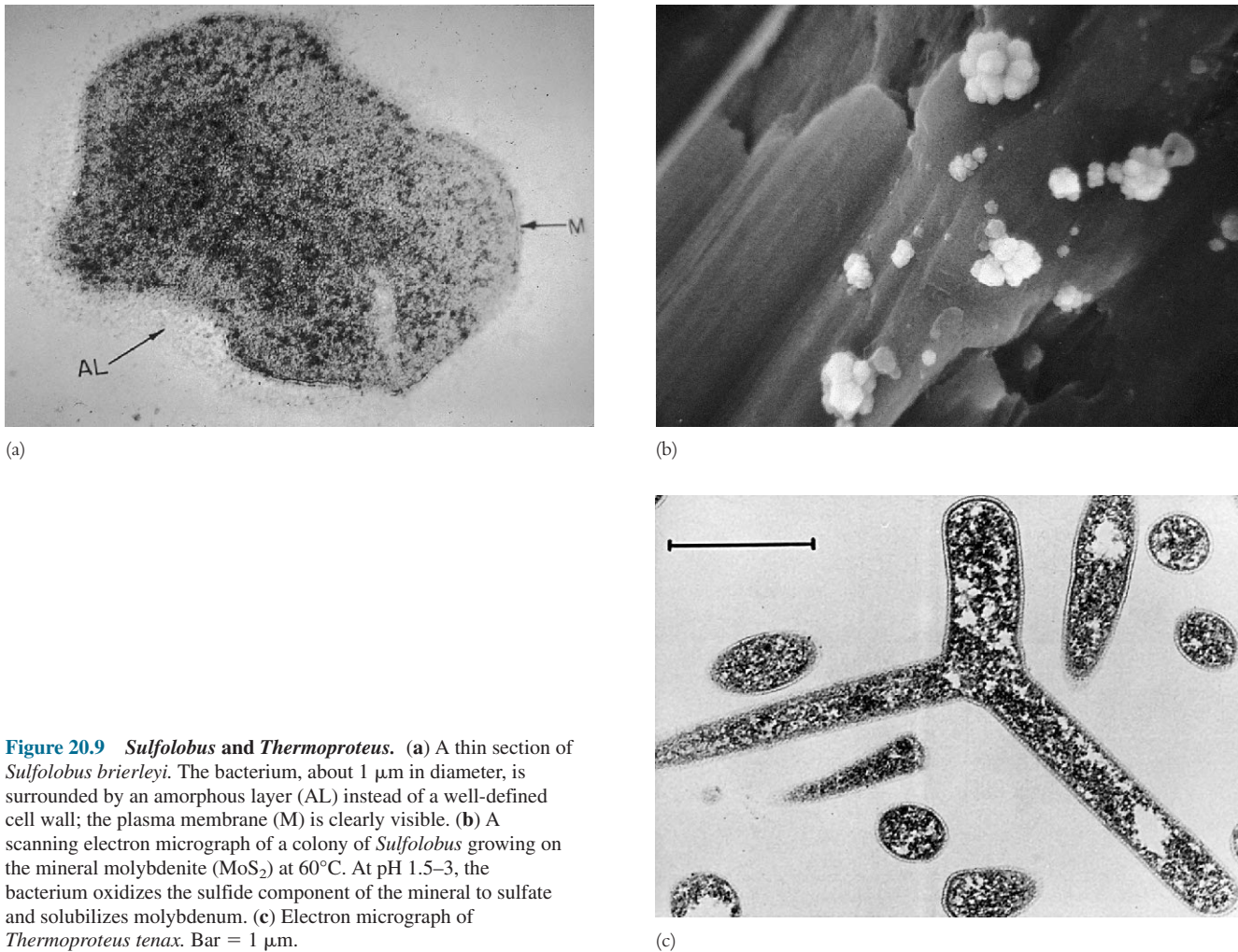


Figure 20.9 *Sulfolobus* and *Thermoproteus*. (a) A thin section of *Sulfolobus brierleyi*. The bacterium, about 1 μm in diameter, is surrounded by an amorphous layer (AL) instead of a well-defined cell wall; the plasma membrane (M) is clearly visible. (b) A scanning electron micrograph of a colony of *Sulfolobus* growing on the mineral molybdenite (MoS₂) at 60°C. At pH 1.5–3, the bacterium oxidizes the sulfide component of the mineral to sulfate and solubilizes molybdenum. (c) Electron micrograph of *Thermoproteus tenax*. Bar = 1 μm.

20.3 Phylum Euryarchaeota

As mentioned previously, the *Euryarchaeota* is a very diverse phylum with many classes, orders, and families. For the sake of clarity, the five major groups that comprise the euryarchaeotes will be briefly discussed with an emphasis on their physiology and ecology.

The Methanogens

Methanogens are strict anaerobes that obtain energy by converting CO₂, H₂, formate, methanol, acetate, and other compounds to either methane or methane and CO₂. They are autotrophic when growing on H₂ and CO₂. This is the largest group of archaea. There are five orders (*Methanobacteriales*, *Methanococcales*, *Methanomicrobiales*, *Methanosarcinales*, and *Methanopyrales*) and 26 genera, which differ greatly in overall shape, 16S rRNA sequence, cell wall chemistry and structure, membrane lipids, and other features. For example, methanogens construct three different types of cell walls. Several genera have walls with pseudomurein as men-

tioned earlier (figure 20.2). Other walls contain either proteins or heteropolysaccharides. The morphology of typical methanogens is shown in **figure 20.10**, and selected properties of representative genera are presented in **table 20.2**

One of the most unusual methanogenic groups is the class *Methanopyri*. It has one order, *Methanopyrales*, one family and a single genus, *Methanopyrus*. This extremely thermophilic rod-shaped methanogen has been isolated from a marine hydrothermal vent. *Methanopyrus kandleri* has a temperature minimum at 84°C and an optimum of 98°C; it will grow up to 110°C (above the boiling point of water). *Methanopyrus* occupies the deepest and most ancient branch of the euryarchaeotes. Perhaps methanogenic archaea were among the earliest organisms. They certainly seem well adapted to living under conditions similar to those presumed to have existed on a young Earth.

As might be inferred from the methanogens' ability to produce methane anaerobically, their metabolism is unusual. These procaryotes contain several unique cofactors: tetrahydromethanopterin (H₄MPT), methanofuran (MFR), coenzyme M

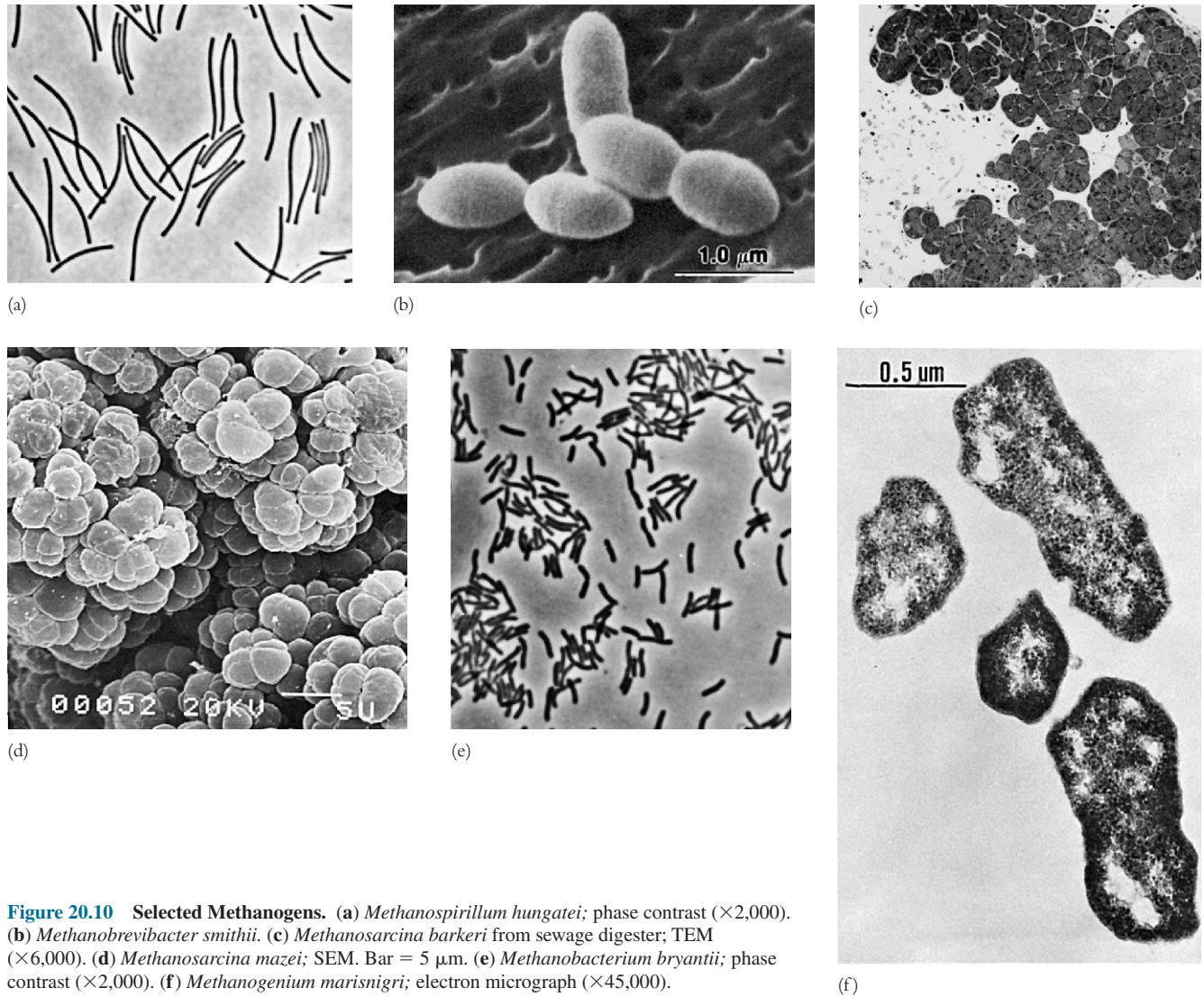


Figure 20.10 Selected Methanogens. (a) *Methanospirillum hungatei*; phase contrast ($\times 2,000$). (b) *Methanobrevibacter smithii*. (c) *Methanosarcina barkeri* from sewage digester; TEM ($\times 6,000$). (d) *Methanosarcina mazei*; SEM. Bar = 5 μm . (e) *Methanobacterium bryantii*; phase contrast ($\times 2,000$). (f) *Methanogenium marisnigri*; electron micrograph ($\times 45,000$).

Table 20.2 Selected Characteristics of Representative Genera of Methanogens

Genus	Morphology	% G + C	Wall Composition	Gram Reaction	Motility	Methanogenic Substrates Used
Order <i>Methanobacteriales</i>						
<i>Methanobacterium</i>	Long rods or filaments	32–61	Pseudomurein	+ to variable	–	H ₂ + CO ₂ , formate
<i>Methanothermus</i>	Straight to slightly curved rods	33	Pseudomurein with an outer protein S-layer	+	+	H ₂ + CO ₂
Order <i>Methanococcales</i>						
<i>Methanococcus</i>	Irregular cocci	29–34	Protein	–	–	H ₂ + CO ₂ , formate
Order <i>Methanomicrobiales</i>						
<i>Methanomicrobium</i>	Short curved rods	45–49	Protein	–	+	H ₂ + CO ₂ , formate
<i>Methanogenium</i>	Irregular cocci	52–61	Protein or glycoprotein	–	–	H ₂ + CO ₂ , formate
<i>Methanospirillum</i>	Curved rods or spirilla	45–50	Protein	–	+	H ₂ + CO ₂ , formate
<i>Methanosarcina</i>	Irregular cocci, packets	36–43	Heteropolysaccharide or protein	+ to variable	–	H ₂ + CO ₂ , methanol, methylamines, acetate

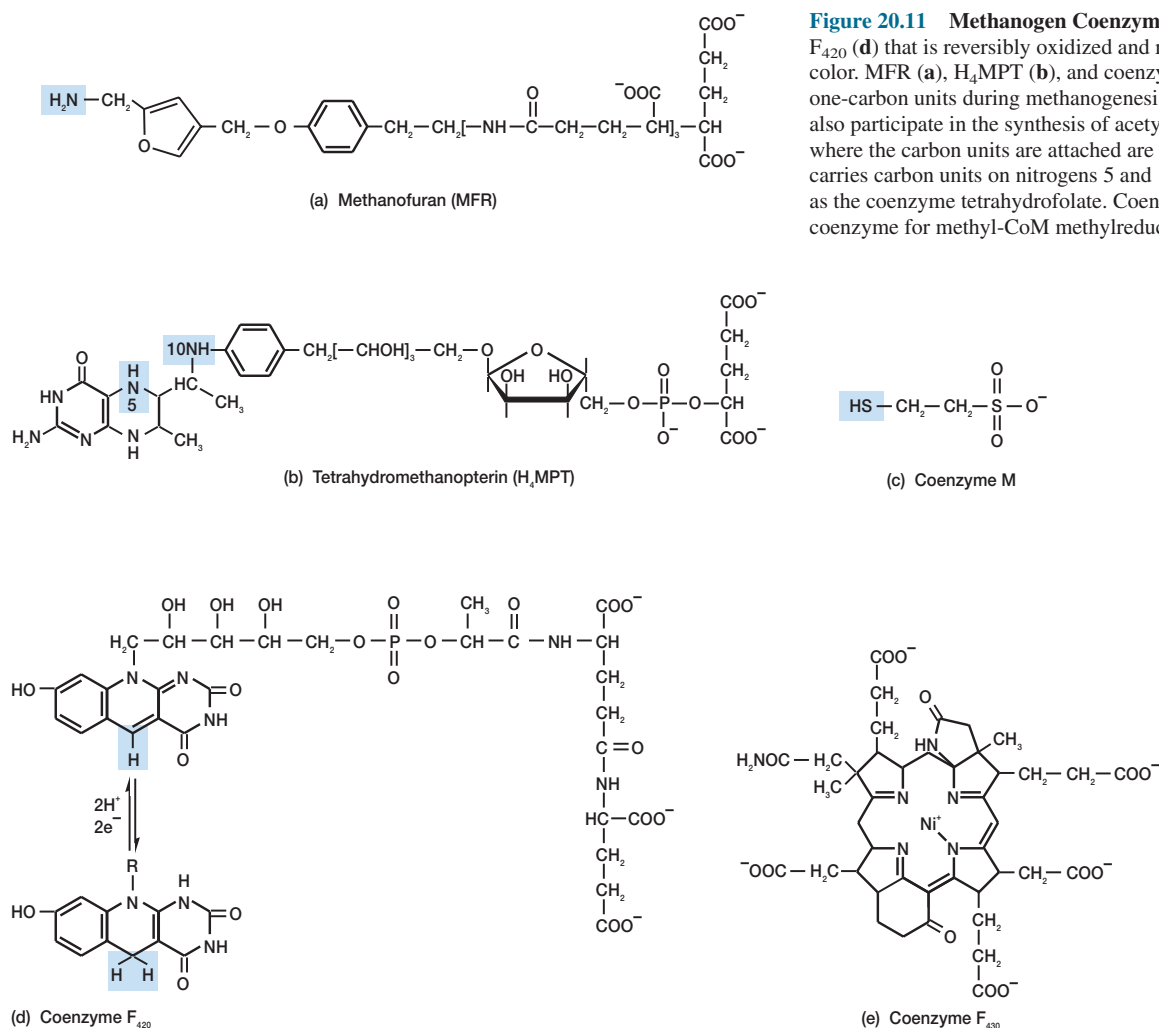


Figure 20.11 Methanogen Coenzymes. The portion of F₄₂₀ (d) that is reversibly oxidized and reduced is shown in color. MFR (a), H₄MPT (b), and coenzyme M (c) carry one-carbon units during methanogenesis (MFR and MPT also participate in the synthesis of acetyl-CoA). The places where the carbon units are attached are in color. H₄MPT carries carbon units on nitrogens 5 and 10 in the same way as the coenzyme tetrahydrofolate. Coenzyme F₄₃₀ (e) is a coenzyme for methyl-CoM methylreductase.

(2-mercaptoethanesulfonic acid), coenzyme F₄₂₀, and coenzyme F₄₃₀ (figure 20.11). The first three cofactors bear the C₁ unit when CO₂ is reduced to CH₄. F₄₂₀ carries electrons and hydrogens, and F₄₃₀ is a nickel tetrapyrrole serving as a cofactor for the enzyme methyl-CoM methylreductase. The pathway for methane synthesis is thought to function as shown in figure 20.12. It appears that ATP synthesis is linked with methanogenesis by electron transport, proton pumping, and a chemiosmotic mechanism (see pp. 187–89). Some methanogens can live autotrophically by forming acetyl-CoA from two molecules of CO₂ and then converting the acetyl-CoA to pyruvate and other products (figure 20.6).

Methanogens thrive in anaerobic environments rich in organic matter: the rumen and intestinal system of animals, freshwater and marine sediments, swamps and marshes, hot springs, anaerobic sludge digesters, and even within anaerobic protozoa. Methanogens often are of ecological significance. The rate of methane production can be so great that bubbles of methane will sometimes rise to the surface of a lake or pond. Rumen methanogens are so active that a cow can belch 200 to 400 liters of methane a day.

Methanogenic archaea are potentially of great practical importance since methane is a clean-burning fuel and an excellent energy source. For many years sewage treatment plants have been using the methane they produce as a source of energy for heat and electricity (see figure 29.28). Anaerobic digester microbes will degrade particulate wastes such as sewage sludge to H₂, CO₂, and acetate. CO₂-reducing methanogens form CH₄ from CO₂ and H₂, while aceticlastic methanogens cleave acetate to CO₂ and CH₄ (about 2/3 of the methane produced by an anaerobic digester comes from acetate). A kilogram of organic matter can yield up to 600 liters of methane. It is quite likely that future research will greatly increase the efficiency of methane production and make methanogenesis an important source of pollution-free energy.

Methanogenesis also can be an ecological problem (see section 30.6). Methane absorbs infrared radiation and thus is a greenhouse gas. There is evidence that atmospheric methane concentrations have been rising over the last 200 years. Methane production may significantly promote future global warming (see Box 30.3). Recently it has been discovered that methanogens can oxidize Fe⁰ and

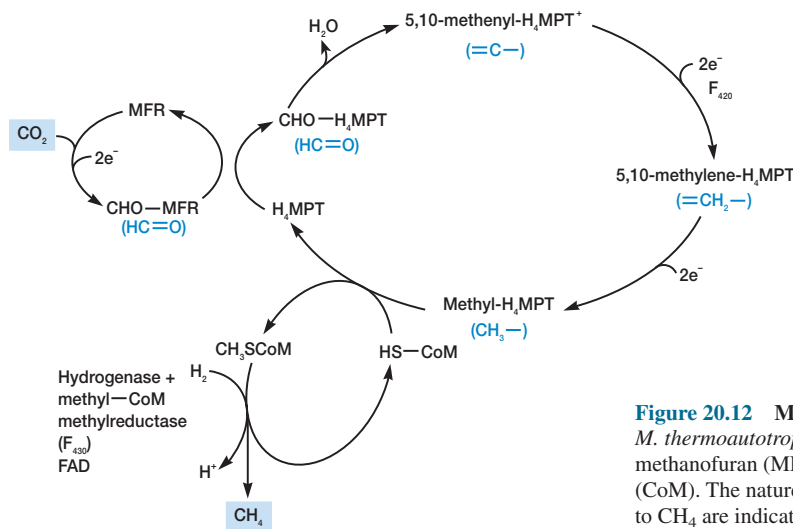


Figure 20.12 Methane Synthesis. Pathway for CH₄ synthesis from CO₂ in *M. thermoautotrophicum*. Cofactor abbreviations: methanopterin (MPT), methanofuran (MFR), and 2-mercaptoethanesulfonic acid or coenzyme M (CoM). The nature of the carbon-containing intermediates leading from CO₂ to CH₄ are indicated in parentheses. See text for further details.

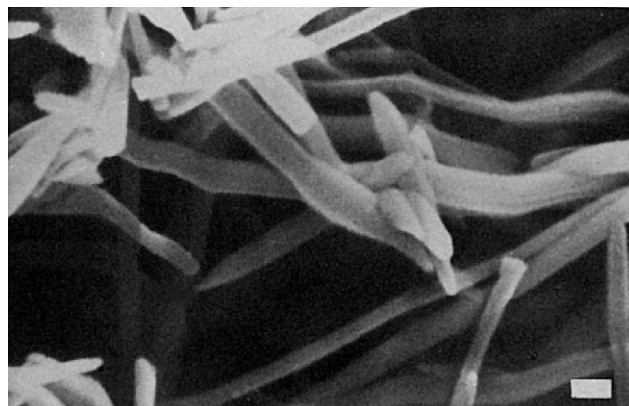
use it to produce methane and energy. This means that methanogens growing around buried or submerged iron pipes and other objects may contribute significantly to iron corrosion.

The Halobacteria

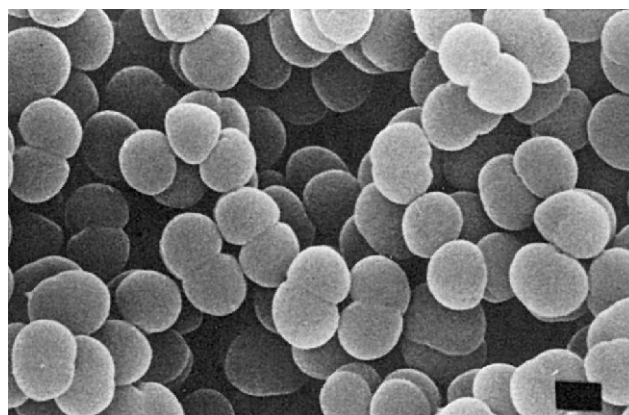
The **extreme halophiles** or **halobacteria**, class *Halobacteria*, are another major group of archaea, currently with 15 genera in one family, the *Halobacteriaceae* (figure 20.13). They are aerobic chemoheterotrophs with respiratory metabolism and require complex nutrients, usually proteins and amino acids, for growth. Species are either nonmotile or motile by lophotrichous flagella.

The most obvious distinguishing trait of this family is its absolute dependence on a high concentration of NaCl. These procaryotes require at least 1.5 M NaCl (about 8%, wt/vol), and usually have a growth optimum at about 3 to 4 M NaCl (17 to 23%). They will grow at salt concentrations approaching saturation (about 36%). *Halobacterium*'s cell wall is so dependent on the presence of NaCl that it disintegrates when the NaCl concentration drops to about 1.5 M. Thus halobacteria only grow in high-salinity habitats such as marine salterns (see figure 28.33a) and salt lakes such as the Dead Sea between Israel and Jordan, and the Great Salt Lake in Utah. They also can grow in food products such as salted fish and cause spoilage. Halobacteria often have red-to-yellow pigmentation from carotenoids that are probably used as protection against strong sunlight (see section 6.4). They can reach such high population levels that salt lakes, salterns, and salted fish actually turn red. [The effect of solutes on halophile growth \(pp. 121–23\)](#)

Probably the best-studied member of the family is *Halobacterium salinarum* (*H. halobium*). This procaryote is unusual because it can trap light energy photosynthetically without the presence of chlorophyll. When exposed to low oxygen concentrations, some strains of *Halobacterium* synthesize a modified cell membrane called the **purple membrane**, which contains the protein **bacteriorhodopsin**. ATP is produced by a unique type of photosynthesis without the participation of bacteriochlorophyll or chlorophyll (**Box 20.1**).



(a)



(b)

Figure 20.13 Examples of Halobacteria. (a) *Halobacterium salinarum*. A young culture that has formed long rods; SEM. Bar = 1 μ m. (b) *Halococcus morrhuae*; SEM. Bar = 1 μ m.

Box 20.1

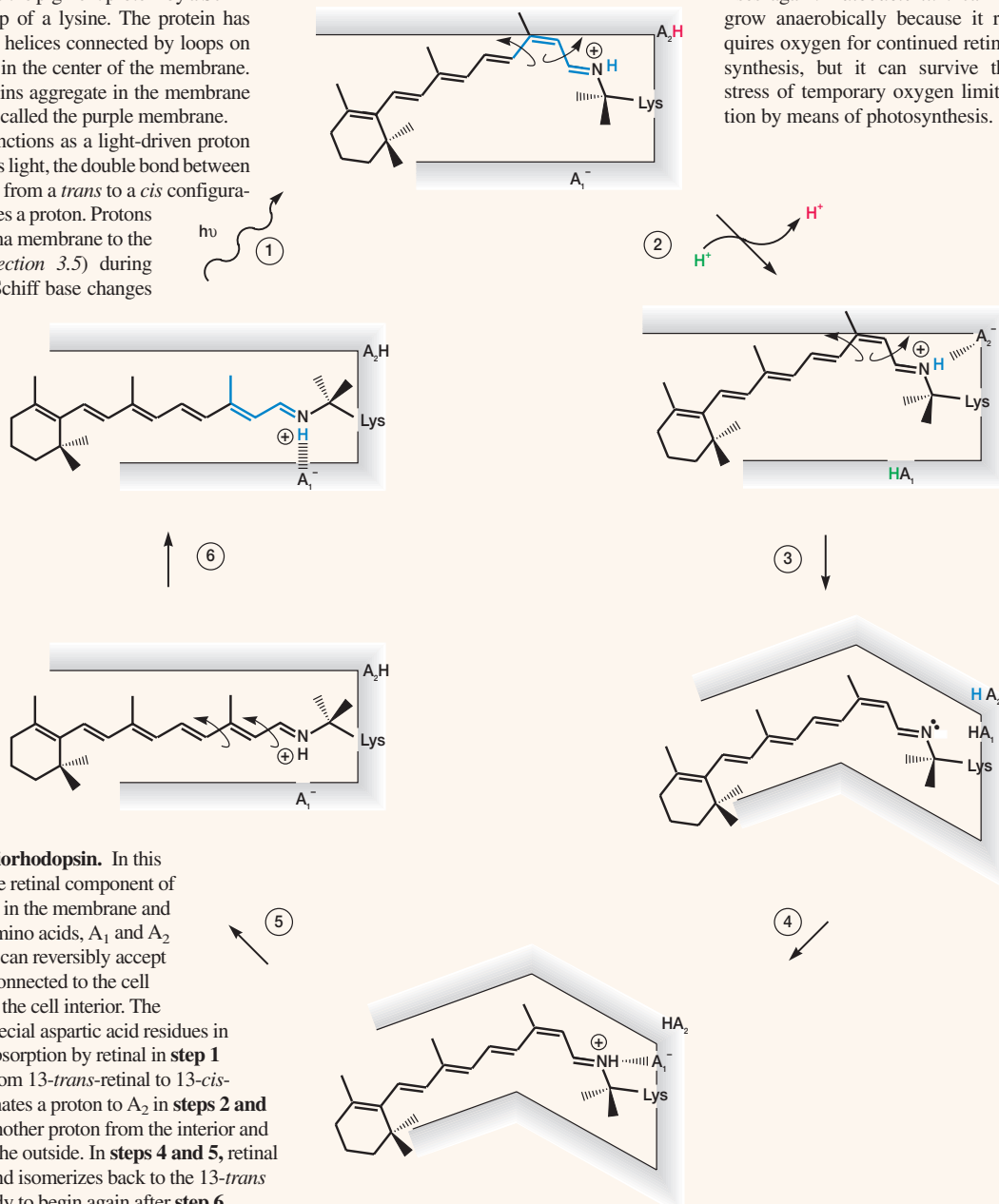
Photosynthesis in *Halobacterium salinarium*

The halophilic bacterium *Halobacterium salinarium* (formerly *H. halobium*) normally depends on respiration for the production of energy. However, under conditions of low oxygen and high light intensity, the bacteria synthesize a deep-purple pigment called bacteriorhodopsin, which closely resembles the sensory pigment rhodopsin from the rods and cones of vertebrate eyes. Bacteriorhodopsin's chromophore is the carotenoid derivative retinal (the aldehyde of vitamin A) that is covalently attached to the pigment protein by a Schiff base with the amino group of a lysine. The protein has seven membrane-spanning helices connected by loops on either side; its retinal rests in the center of the membrane. Individual bacteriorhodopsins aggregate in the membrane to form crystalline patches called the purple membrane.

Bacteriorhodopsin functions as a light-driven proton pump. When retinal absorbs light, the double bond between carbons 13 and 14 changes from a *trans* to a *cis* configuration and the Schiff base loses a proton. Protons are moved across the plasma membrane to the periplasmic space (see section 3.5) during these alterations, and the Schiff base changes are directly involved in this movement (see **Box figure**). The bacteriorhodopsin protein undergoes several conformational changes during the photocycle. These conformational changes also are involved in proton transport. The light-driven proton pumping generates a pH gradient

that can be used to power the synthesis of ATP by a chemiosmotic mechanism (see section 9.5).

This photosynthetic capacity is particularly useful to *Halobacterium* because oxygen is not very soluble in concentrated salt solutions and may decrease to an extremely low level in its habitat. When the surroundings become temporarily anaerobic, the bacterium uses light energy to synthesize sufficient ATP to remain alive until the oxygen level rises again. *Halobacterium* cannot grow anaerobically because it requires oxygen for continued retinal synthesis, but it can survive the stress of temporary oxygen limitation by means of photosynthesis.



The Photocycle of Bacteriorhodopsin. In this hypothetical mechanism the retinal component of bacteriorhodopsin is buried in the membrane and retinal interacts with two amino acids, A_1 and A_2 (aspartates 96 and 85), that can reversibly accept and donate protons. A_2 is connected to the cell exterior, and A_1 is closer to the cell interior. The two amino acids may be special aspartic acid residues in bacteriorhodopsin. Light absorption by retinal in **step 1** triggers an isomerization from 13-*trans*-retinal to 13-*cis*-retinal. The retinal then donates a proton to A_2 in **steps 2 and 3**, while A_1 is picking up another proton from the interior and A_2 is releasing a proton to the outside. In **steps 4 and 5**, retinal obtains a proton from A_1 and isomerizes back to the 13-*trans* form. The cycle is then ready to begin again after **step 6**.

Halobacterium actually has four rhodopsins, each with a different function. As already mentioned, bacteriorhodopsin drives outward proton transport for purposes of ATP synthesis. Halorhodopsin uses light energy to transport chloride ions into the cell and maintain a 4 to 5 M intracellular KCl concentration. Finally there are two rhodopsins that act as photoreceptors, one for red light and one for blue. They control flagellar activity to position the organism optimally in the water column. *Halobacterium* moves to a location of high light intensity, but one in which ultraviolet light is not sufficiently intense to be lethal.

The Thermoplasmas

Prokaryotes in the class *Thermoplasmata* are thermoacidophiles that lack cell walls. At present, only two genera, *Thermoplasma* and *Picrophilus*, are known. They are sufficiently different from one another to be placed in separate families, *Thermoplasmataceae* and *Picrophilaceae*.

Thermoplasma grows in refuse piles of coal mines. These piles contain large amounts of iron pyrite (FeS), which is oxidized to sulfuric acid by chemolithotrophic bacteria. As a result the piles become very hot and acidic. This is an ideal habitat for *Thermoplasma* since it grows best at 55 to 59°C and pH 1 to 2. Although it lacks a cell wall, its plasma membrane is strengthened by large quantities of diglycerol tetraethers, lipopolysaccharides, and glycoproteins. The organism's DNA is stabilized by association with a special histone-like protein that condenses the DNA into particles resembling eucaryotic nucleosomes. At 59°C, *Thermoplasma* takes the form of an irregular filament, whereas at lower temperatures it is spherical (figure 20.14). The cells may be flagellated and motile.

Picrophilus is even more unusual than *Thermoplasma*. It originally was isolated from moderately hot solfataric fields in Japan. Although it lacks a regular cell wall, *Picrophilus* has an S-layer outside its plasma membrane (see section 3.6). The cells grow as irregularly shaped cocci, around 1 to 1.5 µm in diameter and have large cytoplasmic cavities that are not membrane bounded. *Picrophilus* is aerobic and grows between 47 and 65°C with an optimum of 60°C. It is most remarkable in its pH requirements. The organism will grow only below pH 3.5 and has a growth optimum at pH 0.7. Growth actually occurs at about pH 0!

Extremely Thermophilic S⁰-Metabolizers

This physiological group contains the class *Thermococci*, with one order, *Thermococcales*. The *Thermococcales* are strictly anaerobic and can reduce sulfur to sulfide. They are motile by flagella and have optimum growth temperatures around 88 to 100°C. The order contains one family and two genera, *Thermococcus* and *Pyrococcus*.

Sulfate-Reducing Archaea

Archaeal sulfate reducers are found in the class *Archaeoglobi* and the order *Archaeoglobales*. This order has only one family and one genus. *Archaeoglobus* contains gram-negative, irregular coccoid cells with walls consisting of glycoprotein subunits. It can extract electrons from a variety of electron donors (e.g., H₂, lactate, glucose) and reduce sulfate, sulfite, or thiosulfate to sulfide. Elemental sulfur is not used as an acceptor. *Archaeoglobus* is extremely

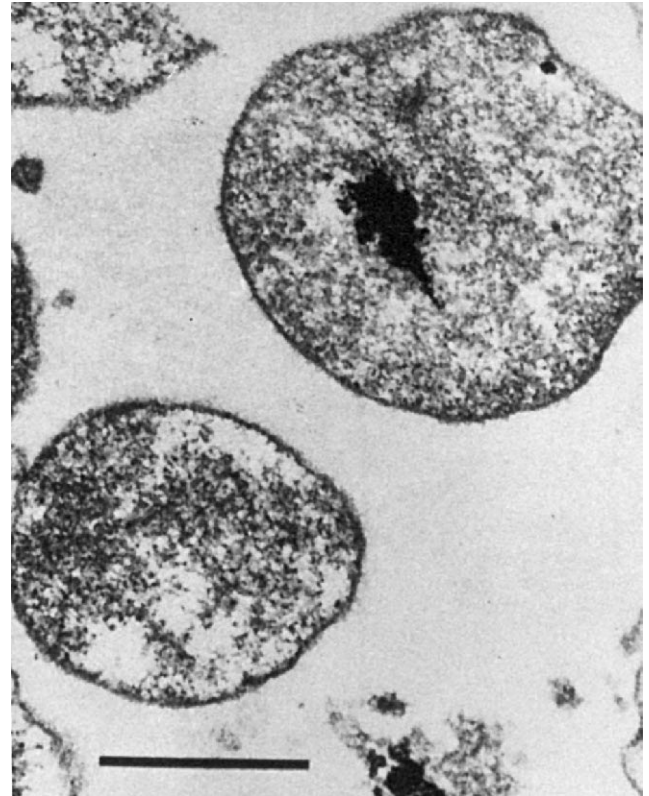


Figure 20.14 *Thermoplasma*. Transmission electron micrograph. Bar = 0.5 µm.

thermophilic (the optimum is about 83°C) and can be isolated from marine hydrothermal vents. The organism is not only unusual in being able to reduce sulfate, unlike other archaea, but it also possesses the methanogen coenzymes F₄₂₀ and methanopterin.

1. What are thermoacidophiles and where do they grow? In what ways do they use sulfur in their metabolism? Briefly describe *Sulfolobus* and *Thermoproteus*.
2. Generally characterize methanogenic archaea and distinguish them from other groups.
3. Briefly describe how methanogens produce methane and the roles of their unique cofactors in this process.
4. Where does one find methanogens? Discuss their ecological and practical importance.
5. Where are the extreme halophiles found and what is unusual about their cell walls and growth requirements?
6. Briefly describe how *Halobacterium* carries out photosynthesis. What is the purple membrane and what pigment does it contain?
7. How is *Thermoplasma* able to live in acidic, very hot coal refuse piles when it lacks a cell wall? How is its DNA stabilized? What is so remarkable about *Picrophilus*?
8. Characterize *Archaeoglobus*. In what way is it similar to the methanogens and how does it differ from other extreme thermophiles?

Summary

1. Organisms may be divided into three major groups: *Archaea*, *Bacteria*, and *Eucarya* or eucaryotes.
2. The *Archaea* are highly diverse with respect to morphology, reproduction, physiology, and ecology. They grow in anaerobic, hypersaline, and high-temperature habitats.
3. Archaeal cell walls do not contain peptidoglycan and differ from bacterial walls in structure. They may be composed of pseudomurein, polysaccharides, or glycoproteins and other proteins.
4. The membrane lipids differ from those of other organisms in having branched chain hydrocarbons connected to glycerol by ether links. Bacterial and eucaryotic lipids have glycerol connected to fatty acids by ester bonds (figure 20.3).
5. Their tRNA, ribosomes, elongation factors, RNA polymerases, and other components distinguish archaea from bacteria and eucaryotes.
6. Although much of archaeal metabolism appears similar to that of other organisms, the archaea do differ with respect to glucose catabolism, pathways for CO₂ fixation, and the ability of some to synthesize methane.
7. Archaea may be divided into five groups: Methanogenic archaea, sulfate reducers, extreme halophiles, cell wall-less archaea, and extremely thermophilic S⁰-metabolizers (table 20.1).
8. The second edition of *Bergey's Manual* divides the *Archaea* into two phyla, the *Crenarchaeota* and *Euryarchaeota*, each with several orders (figure 19.14).
9. The extremely thermophilic S⁰-metabolizers in the phylum *Crenarchaeota* depend on sulfur for growth and are frequently acidophiles. The sulfur may be used as an electron acceptor in anaerobic respiration or as an electron donor by lithotrophs. They are almost always strict anaerobes and grow in geothermally heated soil and water that is rich in sulfur.
10. The phylum *Euryarchaeota* contains five major groups: methanogens, halobacteria, the thermoplasmas, extremely thermophilic S⁰-metabolizers, and sulfate-reducing archaea.
11. Methanogenic archaea are strict anaerobes that can obtain energy through the synthesis of methane. They have several unusual cofactors that are involved in methanogenesis.
12. Extreme halophiles or halobacteria are aerobic chemoheterotrophs that require at least 1.5 M NaCl for growth. They are found in habitats such as salterns, salt lakes, and salted fish.
13. *Halobacterium salinarium* can carry out photosynthesis without chlorophyll or bacteriorhodopsin, which employs retinal to pump protons.
14. The thermophilic archaeon *Thermoplasma* grows in hot, acid coal refuse piles and survives despite the lack of a cell wall. Another thermoplasm, *Picrophilus*, can grow at pH 0.
15. The class *Thermococci* contains extremely thermophilic organisms that can reduce sulfur to sulfide.
16. Sulfate-reducing archaea are placed in the class *Archaeoglobi*. The extreme thermophile *Archaeoglobus* differs from other archaea in using a variety of electron donors to reduce sulfate. It also contains the methanogen cofactors F₄₂₀ and methanopterin.

Key Terms

Archaea 451
bacteriorhodopsin 461
extreme halophiles 461

halobacteria 461
methanogens 458
pseudomurein 452

purple membrane 461
thermoacidophiles 457

Questions for Thought and Review

1. Discuss how the unusual properties of extreme halophiles and thermophiles reflect the habitats in which they grow.
2. Why do we classify *Thermoplasma* as an archaeon rather than as a mycoplasma?
3. Discuss what adaptations extremely thermophilic archaea require to grow at 100°C.
4. Do you think archaea should be separate from the bacteria although both groups are procaryotic? Give your reasoning and evidence.
5. Supposing you wished to isolate procaryotes from a hot spring in Yellowstone Park, how would you go about it?

Critical Thinking Questions

1. Explain why the fixation of CO₂ by *Thermoproteus* and possibly by *Sulfolobus* using a reductive reversal of the TCA cycle is *not* photosynthesis.
2. Often when the temperature increases, many procaryotes change their shapes from elongated rods into spheres. Suggest one reason for this change.
3. Why would ether linkages be more stable in membrane lipids? How would the presence of tetraether linkages stabilize a thermophile's membrane?

Additional Reading

20.1 Introduction to the Archaea

- Balows, A.; Truper, H. G.; Dworkin, M.; Harder, W.; and Schleifer, K.-H. 1992. *The prokaryotes*, 2d ed. New York: Springer-Verlag.
- Bell, S. D., and Jackson, S. P. 1998. Transcription and translation in Archaea: A mosaic of eukaryal and bacterial features. *Trends Microbiol.* 6(6):222–28.
- Bernander, R. 2000. Chromosome replication, nucleoid segregation and cell division in Archaea. *Trends Microbiol.* 8(6):278–83.
- Charlebois, R. L. 1999. Archaea: Whose sister lineage? In *Organization of the prokaryotic genome*, R. L. Charlebois, editor, 63–76. Washington, D.C.: ASM Press.
- Cramer, W. A., and Knaff, D. B. 1991. *Energy transduction in biological membranes*. New York: Springer-Verlag.
- Danson, M. J. 1988. Archaeobacteria: The comparative enzymology of their central metabolic pathways. In *Advances in microbial physiology*, A. H. Rose and D. W. Tempest, editors, 165–231. New York: Academic press.
- Danson, M. J., and Hough, D. W. 1998. Structure, function and stability of enzymes from the Archaea. *Trends Microbiol.* 6(8):307–14.
- Edgell, D., and Doolittle, W. F. 1997. Archaea and the origin(s) of DNA replication proteins. *Cell* 89(7):995–98.
- Fuhrman, J. A., and Davis, A. A. 1997. Widespread archaea and novel bacteria from the deep sea as shown by 16S rRNA gene sequences. *Mar. Ecol. Prog. Ser.* 150:275–85.
- Garrity, G. M., editor-in-chief. 2001. *Bergey's manual of systematic bacteriology*, 2d. ed., vol. 1, D. R. Boone and R. W. Castenholz, editors. New York: Springer-Verlag.
- Kandler, O., and Zillig, W., editors. 1986. *Archaeobacteria '85*. New York: Gustav Fischer Verlag.
- Kates, M.; Kushner, D. J.; and Matheson, A. T. 1993. *The biochemistry of Archaea* (Archaeobacteria). New York: Elsevier.
- Macario, A. J. L.; Lange, M.; Ahring, B. K.; and De Macario, E. C. 1999. Stress genes and proteins in the Archaea. *Micro. Mol. Biol. Rev.* 63(4):923–67.
- Olsen, G. J., and Woese, C. R. 1997. Archaeal genomics: An overview. *Cell* 89(7):991–94.

- Reeve, J. N.; Sandman, K.; and Daniels, C. J. 1997. Archaeal histones, nucleosomes, and transcription initiation. *Cell* 89(7):999–1002.
- Schäfer, G.; Engelhard, M.; and Müller, V. 1999. Bioenergetics of the Archaea. *Micro. Mol. Biol. Rev.* 63(3):570–620.
- Schlegel, H. G., and Bowien, B. 1989. *Autotrophic bacteria*. Madison, Wis.: Science Tech Publishers.
- Sowers, K. R., and Schreier, H. J. 1999. Gene transfer systems for the Archaea. *Trends Microbiol.* 7(5):212–19.
- Vetriani, C., and Reysenbach, A.-L. 2000. Archaea. In *Encyclopedia of microbiology*, 2d ed., vol. 1, J. Lederberg, editor-in-chief, 319–31. San Diego: Academic Press.
- Woese, C. R. 1981. Archaeobacteria. *Sci. Am.* 244(6):98–122.
- Woese, C. R., and Wolfe, R. S., editors. 1985. *Archaeobacteria. Volume VIII of The bacteria: A treatise on structure and function*. New York: Academic Press.
- Woese, C. R.; Kandler, O.; and Wheelis, M. L. 1990. Toward a natural system of organisms: Proposal for the domains Archaea, Bacteria, and Eucarya. *Proc. Nat. Acad. Sci.* 87:4576–79.
- Wood, H. G.; Radsdale, S. W.; and Pezacka, E. 1986. The acetyl-CoA pathway: A newly discovered pathway of autotrophic growth. *Trends Biochem. Sci.* 11(1):14–17.
- Zillig, W.; Palm, P.; Reiter, W.-D.; Gropp, F.; Pühler, G.; and Klenk, H.-P. 1988. Comparative evaluation of gene expression in archaeobacteria. *Eur. J. Biochem.* 173:473–82.

20.2 Phylum Crenarchaeota

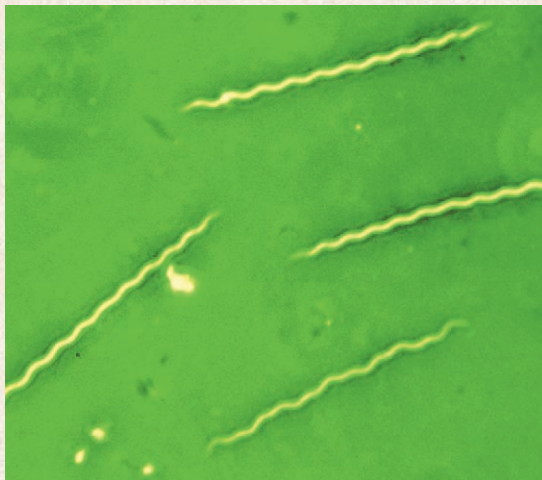
- Burggraf, S.; Huber, H.; and Stetter, K. O. 1997. Reclassification of the crenarchaeal orders and families in accordance with 16S rRNA sequence data. *Int. J. Syst. Bacteriol.* 47(3):657–60.
- Pley, U.; Schipka, J.; Gambacorta, A.; Jannasch, H. W.; Fricke, H.; Rachel, R.; and Stetter, K. O. 1991. *Pyrodictium abyssi* sp. nov. represents a novel heterotrophic marine archaeal hyperthermophile growing at 110°C. *System. Appl. Microbiol.* 14:245–53.

20.3 Phylum Euryarchaeota

- Balch, W. E.; Fox, G. E.; Magrum, L. J.; Woese, C. R.; and Wolfe, R. S. 1979. Methanogens: Reevaluation of a unique biological group. *Microbiol. Rev.* 43:260–96.
- Birge, R. R. 1990. Nature of the primary photochemical events in rhodopsin and bacteriorhodopsin. *Biochem. Biophys. Acta* 1016:293–327.
- Bult, C. J., et al. 1996. Complete genome sequence of the methanogenic archaeon, *Methanococcus jannaschii*. *Science* 273:1058–1107.
- DiMarco, A. A.; Bobik, T. A.; and Wolfe, R. S. 1990. Unusual coenzymes of methanogenesis. *Ann. Rev. Biochem.* 59:355–94.
- Jones, W. J.; Nagle, D. P., Jr.; and Whitman, W. B. 1987. Methanogens and the diversity of archaeobacteria. *Microbiol. Rev.* 51:135–77.
- Luecke, H.; Schobert, B.; Richter, H.-T.; Cartailier, J.-P.; and Lanyi, J. K. 1999. Structural changes in bacteriorhodopsin during ion transport at 2 angstrom resolution. *Science* 286:255–60.
- Norton, C. F. 1992. Rediscovering the ecology of halobacteria. *ASM News* 58(7):363–67.
- Oesterhelt, D., and Tittor, J. 1989. Two pumps, one principle: Light-driven ion transport in halobacteria. *Trends Biochem. Sci.* 14:57–61.
- Oren, A. 1999. Bioenergetic aspects of halophilism. *Micro. Mol. Biol. Rev.* 63(2):334–48.
- Schleper, C.; Puehler, G.; Holz, I.; Gambacorta, A.; Janekovic, D.; Santarius, U.; Klenk, H.-P.; and Zillig, W. 1995. *Picrophilus* gen. nov., fam. nov.: A novel aerobic, heterotrophic, thermoacidophilic genus and family comprising archaea capable of growth around pH 0. *J. Bacteriol.* 177(24):7050–59.
- Spudich, J. L. 1993. Color sensing in the Archaea: a eukaryotic-like receptor coupled to a procaryotic transducer. *J. Bacteriol.* 175(24):7755–61.
- Stoeckenius, W. 1976. The purple membrane of salt-loving bacteria. *Sci. Am.* 234(6):38–46.
- Zinder, S. H. 1984. Microbiology of anaerobic conversion of organic wastes to methane: Recent developments. *ASM News* 50(7):294–98.

CHAPTER 21

Bacteria: The Deinococci and Nonproteobacteria Gram Negatives



Spirochetes are distinguished by their structure and mechanism of motility. *Treponema pallidum* shown here causes syphilis.

Outline

- 21.1 *Aquificae* and *Thermotogae* 467
- 21.2 *Deinococcus-Thermus* 468
- 21.3 Photosynthetic Bacteria 468
 - Phylum *Chloroflexi* 470
 - Phylum *Chlorobi* 470
 - Phylum *Cyanobacteria* 471
- 21.4 Phylum *Planctomycetes* 477
- 21.5 Phylum *Chlamydiae* 477
- 21.6 Phylum *Spirochaetes* 479
- 21.7 Phylum *Bacteroidetes* 481

Concepts

1. The first edition of *Bergey's Manual* takes a largely phenetic approach to classification and relates bacteria based on their overall similarity. The second edition classifies bacteria according to phylogenetic relationships with emphasis on 16S rRNA sequence comparisons.
2. Some bacterial groups, such as those represented by the hyperthermophiles *Aquifex* and *Thermotoga*, are deeply branching and very old; other bacterial taxa have arisen more recently.
3. Because of its emphasis on phylogenetic relationships, the second edition of *Bergey's Manual* has substantially rearranged bacterial groups and taxonomic categories. For example, the second edition places the gram-positive deinococci in volume 1, which otherwise contains gram-negative bacteria. Bacteria such as the rickettsias and chlamydiae are separated into different sections despite their similar life-styles. The thermotogas and many other completely new groups have been added.

Concepts *(Continued)*

- Although most of the photosynthetic bacteria are located in volume 1 of the second edition, the purple bacteria have been moved to the proteobacteria in volume 2. The cyanobacteria are separated from other photosynthetic bacteria because they resemble eucaryotic phototrophs in having photosystem II and carrying out oxygenic photosynthesis. Their rRNA sequences also indicate that they are different from other photosynthetic bacteria.
- Bacteria such as the chlamydiae that are obligately intracellular parasites have relinquished some of their metabolic independence through loss of metabolic pathways. They use their host's energy supply and/or cell constituents.
- Gliding motility is widely distributed among bacteria and is very useful to organisms that digest insoluble nutrients or move over the surfaces of moist, solid substrata.

There are wide areas of the bacteriological landscape in which we have so far detected only some of the highest peaks, while the rest of the beautiful mountain range is still hidden in the clouds and the morning fogs of ignorance. The gold is still lying on the ground, but we have to bend down to grasp it.

Preface to The Prokaryotes

Chapter 20 surveyed the archaea, which are located in volume 1 of the second edition of *Bergey's Manual*. Volumes 1 and 5 of the new *Bergey's Manual* also will describe a wide variety of other procaryotic groups that are members of the second domain: *Bacteria*. Chapter 21 is devoted to 10 of the more interesting and important bacterial phyla from volumes 1 and 5. Their phylogenetic locations are depicted in the dendrogram on the top of this page. The general organization and perspective of the second edition of *Bergey's Manual* will be followed in most cases.

Although the treatment of each bacterial group varies somewhat from that of the others, usually an attempt is made to provide a brief review of the group's biology. Such aspects as distinguishing characteristics, morphology, reproduction, physiology, metabolism, and ecology are included. The taxonomy of each major group is summarized, and representative species are discussed. This approach should help one appreciate bacteria as living organisms rather than simply as agents of disease with little interest or importance in other contexts.

21.1 *Aquificae* and *Thermotogae*

The phylum *Aquificae*, which is thought to represent the deepest or oldest branch of bacteria, contains one class, one order, and five genera. Two of the best-studied genera are *Aquifex* and *Hydrogenobacter*. *Aquifex pyrophilus* is a gram-negative, microaerophilic rod. It is a hyperthermophile with a temperature optimum of 85°C and a maximum of 95°C. *Aquifex* is an autotroph

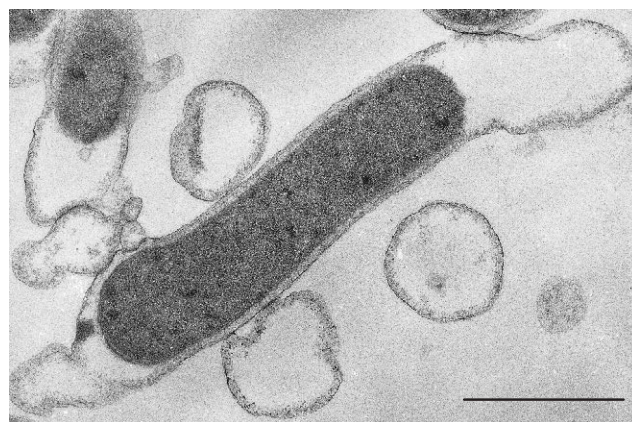
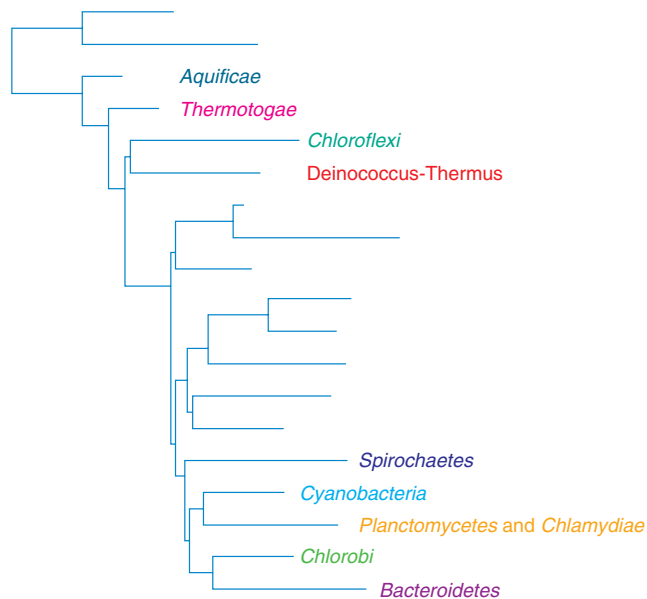


Figure 21.1 *Thermotoga maritima*. Note the loose sheath extending from each end of the cell. Bar = 1 μ m.

and generates energy by oxidizing donors such as hydrogen, thio-sulfate, and sulfur with oxygen as the acceptor. Because *Aquifex* and *Hydrogenobacter* are both thermophilic chemolithoautotrophs, it has been suggested that the bacterial ancestor was probably thermophilic and chemolithoautotrophic.

The second oldest or deepest branch is the phylum *Thermotogae*, which also has one class, one order, and five genera. The members of the genus *Thermotoga* [Greek *therme*, heat; Latin *toga*, outer garment], like *Aquifex*, are hyperthermophiles with a growth optimum of 80°C and a maximum of 90°C. They are gram-negative rods with an outer sheathlike envelope (like a toga) that can extend or balloon out from the ends of the cell (**figure 21.1**). They grow in

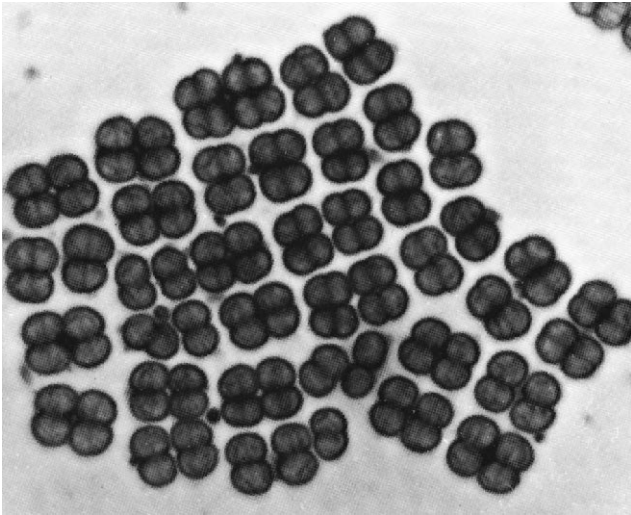


Figure 21.2 The Deinococci. A *Deinococcus radiodurans* microcolony showing cocci arranged in tetrads (average cell diameter 2.5 μm).

active geothermal areas, both marine hydrothermal systems and terrestrial solfataric springs. In contrast to *Aquifex*, *Thermotoga* is a chemoheterotroph with a functional glycolytic pathway and can grow anaerobically on carbohydrates and protein digests.

The genomes of *Aquifex aeolicus* and *Thermotoga maritima* have been sequenced. The *Aquifex* genome is about a third the size of the *E. coli* genome, but still contains the genes required for chemolithoautotrophy. The small genome does seem to reduce its metabolic flexibility; it will not grow on common organic substrates such as sugars and amino acids. The *Thermotoga* genome is somewhat larger and has genes for sugar degradation as already mentioned. About 24% of its coding sequences are similar to archaeal genes; this proportion is greater than that of other bacteria, including *Aquifex* (16% similarity) and may be due to horizontal gene transfer (see pp. 352–53).

21.2 Deinococcus-Thermus

The phylum Deinococcus-Thermus contains the class *Deinococci* and the orders *Deinococcales* and *Thermales*. There are only three genera in the phylum; the genus *Deinococcus* is best studied. Deinococci are spherical or rod-shaped with distinctively different 16S rRNA. They often are associated in pairs or tetrads (figure 21.2) and are aerobic, mesophilic, and catalase positive; usually they can produce acid from only a few sugars. Although they stain gram positive, their cell wall is layered and has an outer membrane like the gram-negative bacteria. They also differ from gram-positive cocci in having L-ornithine in their peptidoglycan, lacking teichoic acid, and having a plasma membrane with large amounts of palmitoleic acid rather than phosphatidylglycerol phospholipids. Almost all strains are extraordinarily resistant to both desiccation and radiation; they can survive as much as 3 to 5 million rad of radiation (an exposure of 100 rad can be lethal to humans).

Much remains to be discovered about the biology of these bacteria. Deinococci can be isolated from ground meat, feces, air, freshwater, and other sources, but their natural habitat is not yet known. Recent research indicates that their great resistance to desiccation and radiation may result from their ability to repair severely damaged chromosomes. The genome consists of two circular chromosomes, a megaplasmid, and a small plasmid. If the bacteria are exposed to enough radiation, their chromosomes are shattered into many fragments. Within 12 to 24 hours, they splice the fragments back together. A major reason for their ability to repair DNA is an unusually efficient RecA protein (see p. 255). However, it is thought that their chromosome structure also must be different for them to survive fragmentation. Recently the genome of *D. radiodurans* has been sequenced (see p. 351). The results show that the bacterium has an ample array of DNA repair mechanisms and many repeat sequences.

21.3 Photosynthetic Bacteria

There are three groups of photosynthetic bacteria: the purple bacteria, the green bacteria, and the cyanobacteria (table 21.1). The cyanobacteria differ most fundamentally from the green and purple photosynthetic bacteria in being able to carry out **oxygenic photosynthesis**. They use water as an electron donor and generate oxygen during photosynthesis. In contrast, purple and green bacteria use **anoxygenic photosynthesis**. Because they are unable to use water as an electron source, they employ reduced molecules such as hydrogen sulfide, sulfur, hydrogen, and organic matter as their electron source for the generation of NADH and NADPH. Consequently, purple and green bacteria do not produce oxygen but many form sulfur granules. Purple sulfur bacteria accumulate granules within their cells, whereas green sulfur bacteria deposit the sulfur granules outside their cells. The purple nonsulfur bacteria use organic molecules as an electron source. There also are differences in photosynthetic pigments, the organization of photosynthetic membranes, nutritional requirements, and oxygen relationships. [The mechanism of bacterial photosynthesis \(pp. 195–201\)](#)

The connection between photosynthetic pigments, oxygen relationships, and ecological distribution should be noted here. As table 21.1 indicates, the purple and green bacteria differ from the cyanobacteria in having bacteriochlorophylls rather than chlorophyll *a* (see figure 9.26). This proves to be quite useful because of the distinctive absorption spectra of the bacteriochlorophylls and accessory pigments. Normally green and purple bacteria are anaerobic and use H_2S and other reduced electron donors during photosynthesis (table 21.1). Because these bacteria grow best in deeper anaerobic zones of aquatic habitats, they cannot effectively use parts of the visible spectrum normally employed by photosynthetic organisms. There often is a dense surface layer of cyanobacteria and algae in lakes and ponds that absorbs a large amount of blue and red light. The bacteriochlorophyll pigments of purple and green bacteria absorb longer wavelength, far-red light (table 21.2) not used by other photosynthesizers (figure 21.3). In addition, the bacteriochlorophyll absorption peaks at about 350 to 550 nm enable

Table 21.1 Characteristics of the Major Groups of Photosynthetic Bacteria

Characteristic	<i>Anoxygenic Photosynthetic Bacteria</i>				<i>Oxygenic Photosynthetic Bacteria</i>
	Green Sulfur ^a	Green Nonsulfur ^b	Purple Sulfur	Purple Nonsulfur	Cyanobacteria
Major photosynthetic pigments	Bacteriochlorophylls <i>a</i> plus <i>c</i> , <i>d</i> , or <i>e</i> (the major pigment)	Bacteriochlorophylls <i>a</i> and <i>c</i>	Bacteriochlorophyll <i>a</i> or <i>b</i>	Bacteriochlorophyll <i>a</i> or <i>b</i>	Chlorophyll <i>a</i> plus phycobiliproteins
Morphology of photosynthetic membranes	Photosynthetic system partly in chlorosomes that are independent of the plasma membrane	Chlorosomes present when grown anaerobically	Photosynthetic system contained in spherical or lamellar membrane complexes that are continuous with the plasma membrane	Photosynthetic system contained in spherical or lamellar membrane complexes that are continuous with the plasma membrane	Membranes lined with phycobilisomes
Photosynthetic electron donors	H ₂ , H ₂ S, S	Photoheterotrophic donors—a variety of sugars, amino acids, and organic acids; photoautotrophic donors—H ₂ S, H ₂	H ₂ , H ₂ S, S	Usually organic molecules; sometimes reduced sulfur compounds or H ₂	H ₂ O
Sulfur deposition	Outside of the cell		Inside the cell ^c	Outside of the cell	
Nature of photosynthesis	Anoxygenic	Anoxygenic	Anoxygenic	Anoxygenic	Oxygenic (sometimes facultatively anoxygenic)
General metabolic type	Obligately anaerobic photolithoautotrophs	Usually photoheterotrophic; sometimes photoautotrophic or chemoheterotrophic (when aerobic and in the dark)	Obligately anaerobic photolithoautotrophs	Usually anaerobic photoorganoheterotrophs; some facultative photolithoautotrophs (in dark, chemoorganoheterotrophs)	Aerobic photolithoautotrophs
Motility	Nonmotile; some have gas vesicles	Gliding	Motile with polar flagella; some are peritrichously flagellated	Motile with polar flagella or nonmotile; some have gas vesicles	Nonmotile or with gliding motility; some have gas vesicles
Percent G + C	48–58	53–55	45–70	61–72	35–71

^aCharacteristics of *Chlorobi*.^bCharacteristics of *Chloroflexus*.^cWith the exception of *Ectothiorhodospira*.**Table 21.2** Prokaryotic Bacteriochlorophyll and Chlorophyll Absorption Maxima

Pigment	<i>Long Wavelength Maxima (nm)</i>	
	In Ether or Acetone	Approximate Range of Values in Cells
Chlorophyll <i>a</i>	665	680–685
Bacteriochlorophyll <i>a</i>	775	850–910 (purple bacteria) ^a
Bacteriochlorophyll <i>b</i>	790	1,020–1,035
Bacteriochlorophyll <i>c</i>	660	745–760
Bacteriochlorophyll <i>d</i>	650	725–745
Bacteriochlorophyll <i>e</i>	647	715–725

^aThe spectrum of bacteriochlorophyll *a* in green bacteria has a different maximum, 805–810 nm.

them to grow at greater depths because shorter wavelength light can penetrate water farther. As a result, when the water is sufficiently clear, a layer of green and purple bacteria develops in the anaerobic, hydrogen sulfide-rich zone (**figure 21.4**). [The microbial ecology of lakes \(pp. 648–49\)](#)

The second edition of *Bergey's Manual* places photosynthetic bacteria into six major groups. The phylum *Chloroflexi* contains the green nonsulfur bacteria, and the phylum *Chlorobi*, the green sulfur bacteria. The cyanobacteria are placed in their own phylum, *Cyanobacteria*. Purple bacteria are divided between three groups. Purple sulfur bacteria are placed in the γ -proteobacteria, families *Chromatiaceae* and *Ectothiorhodospiraceae*. The purple nonsulfur bacteria are distributed between the α -proteobacteria (five different families) and one family of the β -proteobacteria. [The purple photosynthetic bacteria \(pp. 488, 500–1\)](#)

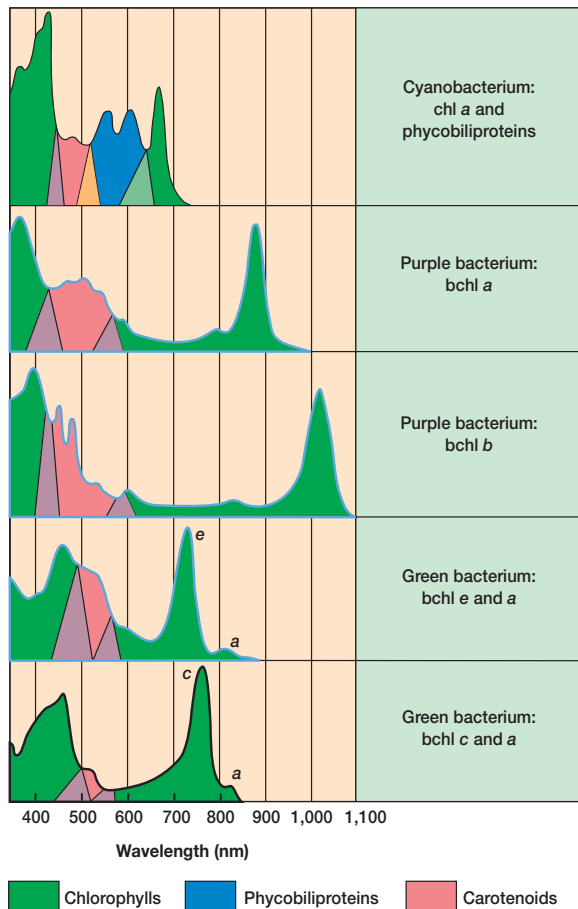


Figure 21.3 Photosynthetic Pigments. Absorption spectra of five photosynthetic bacteria showing the differences in absorption maxima and the contributions of various accessory pigments.

Phylum *Chloroflexi*

The phylum *Chloroflexi* has both photosynthetic and nonphotosynthetic members. *Chloroflexus* is the major representative of the photosynthetic **green nonsulfur bacteria**. It is a filamentous, gliding, thermophilic bacterium that often is isolated from neutral to alkaline hot springs where it grows in the form of orange-reddish mats, usually in association with cyanobacteria. Although it resembles the green bacteria in ultrastructure and photosynthetic pigments, its metabolism is more similar to that of the purple nonsulfur bacteria (p. 488). *Chloroflexus* can carry out anoxygenic photosynthesis with organic compounds as carbon sources or grow aerobically as a chemoheterotroph. It doesn't appear closely related to any bacterial group based on 16S rRNA studies and is a deep and ancient branch of the bacterial tree.

Nutritional types (section 5.3)

The nonphotosynthetic, gliding, rod-shaped or filamentous bacterium *Herpetosiphon* also is included in this phylum. *Herpetosiphon* is an aerobic chemoorganotroph with respiratory me-

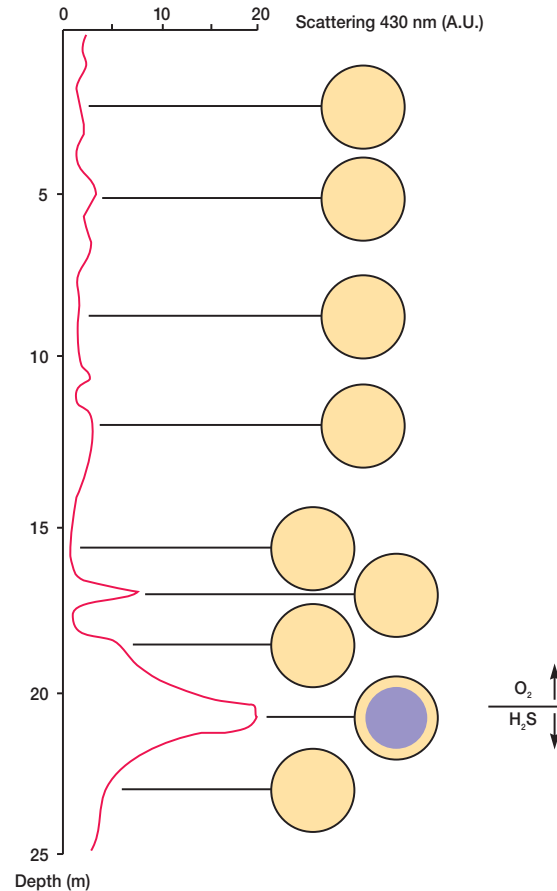


Figure 21.4 The Distribution of Photosynthetic Microorganisms in a Norwegian Fjord. Fifty-milliliter samples taken at various depths have been filtered and the filters mounted on a graph showing the light scattering (an indication of turbidity due to bacterial growth) as a function of depth. The peak at 20.8 m is caused by the purple bacteria *Chromatium* spp. growing at the interface of the aerobic zone and the anaerobic, sulfide-rich layer. The smaller peak at 17.1 m is caused by dinoflagellate growth. A.U. stands for absorbance units.

tabolism and oxygen as the electron acceptor. It can be isolated from freshwater and soil habitats.

Phylum *Chlorobi*

The phylum *Chlorobi* has only one class (*Chlorobia*), order (*Chlorobiales*), and family (*Chlorobiaceae*). The **green sulfur bacteria** are a small group of obligately anaerobic photolithoautotrophs that use hydrogen sulfide, elemental sulfur, and hydrogen as electron sources. The elemental sulfur produced by sulfide oxidation is deposited outside the cell. Their photosynthetic pigments are located in ellipsoidal vesicles called **chlorosomes** or chlorobium vesicles, which are attached to the plasma membrane but are not continuous with it. The chlorosome membrane is not a normal lipid bilayer or unit membrane (see section 3.2). Chloro-

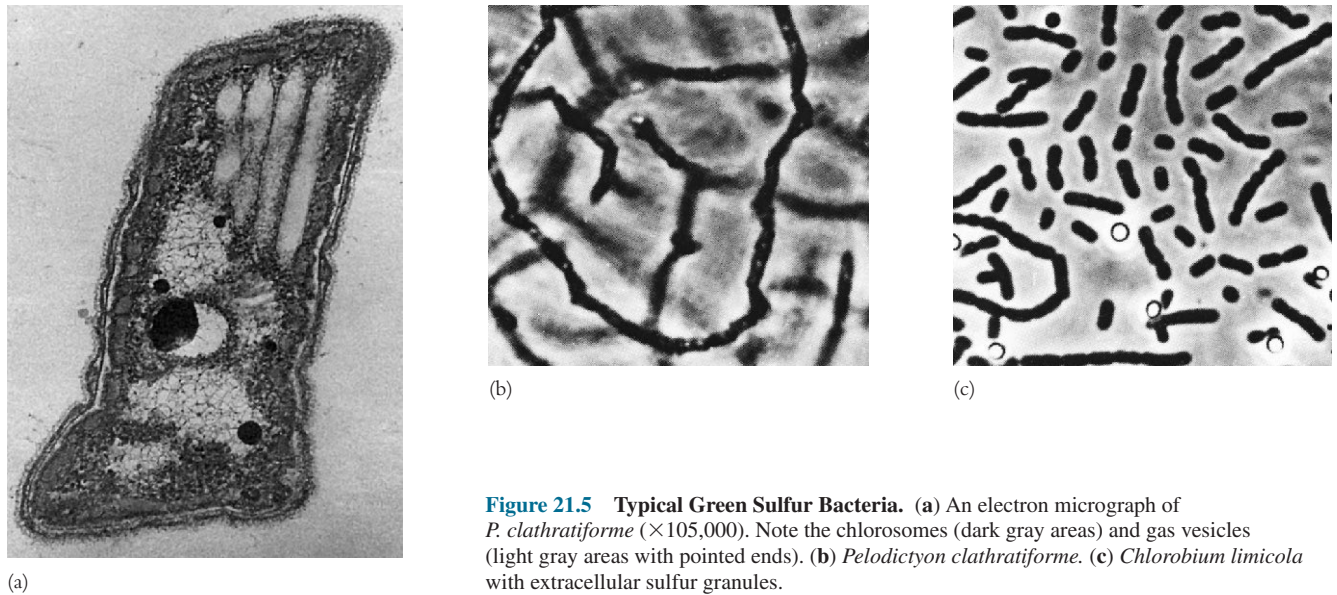


Figure 21.5 Typical Green Sulfur Bacteria. (a) An electron micrograph of *P. clathratiforme* ($\times 105,000$). Note the chlorosomes (dark gray areas) and gas vesicles (light gray areas with pointed ends). (b) *Pelodictyon clathratiforme*. (c) *Chlorobium limicola* with extracellular sulfur granules.

somes contain accessory bacteriochlorophyll pigments, but the reaction center bacteriochlorophyll is located in the plasma membrane and must be able to obtain energy from chlorosome pigments. These bacteria flourish in the anaerobic, sulfide-rich zones of lakes. Although they lack flagella and are nonmotile, some species have gas vesicles (figure 21.5a) to adjust their depth for optimal light and hydrogen sulfide. Those forms without vesicles are found in sulfide-rich muds at the bottom of lakes and ponds.

The green sulfur bacteria are very diverse morphologically. They may be rods, cocci, or vibrios; some grow singly, and others form chains and clusters (figure 21.5b,c). They are either grass-green or chocolate-brown in color. Representative genera are *Chlorobium*, *Prosthecochloris*, and *Pelodictyon*.

1. Give the major distinguishing characteristics of *Aquifex*, *Thermotoga*, and the deinococci. What is thought to contribute to the desiccation and radiation resistance of the deinococci?
2. How do oxygenic and anoxygenic photosynthesis differ from each other and why?
3. In general terms give the major characteristics of the following groups: purple sulfur bacteria, purple nonsulfur bacteria, and green sulfur bacteria. How do purple and green sulfur bacteria differ?
4. Compare the green nonsulfur (*Chloroflexi*) and green sulfur bacteria (*Chlorobi*).
5. What are chlorosomes or chlorobium vesicles?

Phylum Cyanobacteria

The **cyanobacteria** are the largest and most diverse group of photosynthetic bacteria. There is little agreement about the number of cyanobacterial species. Older classifications had as many as 2,000 or



Figure 21.6 Cyanobacterial Thylakoids and Phycobilisomes. *Synechococcus lividus* with an extensive thylakoid system. The phycobilisomes lining these thylakoids are clearly visible as granules at location t ($\times 85,000$).

more species. In one recent system this has been reduced to 62 species and 24 genera. The second edition of *Bergey's Manual of Determinative Bacteriology* describes 56 genera in some detail. The G + C content of the group ranges from 35 to 71%. Although cyanobacteria are true procaryotes, their photosynthetic system closely resembles that of the eucaryotes because they have chlorophyll *a* and photosystem II, and carry out oxygenic photosynthesis. Like the red algae, cyanobacteria use phycobiliproteins as accessory pigments. Photosynthetic pigments and electron transport chain components are located in thylakoid membranes lined with particles called **phycobilisomes** (figure 21.6). These contain phycobilin pigments, particularly

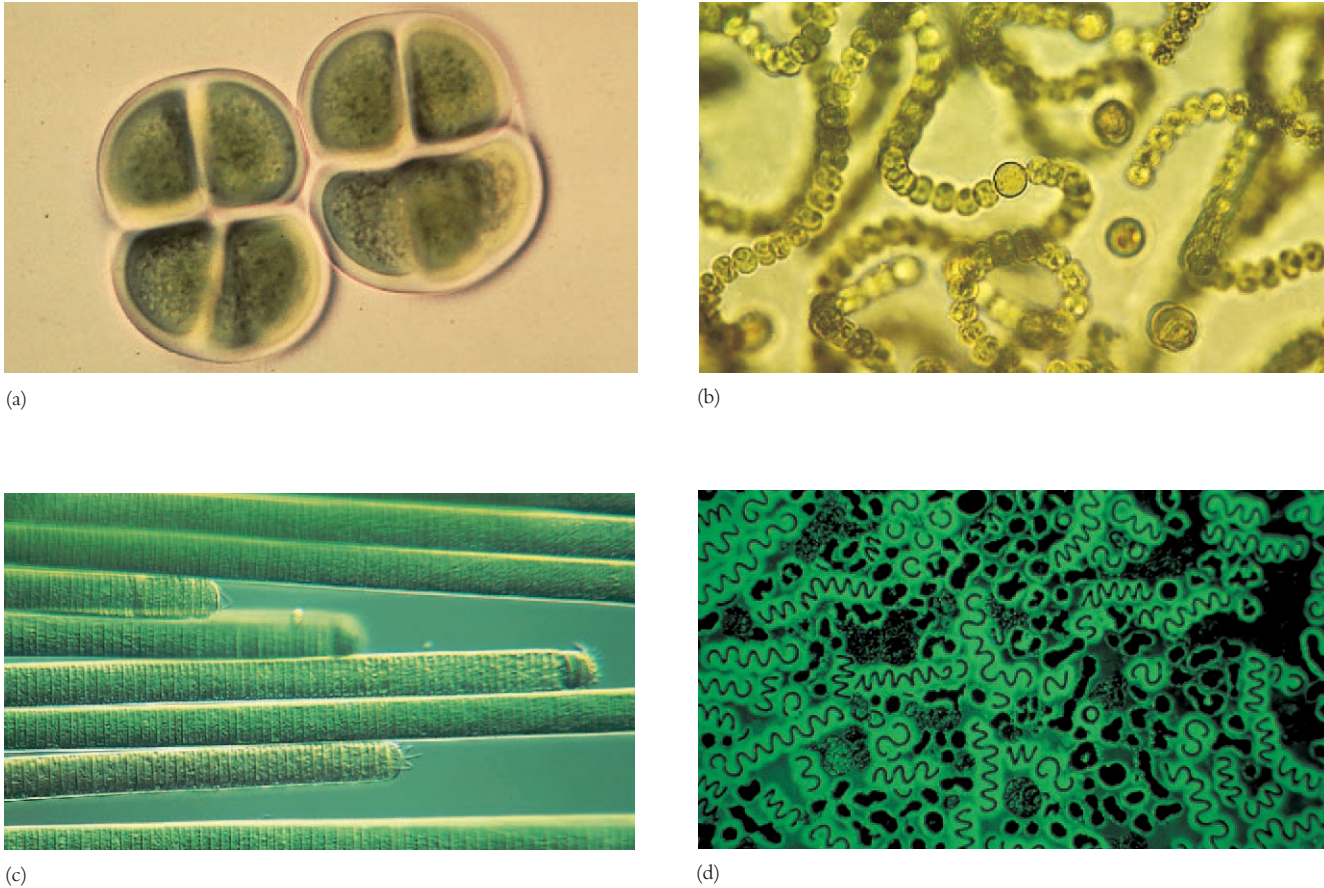


Figure 21.7 Oxygenic Photosynthetic Bacteria. Representative cyanobacteria. (a) *Chroococcus turgidus*, two colonies of four cells each ($\times 600$). (b) *Nostoc* with heterocysts ($\times 550$). (c) *Oscillatoria* trichomes seen with Nomarski interference-contrast optics ($\times 250$). (d) The cyanobacteria *Anabaena spiroides* and *Microcystis aeruginosa*. The spiral *A. spiroides* is covered with a thick gelatinous sheath ($\times 1,000$).

phycocyanin, and transfer energy to photosystem II. Carbon dioxide is assimilated through the Calvin cycle, and the reserve carbohydrate is glycogen. Sometimes they will store extra nitrogen as polymers of arginine or aspartic acid in cyanophycin granules. Since cyanobacteria lack the enzyme α -ketoglutarate dehydrogenase, they do not have a fully functional citric acid cycle. The pentose phosphate pathway plays a central role in their carbohydrate metabolism. Although many cyanobacteria are obligate photolithoautotrophs, some can grow slowly in the dark as chemoheterotrophs by oxidizing glucose and a few other sugars. Under anaerobic conditions *Oscillatoria limnetica* oxidizes hydrogen sulfide instead of water and carries out anoxygenic photosynthesis much like the green photosynthetic bacteria. As these examples illustrate, cyanobacteria are capable of considerable metabolic flexibility.

Cyanobacteria also vary greatly in shape and appearance. They range in diameter from about 1 to 10 μm and may be unicellular, exist as colonies of many shapes, or form filaments called trichomes (**figure 21.7**; see also figures 3.12 and 3.13). A **trichome** is a row of bacterial cells that are in close contact with one another over a large area. In contrast, adjacent cells in a simple chain (such as those commonly found in the genus *Bacillus*) associate by only a small area of contact. Although most appear blue-green because of phycocyanin, a few are red or brown in color because of the red pigment phycoerythrin. Despite this variety, cyanobacteria have typical procaryotic cell structures and a normal gram-negative type cell wall (**figure 21.8**). They often use gas vesicles to move vertically in the water, and many filamentous species have gliding motility

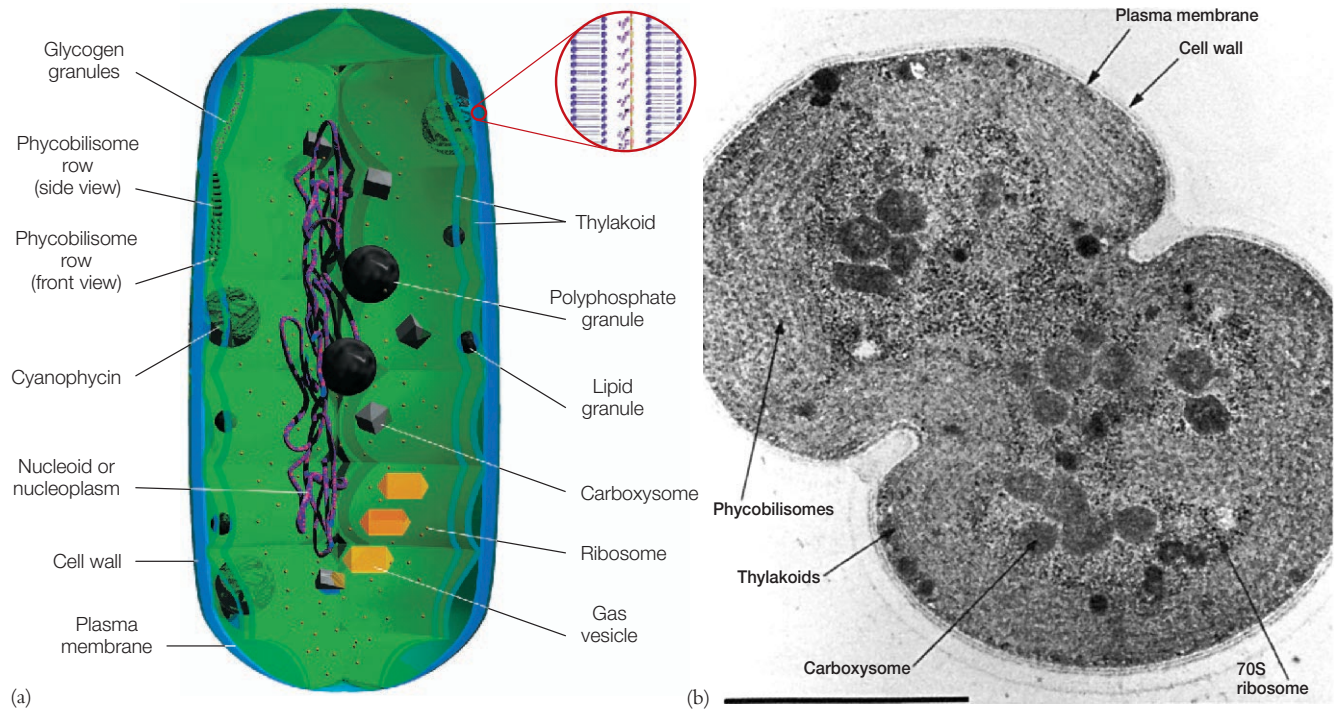


Figure 21.8 Cyanobacterial Cell Structure. (a) Schematic diagram of a vegetative cell. The insert shows an enlarged view of the envelope with its outer membrane and peptidoglycan. (b) Thin section of *Synechocystis* during division. Bar = 1 μm . Many structures are visible. (a) Illustration copyright © Hartwell T. Crim, 1998.

(**Box 21.1**; see also section 3.6). Although cyanobacteria lack flagella, several strains of the marine genus *Synechococcus* are able to move at rates of up to 25 $\mu\text{m}/\text{second}$ by means of an unknown mechanism.

Cyanobacteria show great diversity with respect to reproduction and employ a variety of mechanisms: binary fission, budding, fragmentation, and multiple fission. In the last process a cell enlarges and then divides several times to produce many smaller progeny, which are released upon the rupture of the parental cell. Fragmentation of filamentous cyanobacteria can generate small, motile filaments called **hormogonia**. Some species develop **akinetes**, specialized, dormant, thick-walled resting cells that are resistant to desiccation. Often these germinate to form new filaments.

Many filamentous cyanobacteria fix atmospheric nitrogen by means of special cells called **heterocysts** (figure 21.9). Around 5 to 10% of the cells can develop into heterocysts when cyanobacteria are deprived of both nitrate and ammonia, their preferred nitrogen sources. When transforming themselves into heterocysts, cyanobacterial cells synthesize a very thick new wall, reorganize their photosynthetic membranes, discard their phycobiliproteins and photosystem II, and synthesize the nitrogen-fixing enzyme nitrogenase. Photosystem I is still functional and produces ATP, but no oxygen arises from noncyclic photophosphorylation be-

cause photosystem II is absent. This inability to generate O_2 is critical because the nitrogenase is extremely oxygen sensitive. The heterocyst wall slows or prevents O_2 diffusion into the cell, and any O_2 present is consumed during respiration. The structure and physiology of the heterocyst ensures that it will remain anaerobic; it is dedicated to nitrogen fixation. It obtains nutrients from adjacent vegetative cells and contributes fixed nitrogen in the form of the amino acid glutamine. It should be noted that nitrogen fixation also is carried out by cyanobacteria that lack heterocysts. Some fix nitrogen under dark, anoxic conditions in microbial mats. Planktonic forms such as *Trichodesmium* can fix nitrogen as well. [The biochemistry of nitrogen fixation \(pp. 212–14\)](#)

The classification of cyanobacteria is still in an unsettled state, partly due to the lack of pure cultures. At present all taxonomic schemes must be considered tentative. The second edition of *Bergey's Manual* divides the cyanobacteria into five subsections with 56 genera. **Table 21.3** briefly summarizes the major characteristics of the five subsections. These are distinguished using colony or trichome morphology and reproductive patterns. Some other properties important in cyanobacterial characterization are cell morphology, ultrastructure, genetic characteristics, physiology and biochemistry, and habitat/ecology (preferred habitat and growth habit). The authors consider genera as tentative and often do not provide species names.

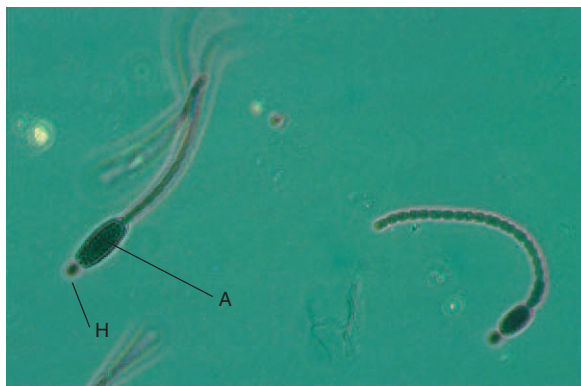
Box 21.1

The Mechanism of Gliding Motility

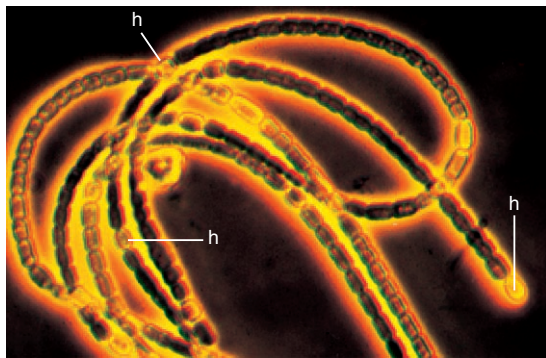
Gliding motility varies greatly in rate (from about 2 μm per minute to over 600 μm per minute) and in the nature of the motion. Bacteria such as *Myxococcus* and *Flexibacter* glide along in a direction parallel to the longitudinal axis of their cells. Others (*Saprospira*) travel with a screwlike motion or even move in a direction perpendicular to the long axis of the cells in their trichome (*Simonsiella*). *Beggiatoa*, cyanobacteria, and some other bacteria rotate around their longitudinal axis while gliding, but this is not always seen. Many will flex or twitch as well as glide. Such diversity in gliding movement may indicate that more than one mechanism for motility exists. This conclusion is supported by the observation that some gliders (e.g., *Cytophaga*, *Flexibacter*, and *Flavobacterium*) move attached latex beads over their surface, whereas others such as *Myxococcus* either do not move beads or move them very slowly. (That is, not all gliding bacteria have rapidly moving cell-surface components.) Although slime is required for gliding, it does not appear to

propel bacteria directly; rather, it probably attaches them to the substratum and lubricates the surface for more efficient movement.

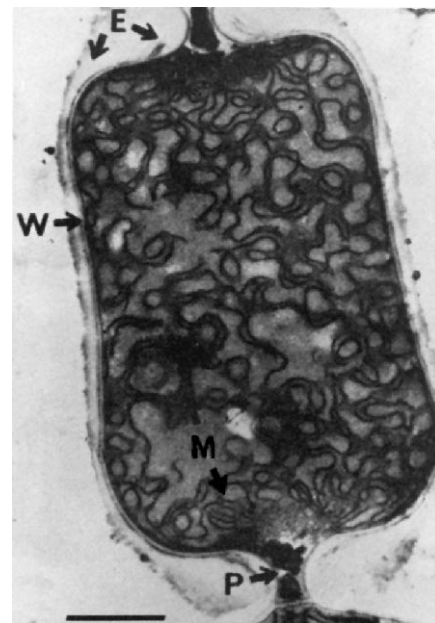
A variety of mechanisms for gliding motility have been proposed. Cytoplasmic fibrils or filaments are associated with the envelope of many gliding bacteria. In *Oscillatoria* they seem to be contractile and may produce waves in the outer membrane, resulting in movement. Ringlike protein complexes or rotary assemblies resembling flagellar basal bodies are present in some envelopes. These assemblies may spin and move the bacterium along. Pili that can extend and retract may be involved in several species. There is some evidence that differences in surface tension can propel *Myxococcus xanthus*. *Myxococcus* may secrete a surfactant at its posterior end (the end opposite the direction of movement) that lowers the surface tension at the rear end of the rod. The cell would be pulled forward by the greater surface tension exerted on its anterior end. *Myxococcus* does use pili when it is gliding as part of a moving group of cells.



(a)



(b)



(c)

Figure 21.9 Examples of Heterocysts and Akinetes. (a) *Cylindrospermum* with terminal heterocysts (H) and subterminal akinetes (A) ($\times 500$). (b) *Anabaena*, with heterocysts. (c) An electron micrograph of an *Anabaena* heterocyst. Bar = 1 μm . Note the cell wall (W), additional outer walls (E), membrane system (M), and a pore channel to the adjacent cell (P).

Table 21.3 Characteristics of the Cyanobacterial Subsections

Subsection	General Shape	Reproduction and Growth	Heterocysts	% G + C	Other Properties	Representative Genera
I	Unicellular rods or cocci; nonfilamentous aggregates	Binary fission, budding	–	31–71	Almost always nonmotile	<i>Chamaesiphon</i> <i>Chroococcus</i> <i>Gloeotheca</i> <i>Gleocapsa</i> <i>Prochloron</i>
II	Unicellular rods or cocci; may be held together in aggregates	Multiple fission to form baeocytes	–	40–46	Only some baeocytes are motile	<i>Pleurocapsa</i> <i>Dermocarpa</i> <i>Chroococcidiopsis</i>
III	Filamentous, unbranched trichome with only vegetative cells	Binary fission in a single plane, fragmentation	–	34–67	Usually motile	<i>Lyngbya</i> <i>Oscillatoria</i> <i>Prochlorothrix</i> <i>Spirulina</i> <i>Pseudanabaena</i>
IV	Filamentous, unbranched trichome may contain specialized cells	Binary fission in a single plane, fragmentation to form hormogonia	+	38–47	Often motile, may produce akinetes	<i>Anabaena</i> <i>Cylindrospermum</i> <i>Aphanizomenon</i> <i>Nostoc</i> <i>Scytonema</i> <i>Calothrix</i>
V	Filamentous trichomes either with branches or composed of more than one row of cells	Binary fission in more than one plane, hormogonia formed	+	42–44	May produce akinetes, greatest morphological complexity and differentiation in cyanobacteria	<i>Fischerella</i> <i>Stigonema</i> <i>Geitleria</i>

Cyanobacterial taxonomy in the second edition of *Bergey's Manual* is very similar to that in the first edition. Probably the major difference is that the order *Prochlorales* is separate from the cyanobacteria in the first edition, while in the second edition the order is dropped and the genera are placed within the phylum *Cyanobacteria*.

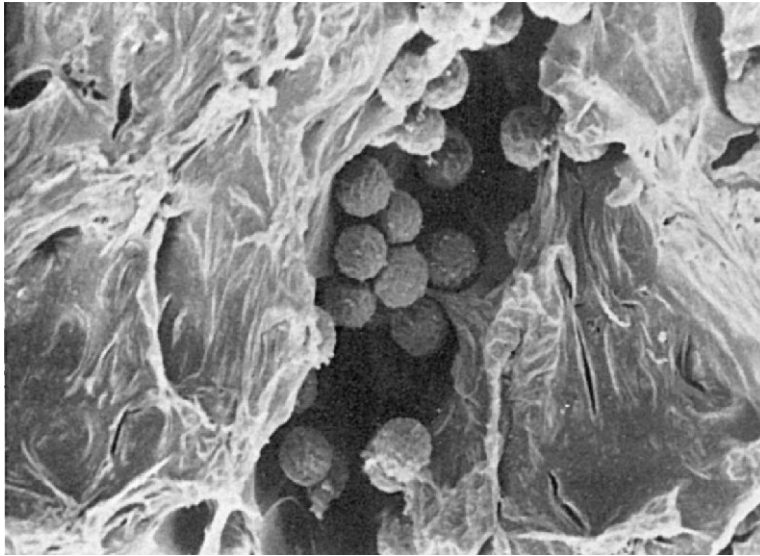
Prochlorophytes are oxygenic phototrophic procaryotes that have both chlorophyll *a* and *b* but lack phycobilins. Thus although they resemble other cyanobacteria with respect to chlorophyll *a*, they differ in also possessing chlorophyll *b*, the only procaryotes to do so. Because prochlorophytes lack phycobilin pigments and phycobilisomes, they are grass-green in color. They resemble chloroplasts in their pigments and thylakoid structure, but their 5S and 16S rRNAs place them within the cyanobacteria. The genera are located in subsections I and III.

Although the prochlorophytes have been classified separately in the past, the second edition of *Bergey's Manual* places them within the cyanobacteria. The three recognized prochlorophyte genera are quite different from one another. *Prochloron* was first discovered as an extracellular symbiont growing either on the surface or within the cloacal cavity of marine colonial ascidian invertebrates (figure 21.10). These bacteria are single-celled, spherical, and from 8 to 30 μm in diameter. Their mol% of G + C is 31 to 41. *Prochlorothrix* is free living, has cylindrical cells

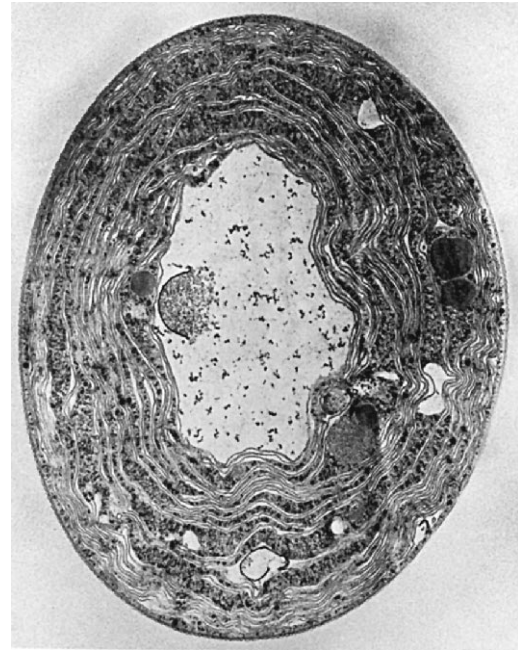
that form filaments, and has been found in Dutch lakes. Its DNA has a higher G + C content (53 mol%). Unlike *Prochloron* it has been cultured in the laboratory.

Prochlorococcus marinus, less than 1 μm in diameter, has recently been discovered flourishing about 100 meters below the ocean surface. It differs from other prochlorophytes in having divinyl chlorophyll *a* and α -carotene instead of chlorophyll *a* and β -carotene. During the summer, it reaches concentrations of 5×10^5 cells per milliliter. It is one of the most numerous of the marine plankton and a significant component of the marine microbial food web.

As mentioned earlier, the five subsections differ in morphology and reproduction (table 21.3). Subsection I contains unicellular rods or cocci that are almost always nonmotile and reproduce by binary fission or budding. Organisms in subsection II are also unicellular, though several individual cells may be held together in an aggregate by an outer wall. Members of this group reproduce by multiple fission to form spherical, very small, reproductive cells, often called **baeocytes**, which escape when the outer wall ruptures. Some baeocytes disperse through gliding motility. The other three subsections contain filamentous cyanobacteria. Usually the trichome is unbranched and often is surrounded by a sheath or slime layer. Cyanobacteria in subsection III form unbranched trichomes composed only of vegetative



(a)



(b)

Figure 21.10 Prochloron. (a) A scanning electron micrograph of *Prochloron* cells on the surface of a *Didemnum candidum* colony. (b) *Prochloron didemni* section; transmission electron micrograph ($\times 23,500$).

cells, whereas the other two subsections produce heterocysts in the absence of an adequate nitrogen source (they also may form akinetes). Heterocystous cyanobacteria are subdivided into those that form linear filaments (subsection IV) and cyanobacteria that divide in a second plane to produce branches or aggregates (subsection V).

Cyanobacteria are very tolerant of environmental extremes and are present in almost all waters and soils. Thermophilic species may grow at temperatures of up to 75°C in neutral to alkaline hot springs. Because these photoautotrophs are so hardy, they are primary colonizers of soils and surfaces that are devoid of plant growth. Some unicellular forms even grow in the fissures of desert rocks. In nutrient-rich warm ponds and lakes, surface cyanobacteria such as *Anacystis* and *Anabaena* can reproduce rapidly to form blooms (figure 21.11). The release of large amounts of organic matter upon the death of the bloom microorganisms stimulates the growth of chemoheterotrophic bacteria that subsequently depletes the available oxygen. This kills fish and other organisms (see section 29.4). Some species can produce toxins that kill livestock and other animals that drink the water. Other cyanobacteria, for example, *Oscillatoria*, are so pollution resistant and characteristic of freshwater with high organic matter content that they are used as water pollution indicators.

Cyanobacteria are particularly successful in establishing symbiotic relationships with other organisms. They are the pho-



Figure 21.11 Bloom of Cyanobacteria and Algae in a Eutrophic Pond.

tosynthetic partner in most lichen associations. Cyanobacteria are symbionts with protozoa and fungi, and nitrogen-fixing species form associations with a variety of plants (liverworts, mosses, gymnosperms, and angiosperms). [Types of symbiotic relationships \(section 28.2\); Lichens \(pp. 598–99\)](#)

1. Summarize the major characteristics of the cyanobacteria that distinguish them from other photosynthetic organisms.
2. Define or describe the following: phycobilisomes, hormogonia, akinetes, heterocysts, and baeocytes.
3. What is a trichome and how does it differ from a simple chain of cells?
4. Briefly discuss the ways in which cyanobacteria reproduce.
5. How are heterocysts modified to carry out nitrogen fixation? When do cyanobacteria develop heterocysts?
6. Give the features of the five major cyanobacterial groups.
7. Compare the prochlorophytes with other cyanobacteria and chloroplasts. Where does one find them?
8. List some important positive and negative impacts cyanobacteria have on humans and the environment.

21.4 Phylum *Planctomycetes*

The phylum *Planctomycetes* contains one class, one order, and four genera. Members of the phylum are spherical or oval, budding bacteria that lack peptidoglycan and have distinctive crateriform structures or pits in their walls. The nuclear body is membrane-bounded in the planctomycetes *Gemmata obscuriglobus* and *Pirullela*, something not seen in other bacteria (figure 21.12). The genus *Planctomyces* attaches to surfaces through a stalk and holdfast; the other genera in the order lack stalks. Most of these eubacteria have life cycles in which sessile cells bud to produce motile swarmer cells. The swarmer cells are flagellated and swim for a while before settling down to attach and begin reproduction.

21.5 Phylum *Chlamydiae*

The first edition of *Bergey's Manual* places the chlamydiae with the rickettsias because both gram-negative groups are obligately intracellular parasites: they grow and reproduce only within host cells. Based on 16S rRNA data, the second edition moves the chlamydiae into the phylum *Chlamydiae* and the rickettsias to the α -proteobacteria. Although both chlamydiae and rickettsias are obligately intracellular parasites and not much larger than the poxviruses (see p. 376), they differ from viruses in having both DNA and RNA, a plasma membrane, functioning ribosomes, metabolic pathways, reproduction by binary fission, and other distinctive features.

The phylum *Chlamydiae* has one class, one order, four families, and only five genera. The genus *Chlamydia* is by far the most important and best studied; it will be the focus of our attention. **Chlamydiae** are nonmotile, coccoid, gram-negative bacteria, ranging in size from 0.2 to 1.5 μm . They can reproduce only within cytoplasmic vesicles of host cells by a unique developmental cycle involving the formation of elementary bodies and reticulate bodies. Although their envelope resembles that of other gram-negative bacteria, the wall differs in lacking muramic acid

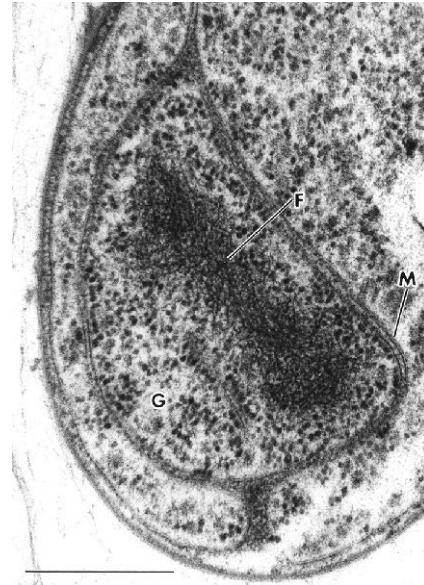


Figure 21.12 *Gemmata obscuriglobus* Nuclear Body. An electron micrograph showing the membrane-bounded nuclear region (Bar = 0.5 μm). Note the central fibrillar area (F), the outer granular area (G), and the outer double-membrane structure (M).

and a peptidoglycan layer. The elementary bodies achieve osmotic stability by cross-linking their outer membrane proteins, and possibly periplasmic proteins, with disulfide bonds. Chlamydiae are extremely limited metabolically and are obligately intracellular parasites of mammals and birds. (However, chlamydia-like bacteria have recently been isolated from spiders, clams, and freshwater invertebrates.) The size of their genome is 4 to 6×10^8 daltons, one of the smallest of all procaryotes, and the G + C content is 41 to 44%.

Chlamydial reproduction begins with the attachment of an **elementary body (EB)** to the cell surface (figure 21.13). Elementary bodies are 0.2 to 0.4 μm in diameter, contain electron-dense nuclear material and a rigid cell wall, and are infectious (figure 21.14). The host cell phagocytoses the EB, which then prevents the fusion of lysosomes with the phagosome and begins to reorganize itself to form a **reticulate body (RB)** or **initial body**. The RB is specialized for reproduction rather than infection. Reticulate bodies are 0.6 to 1.5 μm in diameter and have less dense nuclear material and more ribosomes than EBs; their walls are also more flexible. About 8 to 10 hours after infection, the reticulate body begins to divide and RB reproduction continues until the host cell dies. A chlamydia-filled vacuole or inclusion can become large enough to be seen in a light microscope and even fill the host cytoplasm. After 20 to 25 hours, RBs begin changing back into infectious EBs and continue this process until the host cell lyses and releases the chlamydiae 48 to 72 hours after infection.

Chlamydial metabolism is very different from that of other gram-negative bacteria. It has been thought that chlamydiae cannot catabolize carbohydrates or other substances and synthesize ATP. *Chlamydia psittaci*, one of the best-studied species, lacks both flavo-protein and cytochrome electron transport chain carriers, but does

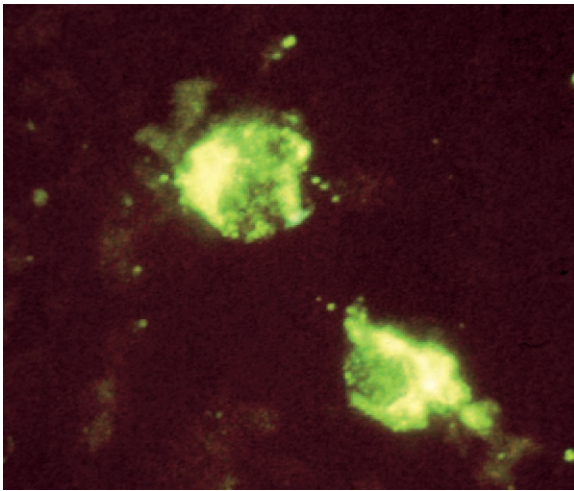
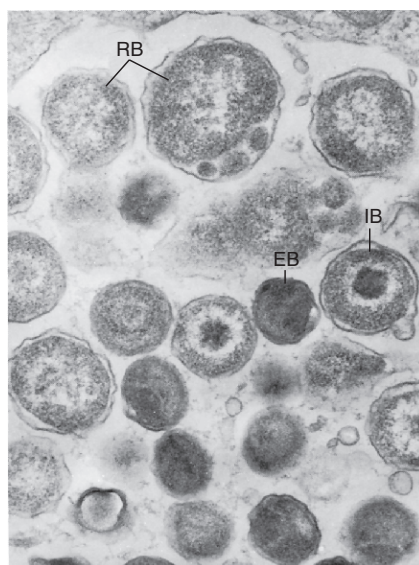


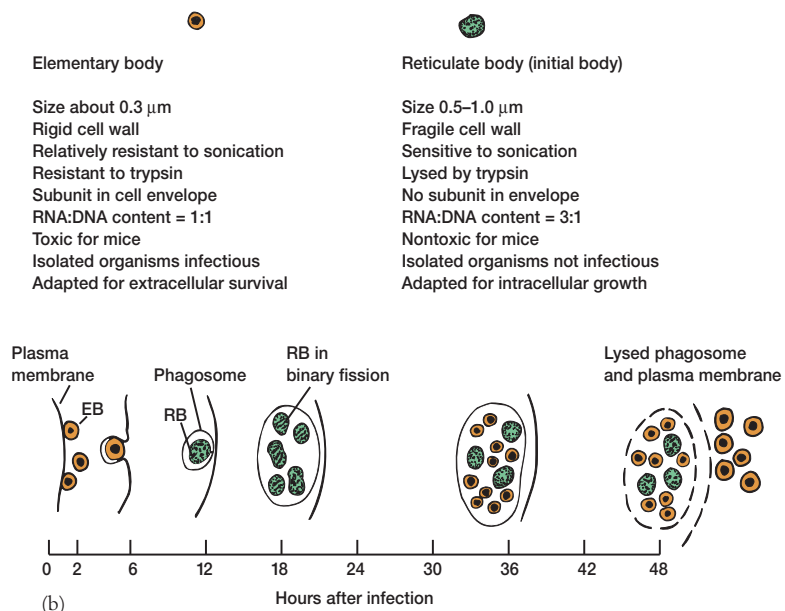
Figure 21.13 The Chlamydial Elementary Body. The infected cells glow a bright green because of the chlamydiae that are stained with fluorescently labeled monoclonal antibodies. The small yellow-green dots are the chlamydiae.

have a membrane translocase that acquires host ATP in exchange for ADP. Thus chlamydiae seem to be energy parasites that are completely dependent on their hosts for ATP. However, this might not be the complete story. The complete sequence of the *C. trachomatis* genome (see p. 351) indicates that the bacterium may be able to synthesize at least some ATP. Although there are two genes for ATP/ADP translocases, there also are genes for substrate-level phosphorylation, electron transport, and oxidative phosphorylation. When supplied with precursors from the host, RBs can synthesize DNA, RNA, glycogen, lipids, and proteins. Presumably the RBs have porins and active membrane transport proteins, but little is known about these. They also can synthesize at least some amino acids and coenzymes. The EBs have minimal metabolic activity and cannot take in ATP or synthesize proteins. They seem to be dormant forms concerned exclusively with transmission and infection.

Three chlamydial species are important pathogens of humans and other warm-blooded animals. *C. trachomatis* infects humans and mice. In humans it causes trachoma, nongonococcal urethritis, and other diseases (see section 39.3). *C. psittaci* causes psittacosis in humans. However, unlike *C. trachomatis*, it also infects many other animals (e.g., parrots, turkeys, sheep, cattle, and cats) and invades the intestinal, respiratory, and genital tracts; the placenta and fetus; the eye; and the synovial fluid of joints. *Chlamydia pneumoniae* is a common cause of human pneumonia. There is now indirect evidence that infections by *C. pneumoniae* may be associated with the development of atherosclerosis and that chlamydial infections may cause severe heart inflammation and damage. Recently a fourth species, *C. pecorum*, has been recognized.



(a)

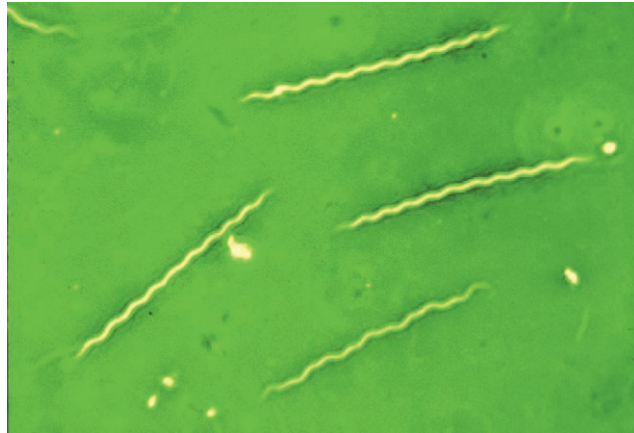


(b)

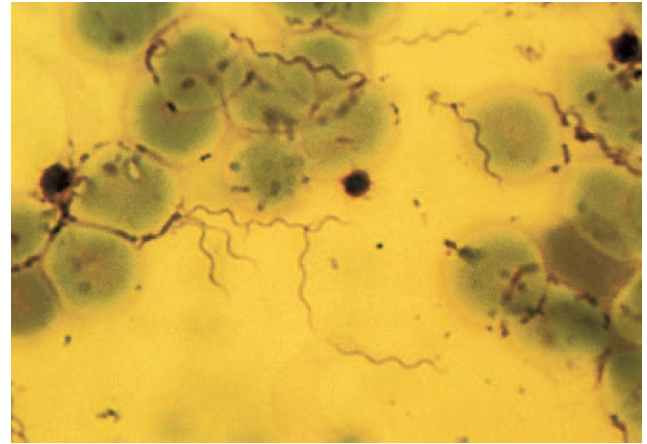
Figure 21.14 The Chlamydial Life Cycle. (a) An electron micrograph of a microcolony of *Chlamydia trachomatis* in the cytoplasm of a host cell ($\times 160,000$). Three developmental stages are visible: the elementary body, EB; reticulate body, RB; and intermediate body, IB, a chlamydial cell intermediate in morphology between the first two forms. (b) A schematic representation of the infectious cycle of chlamydiae.



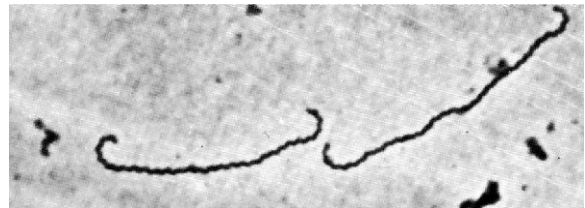
(a)



(b)



(c)



(d)

Figure 21.15 The Spirochetes. Representative examples.

(a) *Cristispira* sp. from the crystalline style of a clam; phase contrast ($\times 2,200$). (b) *Treponema pallidum* ($\times 1,000$). (c) *Borrelia duttonii* from human blood ($\times 500$). (d) *Leptospira interrogans* ($\times 2,200$).

21.6 Phylum *Spirochaetes*

The phylum *Spirochaetes* [Greek *spira*, a coil, and *chaete*, hair] contains gram-negative, chemoheterotrophic bacteria distinguished by their structure and mechanism of motility. They are slender, long bacteria (0.1 to $3.0\ \mu\text{m}$ by 5 to $250\ \mu\text{m}$) with a flexible, helical shape (figure 21.15). Many species are so slim that they are only clearly visible in a light microscope by means of phase-contrast or dark-field optics (see section 2.2). Spirochetes differ greatly from other bacteria with respect to motility and can move through very viscous solutions though they lack external rotating flagella. When in contact with a solid surface, they exhibit creeping or crawling movements. Their unique pattern of motility is due to an unusual morphological structure called the axial filament.

The distinctive features of spirochete morphology are evident in electron micrographs (figure 21.16). The central protoplasmic cylinder contains cytoplasm and the nucleoid, and is bounded by a plasma membrane and gram-negative type cell wall. It corresponds to the body of other gram-negative bacteria. Two to more than a hundred procaryotic flagella, called **axial fibrils**, **periplasmic flagella** or endoflagella, extend from both ends of the cylinder and often overlap one another in the center third of the cell (figure 21.16c,d; see also figure 2.28b). The whole

complex of periplasmic flagella, the **axial filament**, lies inside a flexible outer sheath or outer membrane. The outer sheath contains lipid, protein, and carbohydrate and varies in structure between different genera. Its precise function is unknown, but the sheath is definitely important because spirochetes will die if it is damaged or removed. The outer sheath of *Treponema pallidum* has few proteins exposed on its surface. This allows the syphilis spirochete to avoid attack by host antibodies.

Although the way in which periplasmic flagella propel the cell has not been established, they are responsible for motility because mutants with straight rather than curved flagella are nonmotile. Presumably the periplasmic flagella rotate like the external flagella of other bacteria. This could cause the corkscrew-shaped outer sheath to rotate and move the cell through the surrounding liquid (figure 21.17). Flagellar rotation could also flex or bend the cell and account for the crawling movement seen on solid surfaces.

Spirochetes can be anaerobic, facultatively anaerobic, or aerobic. Carbohydrates, amino acids, long-chain fatty acids, and long-chain fatty alcohols may serve as carbon and energy sources.

The group is exceptionally diverse ecologically and grows in habitats ranging from mud to the human mouth. Members of the genus *Spirochaeta* are free-living and often grow in anaerobic and sulfide-rich freshwater and marine environments. Some species

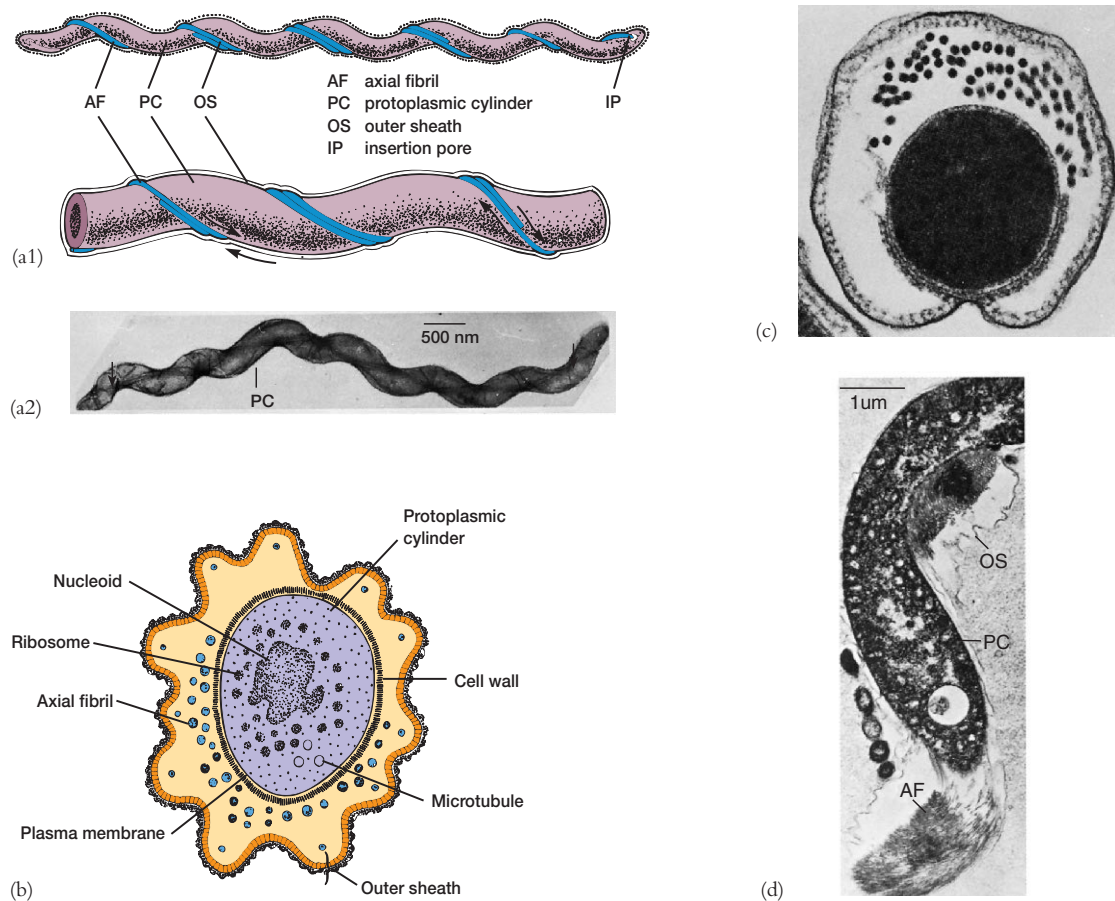


Figure 21.16 Spirochete Morphology. (a1) A surface view of spirochete structure as interpreted from electron micrographs. (a2) A longitudinal view of *T. zuelzeri* with axial fibrils extending most of the cell length. (b) A cross section of a typical spirochete showing morphological details. (c) Electron micrograph of a cross section of *Clevelandina* from the termite *Reticulitermes flavipes* showing the outer sheath, protoplasmic cylinder, and axial fibrils ($\times 70,000$). (d) Longitudinal section of *Cristispira* showing the outer sheath (OS), the protoplasmic cylinder (PC), and the axial fibrils (AF).

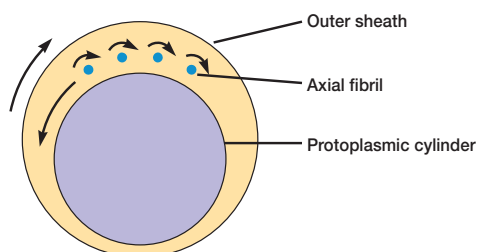
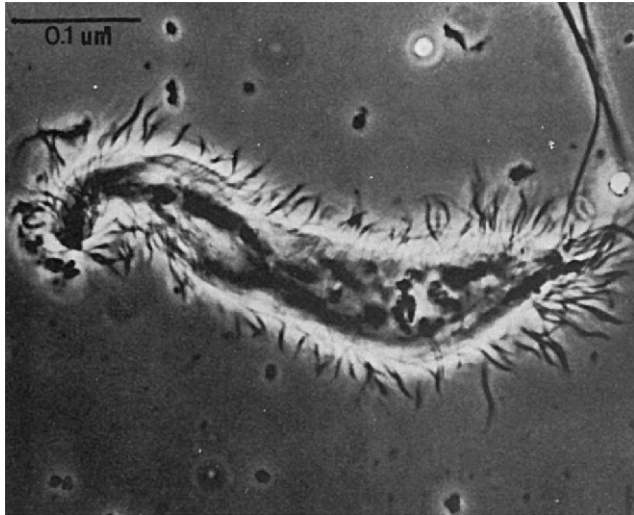


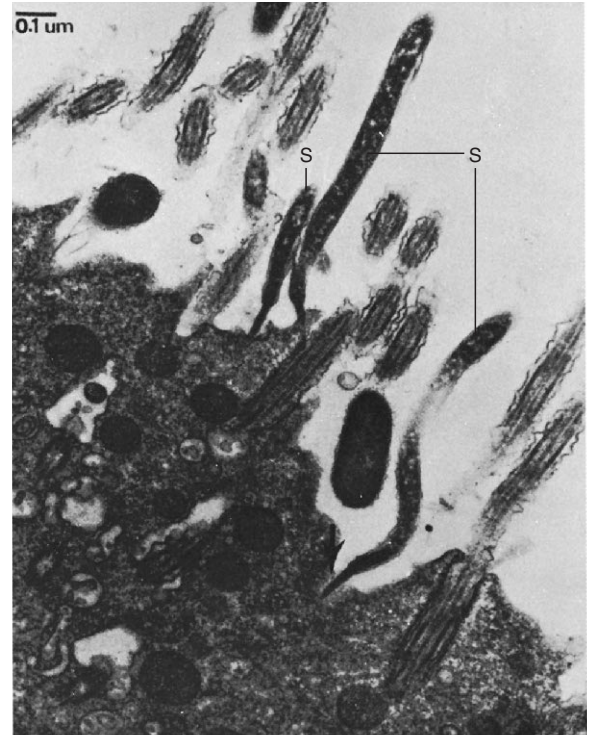
Figure 21.17 Spirochete Motility. A hypothetical mechanism for spirochete motility. See text for details.

of the genus *Leptospira* grow in aerobic water and moist soil. In contrast, many spirochetes form symbiotic associations with other organisms and are found in a variety of locations: the hindguts of termites and wood-eating roaches, the digestive tracts of mollusks (*Cristispira*) and mammals, and the oral cavities of animals (*Treponema denticola*, *T. oralis*). Spirochetes coat the surfaces of many protozoa from termite and wood-eating roach hindguts (**figure 21.18**). For example, the flagellate *Myxotricha paradoxa* is covered with slender spirochetes (0.15 by 10 μm in length) that are firmly attached and help move the protozoan. Some members of the genera *Treponema*, *Borrelia*, and *Leptospira* are important pathogens; for example, *Treponema pallidum* (see **figure 2.8a**) causes syphilis (see **section 39.3**), and *Borrelia burgdorferi* (see **figure 39.8**) is responsible for Lyme disease. The study of *Treponema* and its role in syphilis has been hindered by our inability to culture the spirochete outside its human host. The *T. pallidum* genome has now been sequenced (see



(a)

Figure 21.18 Spirochete-Protozoan Associations. The surface spirochetes serve as organs of motility for protozoa (see text). (a) The spirochete-*Personympha* association with the spirochetes projecting from the surface. (b) Electron micrograph of small spirochetes (S) attached to the membrane of the flagellate protozoan *Barbulanympha*.



(b)

pp. 351–52); the results show that this spirochete is metabolically crippled and quite dependent on its host. The *Borrelia burgdorferi* genome also has been sequenced. It contains a linear chromosome of 910,725 base pairs and at least 17 linear and circular plasmids, which constitute another 533,000 base pairs. The plasmids have some genes that are normally found on chromosomes, and plasmid proteins seem to be involved in bacterial virulence.

The second edition of *Bergey's Manual* divides the phylum *Spirochaetes* into one class, one order (*Spirochaetales*), and three families (*Spirochaetaceae*, *Serpulinaceae*, and *Leptospiraceae*). At present, there are 13 genera in the phylum. **Table 21.4** summarizes some of the more distinctive properties of selected genera.

1. Describe the *Planctomycetes* and their more distinctive properties.
2. Compare chlamydiae with viruses.
3. Give the major characteristics of the phylum *Chlamydiae*. What are elementary and reticulate bodies? Briefly describe the steps in the chlamydial life cycle.
4. How does chlamydial metabolism differ from that of other bacteria?
5. List two or three chlamydial diseases of humans.
6. Give the most important characteristics of the spirochetes.
7. Define the following terms: protoplasmic cylinder, axial fibrils or periplasmic flagella, axial filament, outer sheath or outer membrane. Draw and label a diagram of spirochete morphology, locating these structures.
8. How might spirochetes use their axial filament to move?

21.7 Phylum *Bacteroidetes*

The phylum *Bacteroidetes* is a new addition in the second edition of *Bergey's Manual*. It is a very diverse phylum and seems most closely related to the phylum *Chlorobi*. The phylum has three classes (*Bacteroides*, *Flavobacteria*, and *Sphingobacteria*), 12 families, and 50 genera.

The class *Bacteroides* contains anaerobic, gram-negative, nonsporing, motile or nonmotile rods of various shapes. These bacteria are chemoheterotrophic and usually produce a mixture of organic acids as fermentation end products, but they do not reduce sulfate or other sulfur compounds. The genera are identified according to properties such as general shape, motility and flagellation pattern, and fermentation end products. These bacteria grow in habitats such as the oral cavity and intestinal tract of humans and other animals and the rumen (see section 28.2) of ruminants.

Although the difficulty of culturing these anaerobes has hindered the determination of their significance, they are clearly widespread and important. Often they benefit their host. *Bacteroides ruminicola* is a major component of the rumen flora; it ferments starch, pectin, and other carbohydrates. About 30% of the bacteria isolated from human feces are members of the genus *Bacteroides*, and these organisms may provide extra nutrition by degrading cellulose, pectins, and other complex carbohydrates. The family also is involved in human disease. Members of the genus *Bacteroides* are associated with diseases of major organ systems, ranging from the central nervous

Table 21.4 Characteristics of Spirochete Genera

Genus	Dimensions (μm) and Morphology	G + C Content (mol %)	Oxygen Requirement	Carbon + Energy Source	Habitats
<i>Spirochaeta</i>	0.2–0.75 × 5–250; 2–40 periplasmic flagella (almost always 2)	51–65	Facultatively anaerobic or anaerobic	Carbohydrates	Aquatic and free-living
<i>Cristispira</i>	0.5–3.0 × 30–180; ≥ 100 periplasmic flagella	N.A. ^a	Facultatively anaerobic?	N.A. ^a	Mollusk digestive tract
<i>Treponema</i>	0.1–0.4 × 5–20; 2–16 periplasmic flagella	25–53	Anaerobic or microaerophilic	Carbohydrates or amino acids	Mouth, intestinal tract, and genital areas of animals; some are pathogenic (syphilis, yaws)
<i>Borrelia</i>	0.2–0.5 × 3–20; 14–60 periplasmic flagella	27–32	Anaerobic or microaerophilic	Carbohydrates	Mammals and arthropods; pathogens (relapsing fever, Lyme disease)
<i>Leptospira</i>	0.1 × 6–24; 2 periplasmic flagella	35–49 (one strain is 53)	Aerobic	Fatty acids and alcohols	Free-living or pathogens of mammals, usually located in the kidney (leptospirosis)
<i>Leptonema</i>	0.1 × 6–20; 2 periplasmic flagella	51–53	Aerobic	Fatty acids	Mammals
<i>Brachyspira</i>	0.2 × 1.7–6.0; 8 periplasmic flagella	N.A. ^a	Anaerobic	N.A. ^a	Mammalian intestinal tract
<i>Serpulina</i>	0.3–0.4 × 7–9; 16–18 periplasmic flagella	25–26	Anaerobic	Carbohydrates and amino acids	Mammalian intestinal tract

^aN.A., information not available

system to the skeletal system. *B. fragilis* is a particularly common anaerobic pathogen found in abdominal, pelvic, pulmonary, and blood infections.

Another important group in the *Bacteroidetes* is the class *Sphingobacteria*. Besides the similarity in their 16S rRNA sequences, sphingobacteria often have sphingolipids in their cell walls. Some genera in this class are *Sphingobacterium*, *Saprospira*, *Flexibacter*, *Cytophaga*, *Sporocytophaga*, and *Crenothrix*.

The genera *Cytophaga*, *Sporocytophaga*, and *Flexibacter* differ from each other in morphology, life cycle, and physiology. Bacteria of the genus *Cytophaga* are slender rods, often with pointed ends (figure 21.19a). They differ from the gliding, fruiting myxobacteria in lacking fruiting bodies and having a low G + C ratio. *Sporocytophaga* is similar to *Cytophaga* but forms spherical resting cells called microcysts (figure 21.19b,c). *Flexibacter*, produces long, flexible threadlike cells when young (figure 21.19d) and is unable to use complex polysaccharides. Often colonies of these bacteria are colored yellow to orange because of carotenoid or flexirubin pigments. Some of the flexirubins are chlorinated, which is unusual for biological molecules.

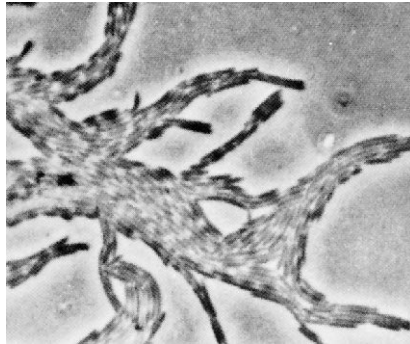
Members of the genera *Cytophaga* and *Sporocytophaga* are aerobes that actively degrade complex polysaccharides. Soil cytophagas digest cellulose; both soil and marine forms attack chitin, pectin, and keratin. Some marine species even degrade agar. Cytophagas play a major role in the mineralization of organic matter and can cause great damage to exposed fish-

ing gear and wooden structures. They also are a major component of the bacterial population in sewage treatment plants and presumably contribute significantly to this waste treatment process.

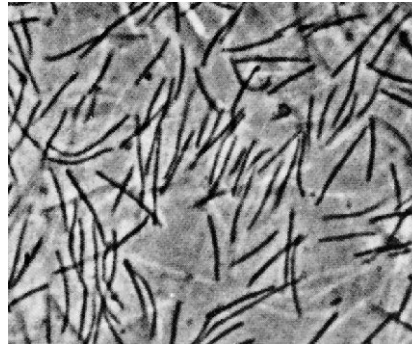
Although most cytophagas are free-living, some can be isolated from vertebrate hosts and are pathogenic. *Cytophaga columnaris* and others cause diseases such as columnaris disease, cold water disease, and fin rot in freshwater and marine fish.

The gliding motility so characteristic of these organisms is quite different from flagellar motility (see section 3.6). **Gliding motility** is present in a wide diversity of taxa: fruiting and nonfruiting aerobic chemoheterotrophs, cyanobacteria, green nonsulfur bacteria, and at least two gram-positive genera (*Helicobacterium* and *Desulfonema*). Gliding bacteria lack flagella and are stationary while suspended in liquid medium. When in contact with a surface, they glide along, leaving a slime trail; the gliding mechanism is unknown (Box 21.1). Movement can be very rapid; some cytophagas travel 150 μm in a minute, whereas filamentous gliding bacteria may reach speeds of more than 600 μm/minute. Young organisms are the most motile, and motility often is lost with age. Low nutrient levels usually stimulate gliding.

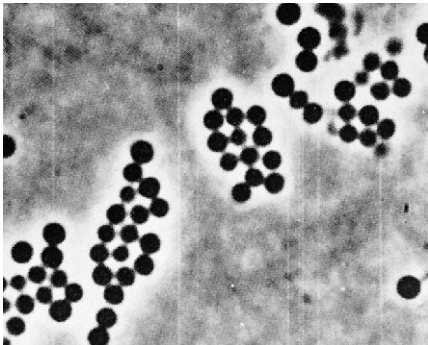
Gliding motility gives a bacterium many advantages. Many aerobic chemoheterotrophic gliding bacteria actively digest insoluble macromolecular substrates such as cellulose and chitin, and gliding motility is ideal for searching these out. Because many of the digestive enzymes are cell bound, the bacteria must



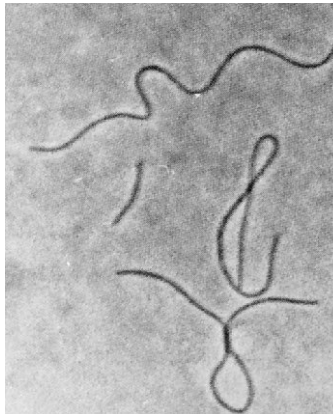
(a)



(b)



(c)



(d)

Figure 21.19 Nonphotosynthetic, Nonfruiting, Gliding Bacteria. Representative members of the order *Cytophagales*. (a) *Cytophaga* sp. ($\times 1,150$). (b) *Sporocytophaga myxococcoides*, vegetative cells on agar ($\times 1,170$). (c) *Sporocytophaga myxococcoides*, mature microcysts ($\times 1,750$). (d) Long thread cells of *Flexibacter elegans* ($\times 1,100$).

be in contact with insoluble nutrient sources; gliding motility makes this possible. Gliding movement is well adapted to drier habitats and to movement within solid masses such as soil, sediments, and rotting wood that are permeated by small channels. Finally, gliding bacteria, like flagellated bacteria, can position themselves at optimal conditions of light intensity, oxygen, hydrogen sulfide, temperature, and other factors that influence growth.

1. Give the major properties of the class *Bacteroides*.
2. How do these bacteria benefit and harm their hosts?
3. Give three advantages of gliding motility.
4. Briefly describe the following genera: *Cytophaga*, *Sporocytophaga*, and *Flexibacter*.
5. Why are the cytophagases ecologically important?

Summary

1. *Aquifex* and *Thermotoga* are hyperthermophilic gram-negative rods that represent the two deepest or oldest phylogenetic branches of the bacteria.
2. Members of the order *Deinococcales* are aerobic, gram-positive cocci and rods that are distinctive in their unusually great resistance to desiccation and radiation.
3. Cyanobacteria carry out oxygenic photosynthesis, and purple and green bacteria use anoxygenic photosynthesis.

4. The four most important groups of purple and green photosynthetic bacteria are the purple sulfur bacteria, the purple nonsulfur bacteria, the green sulfur bacteria, and the green nonsulfur bacteria (table 21.1).
5. The bacteriochlorophyll pigments of purple and green bacteria enable them to live in deeper, anaerobic zones of aquatic habitats.
6. Green nonsulfur bacteria such as *Chloroflexus* are placed in the phylum *Chloroflexi*. *Chloroflexus* is a filamentous, gliding

thermophilic bacterium that is metabolically similar to the purple nonsulfur bacteria.

7. The phylum *Chlorobi* contains the green sulfur bacteria—obligately anaerobic photolithoautotrophs that use hydrogen sulfide, elemental sulfur, and hydrogen as electron sources.
8. Cyanobacteria carry out oxygenic photosynthesis by means of a photosynthetic apparatus similar to that of the eucaryotes. Like the red algae, they have phycobilisomes.

9. Cyanobacteria reproduce by binary fission, budding, multiple fission, and fragmentation to form hormogonia. Some will produce a dormant akinete.
10. Nitrogen-fixing cyanobacteria usually form heterocysts, specialized cells in which nitrogen fixation occurs.
11. The second edition of *Bergey's Manual* divides the cyanobacteria into five subsections and includes the prochlorophytes in the phylum *Cyanobacteria* (table 21.3).
12. The phyla *Planctomycetes* and *Chlamydiae* lack peptidoglycan in their walls.
13. Chlamydiae are nonmotile, coccoid, gram-negative bacteria that reproduce within the cytoplasmic vacuoles of host cells by a life cycle involving elementary bodies (EBs) and reticulate bodies (RBs) (figure 21.14). They are energy parasites.
14. The spirochetes are slender, long, helical, gram-negative bacteria that are motile because of the axial filament underlying an outer sheath or outer membrane (figure 21.16).
15. Members of the class *Bacteroides* are obligately anaerobic, chemoheterotrophic, nonsporing, motile or nonmotile rods of various shapes. Some are important rumen and intestinal symbionts, others can cause disease.
16. Gliding motility is present in a diversity of bacteria, including the sphingobacteria.
17. Cytophagases degrade proteins and complex polysaccharides and are active in the mineralization of organic matter.

Key Terms

akinetes 473

anoxygenic photosynthesis 468

axial fibrils 479

axial filament 479

baecocytes 475

chlamydiae 477

chlorosomes 470

cyanobacteria 471

elementary body (EB) 477

gliding motility 482

green nonsulfur bacteria 470

green sulfur bacteria 470

heterocysts 473

hormogonia 473

initial body 477

oxygenic photosynthesis 468

periplasmic flagella 479

phycobilisomes 471

reticulate body (RB) 477

trichome 472

Questions for Thought and Review

1. Contrast the first and second editions of *Bergey's Manual* with respect to the treatment of the chlamydiae and cytophagases. What does this say about the change in approach between editions?
2. Suppose that you were a microbiologist working on the Mars probe project. How would you determine whether microorganisms are living on the planet, and why?
3. How do cyanobacteria differ from the purple and green photosynthetic bacteria?
4. Relate the physiology of each major group of photosynthetic bacteria to its preferred habitat.
5. Why is nitrogen fixation an oxygen-sensitive process? How are cyanobacteria able to fix nitrogen when they also carry out oxygenic photosynthesis?
6. How might one try to culture a newly discovered rickettsia or chlamydia outside its host?

Critical Thinking Questions

1. Choose a bacterium discussed in this chapter. Write a short 1 to 2 page paper on where and how you would go about isolating the organism. What niche would you sample? What media and selective growth conditions would you use? Be sure to cite references.
2. Many types of movement are employed by bacteria in these phyla. Review them and propose mechanisms by which energy (ATP or proton gradients) might drive the locomotion.

Additional Reading

General

- Balows, A.; Trüper, H. G.; Dworkin, M.; Harder, W.; and Schleifer, K.-H. 1992. *The prokaryotes*, 2d ed. New York: Springer-Verlag.
- Garrity, G. M., editor-in-chief. 2001. *Bergey's manual of systematic bacteriology*, 2d ed., vol. 1, D. R. Boone and R. W. Castenholz, editors. New York: Springer-Verlag.
- Holt, J. G., editor-in-chief. 1984. *Bergey's manual of systematic bacteriology*, vol. 1, N. R. Krieg, editor. Baltimore, Md.: Williams & Wilkins.
- Holt, J. G., editor-in-chief. 1989. *Bergey's Manual of Systematic Bacteriology*, vol. 3, J. T. Staley, M. P. Bryant, and N. Pfennig, editors. Baltimore, Md.: Williams & Wilkins.
- Mayer, F. 1986. *Cytology and morphogenesis of bacteria*. Berlin: Gebrüder Borntraeger.

Nelson, K. E., et al. 1999. Evidence for lateral gene transfer between Archaea and Bacteria from genome sequence of *Thermotoga maritima*. *Nature* 399:323–29.

21.2 Deinococcus-Thermus

- Battista, J. R. 1997. Against all odds: The survival strategies of *Deinococcus radiodurans*. *Annu. Rev. Microbiol.* 51:203–24.
- Battista, J. R.; Earl, A. M.; and Park, M.-J. 1999. Why is *Deinococcus radiodurans* so resistant to ionizing radiation? *Trends Microbiol.* 7(9):362–65.
- Mattimore, V., and Battista, J. R. 1996. Radioreistance of *Deinococcus radiodurans*: Functions necessary to survive ionizing radiation are also necessary to survive prolonged desiccation. *J. Bacteriol.* 178:633–37.
- Rainey, F. A.; Nobre, M. F.; Schumann, R.; Stackebrandt, E.; and Da Costa, M. S. 1997. Phylogenetic diversity of the deinococci as determined by 16S ribosomal DNA sequence comparison. *Int. J. Syst. Bacteriol.* 47(2):510–14.

21.3 Photosynthetic Bacteria

- Adams, D. G. 1992. Multicellularity in cyanobacteria. In *Prokaryotic structure and function*, S. Mohan, C. Dow, and J. A. Coles, editors, 341–84. New York: Cambridge University Press.
- Armstrong, G. A. 1994. Eubacteria show their true colors: Genetics of carotenoid pigment biosynthesis from microbes to plants. *J. Bacteriol.* 176(16):4795–802.
- Bullerjahn, G. S., and Post, A. F. 1993. The prochlorophytes: Are they more than just chlorophyll a/b-containing cyanobacteria? *Crit. Rev. Microbiol.* 19(1):43–59.
- Carmichael, W. W. 1994. The toxins of cyanobacteria. *Sci. Am.* 270(1):78–86.
- Garcia-Pichel, F. 2000. Cyanobacteria. In *Encyclopedia of microbiology*, 2d ed., vol. 1, J. Lederberg, editor-in-chief, 907–29. San Diego: Academic Press.
- Glazer, A. N. 1983. Comparative biochemistry of photosynthetic light-harvesting systems. *Annu. Rev. Biochem.* 52:125–57.

21.1 Aquificae and Thermotogae

- Deckert, G., et al. 1998. The complete genome of the hyperthermophilic bacterium *Aquifex aeolicus*. *Nature* 392:353–58.

- Grossman, A. R.; Schaefer, M. R.; Chiang, G. G.; and Collier, J. L. 1993. The phycobilisome, a light-harvesting complex responsive to environmental conditions. *Microbiol. Rev.* 57(3):725–49.
- Grossman, A. R.; Bhaya, D.; Apt, K. E.; and Kehoe, D. M. 1995. Light-harvesting complexes in oxygenic photosynthesis: Diversity, control, and evolution. *Annu. Rev. Genet.* 29:231–88.
- Partensky, F.; Hess, W. R.; and Vaulot, D. 1999. *Prochlorococcus*, a marine photosynthetic prokaryote of global significance. *Micro. Mol. Biol. Rev.* 63(1): 106–27.
- Peters, G. A. 1978. Blue-green algae and algal associations. *BioScience* 28(9):580–85.
- Rogers, L. J., and Gallon, J. R., editors, 1988. *Biochemistry of the algae and cyanobacteria*. New York: Oxford University Press.
- Stanier, R. Y., and Cohen-Bazire, G. 1977. Phototrophic prokaryotes: The cyanobacteria. *Annu. Rev. Microbiol.* 31:225–74.
- Wolk, C. P. 1996. Heterocyst formation. *Annu. Rev. Genet.* 30:59–78.

21.4 Phylum Planctomycetes

- Fuerst, J. A., and Webb, R. I. 1991. Membrane-bounded nucleoid in the eubacterium *Gemmata obscuriglobus*. *Proc. Natl. Acad. Sci.* 88:8184–88.

21.5 Phylum Chlamydiae

- Beatty, W. L.; Morrison, R. P.; and Byrne, G. I. 1994. Persistent chlamydiae: from cell culture to a paradigm for chlamydial pathogenesis. *Microbiol. Rev.* 58(4):686–99.

- Campbell, L. A.; Kuo, C.-C.; and Grayston, J. T. 1998. *Chlamydia pneumoniae* and cardiovascular disease. *Emerg. Infect. Dis.* 4(4):571–79.
- Hackstadt, T.; Fischer, E. R.; Scidmore, M. A.; Rockey, D. D.; and Heinzen, R. A. 1997. Origins and functions of the chlamydial inclusion. *Trends Microbiol.* 5(7):288–93.
- Hatch, T. P. 1996. Disulfide cross-linked envelope proteins: The functional equivalent of peptidoglycan in chlamydiae? *J. Bacteriol.* 178(1):1–5.
- McClarty, G. 1994. Chlamydiae and the biochemistry of intracellular parasitism. *Trends Microbiol.* 2(5):157–64.
- Moulder, J. W. 1991. Interaction of chlamydiae and host cells in vitro. *Microbiol. Rev.* 55(1):143–90.
- Raulston, J. E., and Wyrick, P. B. 2000. Chlamydia. In *Encyclopedia of microbiology*, 2d ed., vol. 1, J. Lederberg, editor-in-chief, 781–88. San Diego: Academic Press.

21.6 Phylum Spirochaetes

- Canale-Parola, E. 1978. Motility and chemotaxis of spirochetes. *Annu. Rev. Microbiol.* 32:69–99.
- Harwood, C. S., and Canale-Parola, E. 1984. Ecology of spirochetes. *Annu. Rev. Microbiol.* 38:161–92.
- Holt, S. C. 1978. Anatomy and chemistry of spirochetes. *Microbiol. Rev.* 42(1):114–60.
- Margulis, L. 2000. Spirochetes. In *Encyclopedia of microbiology*, 2d ed., vol. 4., J. Lederberg, editor-in-chief, 353–63. San Diego: Academic Press.

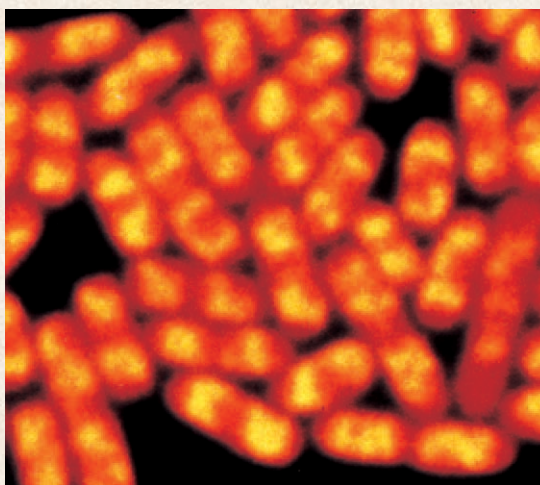
- Radolf, J. D. 1994. Role of outer membrane architecture in immune evasion by *Treponema pallidum* and *Borrelia burgdorferi*. *Trends Microbiol.* 2(9):307–11.
- Radoff J. D.; Steiner, B.; and Shevchenko, D. 1999. *Treponema pallidum*: Doing a remarkable job with what it's got. *Trends Microbiol.* 7(1):7–9.
- Saint Girons, I.; Old, I. G.; and Davidson, B. E. 1994. Molecular biology of the *Borrelia*, bacteria with linear replicons. *Microbiology* 140:1803–16.

21.7 Phylum Bacteroidetes

- Burchart, R. P. 1981. Gliding motility of prokaryotes: Ultrastructure, physiology, and genetics. *Annu. Rev. Microbiol.* 35:497–529.
- McBride, M. J. 2000. Bacterial gliding motility: Mechanisms and mysteries. *ASM News* 66(4):203–10.
- Reichenbach, H. 1981. Taxonomy of the gliding bacteria. *Annu. Rev. Microbiol.* 35:339–64.
- Takeuchi, M.; Sakane, T.; Yanagi, M.; Yamasato, K.; Hamana, K.; and Yokota, A. 1995. Taxonomic study of bacteria isolated from plants: Proposal of *Sphingomonas rosa* sp. nov., *Sphingomonas pruni* sp. nov., *Sphingomonas asaccharolytica* sp. nov., and *Sphingomonas mali* sp. nov. *Int. J. Syst. Bacteriol.* 45(2):334–41.

CHAPTER 22

Bacteria: The Proteobacteria



Salmonella typhi, stained here with fluorescent acridine orange, is a significant human pathogen. It causes typhoid fever.

Outline

22.1	Class <i>Alphaproteobacteria</i> 487 The Purple Nonsulfur Bacteria 487 <i>Rickettsia</i> and <i>Coxiella</i> 488 The <i>Caulobacteraceae</i> and <i>Hyphomicrobiaceae</i> 490 Family <i>Rhizobiaceae</i> 492 Nitrifying Bacteria 493	22.3	Class <i>Gammaproteobacteria</i> 498 The Purple Sulfur Bacteria 498 Order <i>Thiotrichales</i> 501 Order <i>Methylococcales</i> 502 Order <i>Pseudomonadales</i> 503 Order <i>Vibrionales</i> 504 Order <i>Enterobacteriales</i> 505 Order <i>Pasteurellales</i> 507
22.2	Class <i>Betaproteobacteria</i> 495 Order <i>Neisseriales</i> 495 Order <i>Burkholderiales</i> 495 Order <i>Nitrosomonadales</i> 496 Order <i>Hydrogenophilales</i> 496	22.4	Class <i>Deltaproteobacteria</i> 507 Orders <i>Desulfobacteriales</i> , <i>Desulfobacterales</i> , and <i>Desulfuromonadales</i> 507 Order <i>Bdellovibrionales</i> 510 Order <i>Myxococcales</i> 512
		22.5	Class <i>Epsilonproteobacteria</i> 514